

Project Data Page:

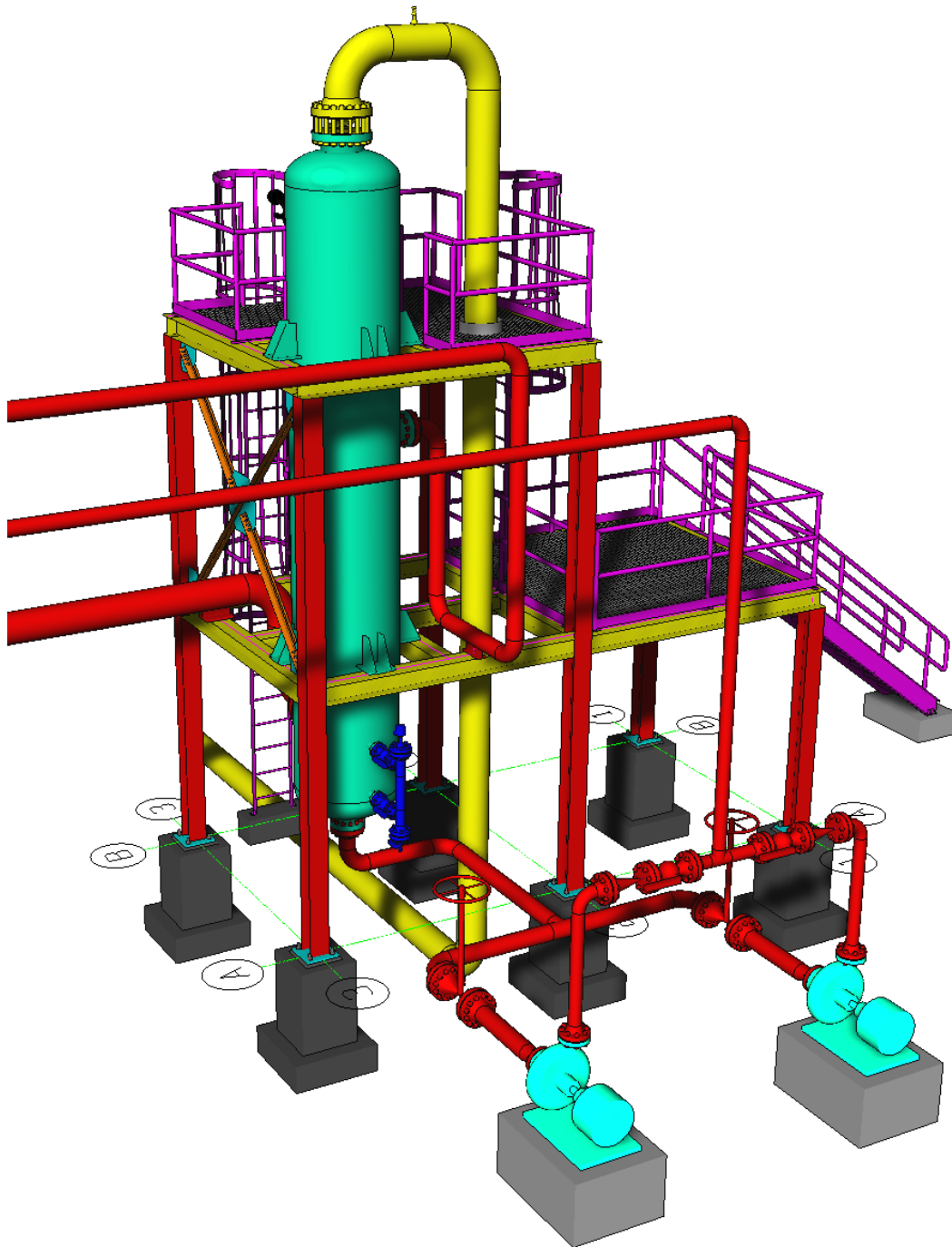




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Mechanical Design for Thermal Reclaimer
Item no: 559-V-105

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\C-17050pe-R0.pvdb

Date of Analysis : Aug 11, 2026 1:48pm

PV Elite 28 U2, April 2026

**Vessel Design Summary:****ASME Code, Section VIII Division 1, 2025**

Diameter Spec : 506.000 mm ID	
Vessel Design Length, Tangent to Tangent	12800.00 mm
Distance of Bottom Tangent above Grade	3500.00 mm
Specified Datum Line Distance	0.00 mm
Internal Design Temperature	225 °C
Internal Design Pressure	0.343 MPa
External Design Temperature	150 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	4 MPa
External Max. Allowable Working Pressure	1 MPa
Shop Test Pressure	4.864 MPa
Field Test Pressure	4.864 MPa
Required Minimum Design Metal Temperature	-1.0 °C
Warmest Computed Minimum Design Metal Temperature	-1.0 °C
Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	
Flange	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	
Nozzle	SA-106 B	...	...	K03006	
Nozzle Flg		...	...		
Nozzle Flg	SA-105	...	...	K03504	
Sup. Lugs	SA-516 70	...	...	K02700	+IT
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Lug Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Bottom head	0.343	0.10	4.000	1.0000	No	No
Shell-1	0.343	0.10	6.000	1.0000	N/A	No
Shell-2	0.343	0.10	6.000	1.0000	N/A	No
Flange-1	0.343	0.10	4.000	1.0000	N/A	No
Flange-2	0.343	0.10	4.000	1.0000	N/A	No
Shell-3	0.343	0.10	7.000	1.0000	N/A	No
Shell-4	0.343	0.10	6.000	1.0000	N/A	No
Flange-3	0.343	0.10	4.000	1.0000	N/A	No
Flange-4	0.343	0.10	4.000	1.0000	N/A	No
Shell-5	0.343	0.10	4.000	1.0000	N/A	No
Flange-5	0.343	0.10	4.000	1.0000	N/A	No
Flange-6	0.343	0.10	4.000	1.0000	N/A	No
Shell-6	0.343	0.10	4.000	1.0000	N/A	No



Top head	0.343	0.10	4.000	1.0000	No	No
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Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	50.0	50.0	10.0	8.0	2.5	2.4	1.00	1.00
Cylinder	3050.0	3000.0	12.0	12.0	2.5	5.4	1.00	1.00
Cylinder	5050.0	2000.0	12.0	12.0	2.5	5.4	1.00	1.00
Body Flg	5165.2	112.0	64.0	62.0	57.9	57.9	1.00	1.00
Body Flg	5277.2	112.0	64.0	62.0	57.9	57.9	1.00	1.00
Cylinder	8277.2	3000.0	14.0	14.0	2.5	5.3	1.00	1.00
Cylinder	9984.8	1707.7	12.0	12.0	2.5	5.3	1.00	1.00
Body Flg	10100.0	112.0	64.0	62.0	57.7	57.7	1.00	1.00
Body Flg	10212.0	112.0	64.0	62.0	57.7	57.7	1.00	1.00
Cylinder	12422.8	2210.8	8.0	8.0	2.5	4.1	1.00	1.00
Body Flg	12538.0	112.0	64.0	62.0	55.1	55.1	1.00	1.00
Body Flg	12650.0	112.0	64.0	62.0	55.1	55.1	1.00	1.00
Cylinder	12750.0	100.0	8.0	8.0	2.5	2.1	1.00	1.00
Ellipse	12800.0	50.0	10.0	8.0	2.5	2.4	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1 2"	WRC-107/537	0.884	n/a	Passed
N10_4"	WRC-297	0.640	Shell	Passed
N2 2"	WRC-297	0.638	Shell	Passed
N3 2"	WRC-297	0.638	Shell	Passed
N11_8"	WRC-297	0.957	Shell	Passed
N4 6"	WRC-297	0.771	Shell	Passed
N5 2"	WRC-297	0.993	Shell	Passed
N6 12"	WRC-107/537	0.513	n/a	Passed

Loads for Foundation/Support Design:

Total Wind Shear on Support	30676.	N
Total Earthquake Shear on Support	37054.	N
Wind Moment on Support	174364528.	N-mm
Earthquake Moment on Support	215528608.	N-mm
Wind Load Case Test Moment on Support	147259280.	N-mm
Earthquake Load Case Test Moment on Support	60077012.	N-mm
Test Case Wind Shear on Support	15338.	N
Maximum Initial Bolt Load	223438.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	6176.4	kgm
Shop Test - Fabricated + water (full)	8721.8	kgm
Shipping - Fab. + rem. intls.+ shipping app.	6176.4	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	6176.4	kgm
Empty - Fab. + intls. + details + wgths.	6335.6	kgm
Operating - Empty + operating liquid (no ca)	14595.9	kgm
Field Test - Empty weight + test medium (full)	8721.8	kgm

**Nozzle Calculation Summary (MPa & mm):**

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1 2"	4	...	4	OK	5.80	OK	No Calc[*]
N7 2"	36	OK	40	OK	2.50	OK	Passed
N10_4"	6	OK	6	OK	6.73	OK	Passed
N2 2"	6	...	6	OK	5.80	OK	No Calc[*]
N8 2"	6	...	6	OK	6.30	OK	No Calc[*]
N3 2"	6	...	6	OK	5.80	OK	No Calc[*]
K1A 2"	6	...	6	OK	5.80	OK	No Calc[*]
K2A 2"	6	...	6	OK	5.80	OK	No Calc[*]
K3A 2"	6	...	6	OK	5.80	OK	No Calc[*]
K4A 2"	6	...	6	OK	6.30	OK	No Calc[*]
K1B 2"	6	...	6	OK	6.30	OK	No Calc[*]
K2B 2"	6	...	6	OK	5.80	OK	No Calc[*]
K3B 2"	6	...	6	OK	6.30	OK	No Calc[*]
K4B 2"	6	...	6	OK	5.80	OK	No Calc[*]
N11_8"	6	OK	6	OK	8.00	OK	Passed
A3 8"	6	OK	6	...		OK	Passed
K6 2"		...	...	OK	5.80	OK	No Calc[*]
P1A 2"	7	...	7	OK	5.80	OK	No Calc[*]
P1B 2"		...	...	OK	5.80	OK	No Calc[*]
N4 6"		...	...	OK	7.22	OK	No Calc[*]
A2 8"		...	...	...		OK	No Calc[*]
N5 2"	4	...	4	OK	5.80	OK	No Calc[*]
K5A 2"	4	...	4	OK	5.80	OK	No Calc[*]
K5B 2"	4	...	4	OK	5.80	OK	No Calc[*]
K5C 2"	4	...	4	OK	5.80	OK	No Calc[*]
P1C 2"	4	...	4	OK	5.80	OK	No Calc[*]
P1D 2"	4	...	4	OK	5.80	OK	No Calc[*]
A1 8"	4	OK	4	...		OK	Passed
N6 12"	4	OK	4	OK	7.87	OK	Passed
N9 2"	4	...	4	OK	5.80	OK	No Calc[*]
K7 2"	4	...	4	OK	5.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 4 Nozzle : A1 8"
 Minimum MAPnc Nozzles : 4 Nozzle : N6 12"
 Minimum MAWP Shells/Flanges : 4 Element : Shell-6
 Minimum MAPnc Shells/Flanges : 4 Element : Flange-4

Computed Vessel M.A.W.P. : 4 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
10	N1 2"	0.000	0.000	105.600
10	N7 2"	0.000	0.000	118.524
20	N10_4"	1550.000	0.000	207.200
20	N2 2"	150.000	0.000	105.600
20	N8 2"	150.000	180.000	107.600
20	N3 2"	2050.000	0.000	105.600
20	K1A 2"	550.000	180.000	106.328
20	K2A 2"	1550.000	180.000	106.328
20	K3A 2"	1050.000	180.000	106.328



20	K4A 2"	1050.000	90.000	107.600
20	K1B 2"	550.000	240.000	107.600
20	K2B 2"	1550.000	240.000	106.328
20	K3B 2"	1050.000	240.000	107.600
20	K4B 2"	550.000	90.000	106.328
20	N11 8"	1050.000	0.000	410.400
20	A3 8"	2550.000	270.000	410.400
60	K6 2"	6777.180	0.000	110.328
60	P1A 2"	6377.180	0.000	110.328
70	P1B 2"	9377.161	0.000	106.328
70	N4 6"	9677.171	180.000	296.659
70	A2 8"	9077.170	180.000	410.400
100	N5 2"	12012.000	0.000	105.600
100	K5A 2"	10312.001	180.000	105.600
100	K5B 2"	10312.001	90.000	105.600
100	K5C 2"	10312.001	270.000	105.600
100	P1C 2"	10312.001	0.000	105.600
100	P1D 2"	11812.000	0.000	105.600
100	A1 8"	12012.000	180.000	410.400
140	N6 12"	0.000	0.000	610.450
140	N9 2"	0.000	0.000	105.600
140	K7 2"	0.000	180.000	105.600

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

Diameter Limit clash between : N9 2" and N6 12".

The Centerline Dist. is 207.00 while the sum of the limit Radii is 358.03 mm.

Diameter Limit clash between : K7 2" and N6 12".

The Centerline Dist. is 207.00 while the sum of the limit Radii is 358.03 mm.

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	206.78	600
N7 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	292.16	300
N2 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	571.55	600
N8 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	571.55	300
N3 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	224.55	600
K1A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K2A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K3A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K4A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K1B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K2B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K3B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K4B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
K6 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	223.91	300
P1A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	223.91	300
P1B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	221.91	300
N5 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	567.55	600
K5A 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	217.91	300
K5B 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	217.91	300
K5C 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	217.91	300
P1C 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	217.91	300
P1D 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	217.91	300
N9 2"	2.000 in	Actual	LWN	3.060	13.462	...	...	219.19	300
K7 2"	2.000 in	Actual	LWN	3.320	16.764	...	...	216.30	300
N10 4"	4.000 in	Actual	WNF	4.630	8.000	...	...	325.25	600
N4 6"	6.000 in	80	WNF	6.625	10.973	...	...	612.00	600
N11 8"	8.000 in	Actual	WNF	8.787	10.000	...	...	268.83	600
A3 8"	8.000 in	Actual	WNF	8.787	10.000	...	...	312.45	300
A2 8"	8.000 in	Actual	WNF	8.787	10.000	...	...	312.45	300
A1 8"	8.000 in	Actual	WNF	8.787	10.000	...	...	308.45	300
N6 12"	12.000 in	40	WNF	12.750	10.312	...	...	293.00	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1 2"	SA-105	6.000	10.000	...	...	...
N7 2"	SA-105	6.000	10.000	...	...	...
N2 2"	SA-105	6.000	10.000	...	...	10.000
N8 2"	SA-105	6.000	10.000	...	...	12.000
N3 2"	SA-105	6.000	10.000	...	...	...
K1A 2"	SA-105	6.000	10.000	...	...	...
K2A 2"	SA-105	6.000	10.000	...	...	...
K3A 2"	SA-105	6.000	10.000	...	...	...
K4A 2"	SA-105	6.000	10.000	...	...	...
K1B 2"	SA-105	6.000	10.000	...	...	...
K2B 2"	SA-105	6.000	10.000	...	...	...
K3B 2"	SA-105	6.000	10.000	...	...	...
K4B 2"	SA-105	6.000	10.000	...	...	...



K6 2"	SA-105	12.000	10.000	...	...	...
P1A 2"	SA-105	12.000	10.000	...	...	...
P1B 2"	SA-105	12.000	10.000	...	...	...
N5 2"	SA-106 B	8.000	10.000	...	...	...
K5A 2"	SA-105	8.000	10.000	...	...	...
K5B 2"	SA-105	8.000	10.000	...	...	...
K5C 2"	SA-105	8.000	10.000	...	...	...
P1C 2"	SA-105	8.000	10.000	...	...	...
P1D 2"	SA-105	8.000	10.000	...	...	...
N9 2"	SA-105	8.000	10.000	...	...	...
K7 2"	SA-105	8.000	10.000	...	...	...
N10_4"	SA-105	6.000	10.000	...	...	...
N4 6"	SA-105	12.000	10.000	...	...	10.000
N11_8"	SA-105	10.000	10.000	...	...	...
A3 8"	SA-105	8.000	10.000	...	...	...
A2 8"	SA-105	12.000	10.000	...	...	...
A1 8"	SA-105	8.000	10.000	...	...	...
N6 12"	SA-105	8.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1 2"	...	0.0	196.85	0.00	Bottom head
N7 2"	...	0.0	250.00	0.00	N1 2"
N2 2"	150.000	0.0	209.55	350.00	Shell-1
N8 2"	150.000	180.0	209.55	350.00	Shell-1
N3 2"	2050.000	0.0	209.55	0.00	Shell-1
K1A 2"	550.000	180.0	206.38	0.00	Shell-1
K2A 2"	1550.000	180.0	206.38	0.00	Shell-1
K3A 2"	1050.000	180.0	206.38	0.00	Shell-1
K4A 2"	1050.000	90.0	206.38	0.00	Shell-1
K1B 2"	550.000	240.0	206.38	0.00	Shell-1
K2B 2"	1550.000	240.0	206.38	0.00	Shell-1
K3B_2"	1050.000	240.0	206.38	0.00	Shell-1
K4B 2"	550.000	90.0	206.38	0.00	Shell-1
K6 2"	6777.180	0.0	206.38	0.00	Shell-3
P1A 2"	6377.180	0.0	206.38	0.00	Shell-3
P1B 2"	9377.160	0.0	206.38	0.00	Shell-4
N5 2"	12012.000	0.0	209.55	350.00	Shell-5
K5A 2"	10312.000	180.0	206.38	0.00	Shell-5
K5B 2"	10312.000	90.0	206.38	0.00	Shell-5
K5C 2"	10312.000	270.0	206.38	0.00	Shell-5
P1C 2"	10312.000	0.0	206.38	0.00	Shell-5
P1D 2"	11811.999	0.0	206.38	0.00	Shell-5
N9 2"	...	0.0	209.55	0.00	Top head
K7 2"	...	180.0	206.38	0.00	Top head
N10_4"	1550.000	0.0	300.00	0.00	Shell-1
N4 6"	9677.170	180.0	250.00	350.00	Shell-4
N11_8"	1050.000	0.0	206.38	0.00	Shell-1
A3 8"	2550.000	270.0	250.00	0.00	Shell-1
A2 8"	9077.170	180.0	250.00	0.00	Shell-4
A1 8"	12012.000	180.0	250.00	0.00	Shell-5
N6 12"	...	0.0	250.00	0.00	Top head

**Bill of Materials:**

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 10.0mm THK X 506.0mm ID X 50.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 506.0mm ID X 2000.0mm	SA-516 70
6	WELD NECK FLANGE: 64.0mm THK X 706.7mm OD	SA-516 70
1	CYLINDER: 14.0mm THK X 506.0mm ID X 3000.0mm	SA-516 70
1	CYLINDER: 12.0mm THK X 506.0mm ID X 1707.7mm	SA-516 70
1	CYLINDER: 8.0mm THK X 506.0mm ID X 2210.8mm	SA-516 70
1	CYLINDER: 8.0mm THK X 506.0mm ID X 100.0mm	SA-516 70
2	LINING: 176.500mm X 3.000mm THK	...
2	LINING: 3000.000mm X 3.000mm THK	...
1	LINING: 2000.000mm X 3.000mm THK	...
6	LINING: 112.000mm X 3.000mm THK	...
1	PACKING: 1800.000mm	...
8	GUSSET PLATES: 200.0mm X 100.0mm X 356.0mm X 30.0mm	SA-516 70
1	TOP RING: 520.0mm X 720.0mm X 32.0mm	SA-516 70
1	BOTTOM RING: 520mm X 920mm X 42mm	SA-516 70
4	LUG BOLTS: 56.0mm TEMA Metric	SA-193 B7
1	PACKING: 1200.000mm	...
1	LINING: 1707.650mm X 3.000mm THK	...
1	PACKING: 1500.000mm	...
1	LINING: 2210.820mm X 3.000mm THK	...
1	LINING: 100.000mm X 3.000mm THK	...
3	CLASS 300 GR 1.1, 8.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	GASKET: 580.3mm OD X 550.7mm ID	...
48	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
96	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
2	GASKET: 580.7mm OD X 550.7mm ID	...
2	CYLINDRICAL SEGMENT 12.0mm THK X 632.5mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm THK X 324.6mm X 3000.0mm	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Flange-1	[11]	!	-1	-21	0.642	12.000	1.00	No
Flange-2	[11]	!	-1	-28	0.546	12.000	1.00	No
Flange-3	[11]	!	-1	-21	0.642	12.000	1.00	No
Flange-4	[11]	!	-1	-6	0.917	12.000	1.00	No
Flange-5	[11]	!	-1	-8	0.874	8.000	1.00	No
Flange-6	[11]	!	-1	-8	0.874	8.000	1.00	No
Bottom head	[10]	B	-29	-30	0.981	8.000	1.00	No
Bottom head	[7]	B	-29	-41	0.778	10.000	1.00	No
Shell-1	[8]	!	-1	-21	0.636	12.000	1.00	No
Shell-2	[8]	!	-1	-21	0.636	12.000	1.00	No
Shell-3	[20]	!	-1	-16	0.723	14.000	1.00	No
Shell-4	[8]	!	-1	-21	0.636	12.000	1.00	No
Shell-5	[8]	!	-1	-1	1.000	8.000	1.00	No
Shell-6	[8]	!	-1	-1	1.000	8.000	1.00	No
Top head	[10]	B	-29	-30	0.981	8.000	1.00	No
Top head	[7]	B	-29	-41	0.778	10.000	1.00	No
N1 2"	[1]	A	-8	-29	0.884	8.000	1.00	No
N7 2"	[1]	A	1	-104	0.058	13.462	1.00	No
N104"	[1]	A	-3	-29	0.636	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48				
N2 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
N8 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
N3 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K1A 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K2A 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K3A 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K4A 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
K1B 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K2B 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K3B2"	[1]	A	-3	-29	0.636	12.000	1.00	No
K4B 2"	[1]	A	-3	-29	0.636	12.000	1.00	No
N118"	[1]	A	-3	-29	0.636	12.000	1.00	No
A3 8"	[1]	!	-1	-21	0.636	12.000	1.00	No
K6 2"	[1]	!	-1	-29	0.538	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
P1A 2"	[1]	!	-1	-29	0.538	14.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
P1B 2"	[1]	!	-1	-21	0.636	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
N4 6"	[1]	!	-1	-21	0.636	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48				
A2 8"	[1]	!	-1	-21	0.636	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
N5 2"	[1]	B	-29	-29	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-48				
K5A 2"	[1]	!	-1	-1	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
K5B 2"	[1]	!	-1	-1	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
K5C 2"	[1]	!	-1	-1	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
P1C 2"	[1]	!	-1	-1	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
P1D 2"	[1]	!	-1	-1	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
A1 8"	[1]	!	-1	-1	1.000	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33				
N6 12"	[1]	B	-8	-37	0.524	9.023	1.00	No
Nozzle Flg	[5]	A	-18	-33				
N9 2"	[1]	B	-29	-30	0.981	8.000	1.00	No



MDMT Summary: Step: 71 1:48pm Aug 11, 2026

Nozzle Flg	[5]	A	-18	-33					
K7 2"	[1]	B	-29	-30	-29	0.981	8.000	1.00	No
Nozzle Flg	[5]	A	-18	-33					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: 1 -1

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -1.0 °C

Notes:

[!] - This was an impact tested material.

[1] - Governing Nozzle Weld.

[4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).

[5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).

[6] - MDMT Calculations at the Shell/Head Joint.

[7] - MDMT Calculations for the Straight Flange.

[8] - Cylinder/Cone/Flange Junction MDMT.

[9] - Calculations in the Spherical Portion of the Head.

[10] - Calculations in the Knuckle Portion of the Head.

[11] - Calculated (Body Flange) Flange MDMT.

[12] - Calculated Flat Head MDMT per UCS-66.3

[13] - Tubesheet MDMT, shell side, if applicable

[14] - Tubesheet MDMT, tube side, if applicable

[15] - Nozzle Material

[16] - Shell or Head Material

[17] - Impact Testing required

[18] - Impact Testing not required, see UCS-66(b)(3)

[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations

[21] - Body Flange Bolting Material

[22] - Nozzle Flange Bolting Material

[23] - Stiffening Ring to Shell Weld

[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note	10	20	The wind diameter multiplier is very big.
Note	20	30	The wind diameter multiplier is very big.
Note	30	40	The wind diameter multiplier is very big.
Note	40	50	The wind diameter multiplier is very big.
Note	50	60	The wind diameter multiplier is very big.
Note	60	70	The wind diameter multiplier is very big.
Note	70	80	The wind diameter multiplier is very big.
Note	80	90	The wind diameter multiplier is very big.
Note	90	100	The wind diameter multiplier is very big.
Note	100	110	The wind diameter multiplier is very big.
Note	110	120	The wind diameter multiplier is very big.
Note	120	130	The wind diameter multiplier is very big.
Note	130	140	The wind diameter multiplier is very big.
Note	140	150	The wind diameter multiplier is very big.

Class From To: Check of Additional Element Data

Note: Lug supports should be modeled on an element of their own whose length is just slightly greater than the height of the lug assembly. The vessel does not have to be constructed with this extra element, just modeled this way. This is to insure there is proper mass distribution for these types of models.

Warning:

This is a vertical vessel on support lugs and the earthquake type selected is a building code. Building codes assume all of the mass is located above the point of support. This makes their use on lug supported vessels questionable. It is strongly suggested that based on the seismic zone and applicable parameters, an appropriate lateral (Gx) and (+-) vertical (Gy) acceleration be applied using the G loading option.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not



obtain proper resolution on some results. This value can be changed in
Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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**Units used in this Analysis (SIASME):**

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Mechanical Design for Thermal Reclaimer

Item no: 559-V-105

Design Internal Pressure (for Hydrotest)	0.3432	MPa
Design Internal Temperature	225.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	Amine	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS



Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	3500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.450
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	Y
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	No
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes



Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/14
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Bottom head
Distance "FROM" to "TO"	50 mm
Inside Diameter	506 mm
Element Thickness	8 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	221 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No
Element From Node	10
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	-126.5 mm
Height/Length of Lining	176.5 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³
Element From Node	10
Detail Type	Nozzle
Detail ID	N1 2"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	352.85 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	10
Detail Type	Nozzle
Detail ID	N7 2"
Dist. from "FROM" Node / Offset dist	150 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	84.382 N



Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105

Element#	2/14
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell-1

Distance "FROM" to "TO"	3000	mm
Inside Diameter	506	mm
Element Thickness	12	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	12	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3432	MPa
Design Temperature Internal Pressure	225	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	2.1400001	
Material Name	SA-516 70	+IT
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N104"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	4	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	226.07	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2 2"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	166.29	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N8 2"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	160.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	N3 2"	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	93.998	N
Grade of Attached Flange	GR 1.11	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K1A 2"	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.928	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K2A 2"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.928	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K3A 2"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.928	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	K4A 2"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	99.928	N
Grade of Attached Flange	GR 1.1	



Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K1B 2
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	240.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.928 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K2B 2"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	240.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	104.65 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K3B2"
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	240.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.928 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	K4B 2"
Dist. from "FROM" Node / Offset dist	500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	104.65 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20
Detail Type	Nozzle
Detail ID	N118"
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	8 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	625.75 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105
Element From Node	20



Detail Type	Nozzle
Detail ID	A3 8"
Dist. from "FROM" Node / Offset dist	2500 mm
Nozzle Diameter	8 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	1072.7 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	20
Detail Type	Weight
Detail ID	REBOILEER
Dist. from "FROM" Node / Offset dist	1250 mm
Miscellaneous Weight	58267 N
Offset from Element Centerline	1500 mm

Element From Node	20
Detail Type	Weight
Detail ID	N11 Fy wt
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	5550.2 N
Offset from Element Centerline	1367 mm

Element From Node	20
Detail Type	Weight
Detail ID	N8 dist wt
Dist. from "FROM" Node / Offset dist	150 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	20
Detail Type	Weight
Detail ID	N8 dist sup wt
Dist. from "FROM" Node / Offset dist	100 mm
Miscellaneous Weight	98.06 N
Offset from Element Centerline	0 mm

Element From Node	20
Detail Type	Weight
Detail ID	A3 davit wt
Dist. from "FROM" Node / Offset dist	2500 mm
Miscellaneous Weight	294.18 N
Offset from Element Centerline	300 mm

Element#	3/14
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	2000 mm
Inside Diameter	506 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0



Weld is pre-Heated	No
Element From Node	30
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	2000 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³
Element From Node	30
Detail Type	Weight
Detail ID	CLEAT WEIGHT
Dist. from "FROM" Node / Offset dist	1450 mm
Miscellaneous Weight	588.36 N
Offset from Element Centerline	0 mm

Element#	4/14
Element From Node	40
Element To Node	50
Element Type	Flange
Description	Flange-1
Distance "FROM" to "TO"	112 mm
Flange Inside Diameter	506 mm
Element Thickness	62 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	64 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	40
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	112 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³

Element#	5/14
Element From Node	50
Element To Node	60
Element Type	Flange
Description	Flange-2
Distance "FROM" to "TO"	112 mm
Flange Inside Diameter	506 mm
Element Thickness	62 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	64 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



Element From Node	50
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	112 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³

Element#	6/14
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	506 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Packing
Detail ID	Pack: [1 of 1]
Dist. from "FROM" Node / Offset dist	1200 mm
Height of Packed Section	1800 mm
Density	0.00018 kg/cm ³
Percent Volume Holdup	10.0
Specific Gravity of Packing Liquid	1.109

Element From Node	60
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	K6 2"
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	101.01 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	60
Detail Type	Nozzle
Detail ID	P1A 2"
Dist. from "FROM" Node / Offset dist	1100 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None



Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	101.01 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	60
Detail Type	Lug
Detail ID	Support Lugs
Dist. from "FROM" Node / Offset dist	1000 mm
Number of Lugs	4
Dist. from OD to Lug Cntrline(dlug)	150 mm
Height of Gusset Plates (hgp)	356 mm
Force Bearing Width (wfb)	46 mm
Weight of Lug	732.64 N
Lug Start Angle (degrees)	0.0

Element From Node	60
Detail Type	Weight
Detail ID	Noz clad wt
Dist. from "FROM" Node / Offset dist	1500 mm
Miscellaneous Weight	294.18 N
Offset from Element Centerline	0 mm

Element From Node	60
Detail Type	Weight
Detail ID	Bed2 sup ring w
Dist. from "FROM" Node / Offset dist	1150 mm
Miscellaneous Weight	196.12 N
Offset from Element Centerline	0 mm

Element From Node	60
Detail Type	Weight
Detail ID	Bed2 sup grid w
Dist. from "FROM" Node / Offset dist	1200 mm
Miscellaneous Weight	196.12 N
Offset from Element Centerline	0 mm

Element#	7/14
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	1707.7 mm
Inside Diameter	506 mm
Element Thickness	12 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	12 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Packing
Detail ID	Bed2
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	1200 mm
Density	0.00018 kg/cm³



Percent Volume Holdup	10.0	
Specific Gravity of Packing Liquid	1.109	
Element From Node	70	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1707.7	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	P1B 2"	
Dist. from "FROM" Node / Offset dist	1100	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	101.01	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	N4 6"	
Dist. from "FROM" Node / Offset dist	1400	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1351.6	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	A2 8"	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	8	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	783.66	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	70	
Detail Type	Weight	
Detail ID	N4 Fy wt	
Dist. from "FROM" Node / Offset dist	1400	mm
Miscellaneous Weight	7678.1	N
Offset from Element Centerline	1339	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	N4 dist wt	
Dist. from "FROM" Node / Offset dist	1400	mm
Miscellaneous Weight	588.36	N
Offset from Element Centerline	0	mm
Element From Node	70	
Detail Type	Weight	
Detail ID	N4 dist sup wt	
Dist. from "FROM" Node / Offset dist	1350	mm



Miscellaneous Weight 196.12 N
Offset from Element Centerline 0 mm

Element From Node 70
Detail Type Weight
Detail ID Bed2 limiter
Dist. from "FROM" Node / Offset dist 1250 mm
Miscellaneous Weight 196.12 N
Offset from Element Centerline 0 mm

Element From Node 70
Detail Type Weight
Detail ID A2 davit wt
Dist. from "FROM" Node / Offset dist 800 mm
Miscellaneous Weight 294.18 N
Offset from Element Centerline 300 mm

Element# 8/14
Element From Node 80
Element To Node 90
Element Type Flange
Description Flange-3
Distance "FROM" to "TO" 112 mm
Flange Inside Diameter 506 mm
Element Thickness 62 mm
Internal Corrosion Allowance 1 mm
Nominal Thickness 64 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 0.3432 MPa
Design Temperature Internal Pressure 225 °C
Design External Pressure 0.1034 MPa
Design Temperature External Pressure 150 °C
Effective Diameter Multiplier 2.1400001
Material Name SA-516 70 +IT
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No

Element From Node 80
Detail Type Lining
Detail ID Clad
Dist. from "FROM" Node / Offset dist 0 mm
Height/Length of Lining 112 mm
Thickness of Lining 3 mm
Density 0.008027 kg/cm³

Element# 9/14
Element From Node 90
Element To Node 100
Element Type Flange
Description Flange-4
Distance "FROM" to "TO" 112 mm
Flange Inside Diameter 506 mm
Element Thickness 62 mm
Internal Corrosion Allowance 1 mm
Nominal Thickness 64 mm
External Corrosion Allowance 0 mm
Design Internal Pressure 0.3432 MPa
Design Temperature Internal Pressure 225 °C
Design External Pressure 0.1034 MPa
Design Temperature External Pressure 150 °C
Effective Diameter Multiplier 2.1400001
Material Name SA-516 70 +IT
Perform Flange Stress Calculation (Y/N) Y
Weld is pre-Heated No



Element From Node	90
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	112 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³

Element#	10/14
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Shell-5
Distance "FROM" to "TO"	2210.8 mm
Inside Diameter	506 mm
Element Thickness	8 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	8 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Packing
Detail ID	Bed 1
Dist. from "FROM" Node / Offset dist	200 mm
Height of Packed Section	1500 mm
Density	0.00018 kg/cm ³
Percent Volume Holdup	10.0
Specific Gravity of Packing Liquid	1.109

Element From Node	100
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	2210.8 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³

Element From Node	100
Detail Type	Nozzle
Detail ID	N5 2"
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	76.031 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	100
Detail Type	Nozzle
Detail ID	K5A 2"
Dist. from "FROM" Node / Offset dist	100 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None



Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.933 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	100
Detail Type	Nozzle
Detail ID	K5B 2"
Dist. from "FROM" Node / Offset dist	100 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.933 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	100
Detail Type	Nozzle
Detail ID	K5C 2"
Dist. from "FROM" Node / Offset dist	100 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.933 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	100
Detail Type	Nozzle
Detail ID	P1C 2"
Dist. from "FROM" Node / Offset dist	100 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.928 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	100
Detail Type	Nozzle
Detail ID	P1D 2"
Dist. from "FROM" Node / Offset dist	1600 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	99.928 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT
Element From Node	100
Detail Type	Nozzle
Detail ID	A1 8"
Dist. from "FROM" Node / Offset dist	1800 mm
Nozzle Diameter	8 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	Y



Weight of Nozzle (Used if > 0)	1060.6	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT

Element From Node	100	
Detail Type	Weight	
Detail ID	Lifting lug1	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	280	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Lifting lug 2	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	280	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	N5 dist wt	
Dist. from "FROM" Node / Offset dist	1800	mm
Miscellaneous Weight	588.36	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	N5 dist sup wt	
Dist. from "FROM" Node / Offset dist	1750	mm
Miscellaneous Weight	196.12	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Bed 1 limiter	
Dist. from "FROM" Node / Offset dist	1700	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Bed1 sup ring	
Dist. from "FROM" Node / Offset dist	150	mm
Miscellaneous Weight	98.06	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	Bed1 sup grid	
Dist. from "FROM" Node / Offset dist	180	mm
Miscellaneous Weight	196.12	N
Offset from Element Centerline	0	mm

Element From Node	100	
Detail Type	Weight	
Detail ID	A1 davit wt	
Dist. from "FROM" Node / Offset dist	1800	mm
Miscellaneous Weight	294.18	N
Offset from Element Centerline	300	mm

Element#	11/14	
Element From Node	110	
Element To Node	120	
Element Type	Flange	
Description	Flange-5	
Distance "FROM" to "TO"	112	mm



Flange Inside Diameter	506	mm
Element Thickness	62	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	64	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3432	MPa
Design Temperature Internal Pressure	225	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	2.1400001	
Material Name	SA-516 70	+IT
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	110	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	112	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm³

Element#	12/14	
Element From Node	120	
Element To Node	130	
Element Type	Flange	
Description	Flange-6	
Distance "FROM" to "TO"	112	mm
Flange Inside Diameter	506	mm
Element Thickness	62	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	64	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3432	MPa
Design Temperature Internal Pressure	225	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	2.1400001	
Material Name	SA-516 70	+IT
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	112	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm³

Element#	13/14	
Element From Node	130	
Element To Node	140	
Element Type	Cylinder	
Description	Shell-6	
Distance "FROM" to "TO"	100	mm
Inside Diameter	506	mm
Element Thickness	8	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	8	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3432	MPa
Design Temperature Internal Pressure	225	°C
Design External Pressure	0.1034	MPa



Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	130
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	100 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm³

Element#	14/14
Element From Node	140
Element To Node	150
Element Type	Elliptical
Description	Top head
Distance "FROM" to "TO"	50 mm
Inside Diameter	506 mm
Element Thickness	8 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3432 MPa
Design Temperature Internal Pressure	225 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	2.1400001
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	140
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	176.5 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm³

Element From Node	140
Detail Type	Nozzle
Detail ID	N6 12"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	12 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	863.04 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-105 +IT

Element From Node	140
Detail Type	Nozzle
Detail ID	N9 2"
Dist. from "FROM" Node / Offset dist	200 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0



Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	140	
Detail Type	Nozzle	
Detail ID	K7 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	133.41	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	140	
Detail Type	Weight	
Detail ID	N6 Fy wt	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	12110	N
Offset from Element Centerline	1415	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	PIPE DAVIT	
Dist. from "FROM" Node / Offset dist	0.004651	mm
Miscellaneous Weight	4903	N
Offset from Element Centerline	280	mm
Element From Node	140	
Detail Type	Weight	
Detail ID	Insert plate wt	
Dist. from "FROM" Node / Offset dist	125	mm
Miscellaneous Weight	294.18	N
Offset from Element Centerline	0	mm
Element From Node	140	
Detail Type	For./Mom.	
Detail ID	N6 F/M	
Dist. from "FROM" Node / Offset dist	125	mm
Force in X Direction	16500	N
Force in Y Direction	24750	N
Force in Z Direction	16500	N
Moment about X Axis	13860000	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	13860000	N-mm
Force/Moment Combination Method	SRSS	

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Bottom head		...	50	...	50
Shell-1		...	3050	...	3000
Shell-2		...	5050	...	2000
Flange-1		...	5165.17	...	112
Flange-2		...	5277.17	...	112
Shell-3		...	8277.17	...	3000
Shell-4		...	9984.83	...	1707.65
Flange-3		...	10100	...	112
Flange-4		...	10212	...	112
Shell-5		...	12422.8	...	2210.82
Flange-5		...	12538	...	112
Flange-6		...	12650	...	112
Shell-6		...	12750	...	100
Top head		...	12800	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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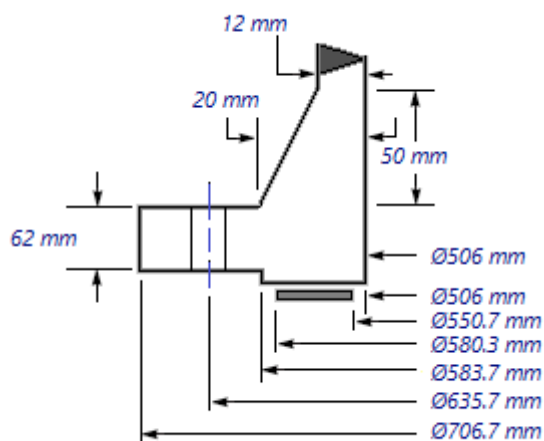


Flange Input Data Values

Description: FLANGE-1 :

Flange-1

Flange Type	Integral Weld Neck		
Design Pressure	P	0.34	MPa
Design Temperature		225	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm
Flange Outside Diameter	A	706.700	mm
Flange Thickness	t	62.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-516 70 +IT	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	635.700	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	583.700	mm
Flange Face Inside Diameter	Fid	506.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	580.300	mm
Gasket Inside Diameter	Gi	550.700	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:



$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.34 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca$$

$$= 1.6327 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (254.0 + 0.6 \cdot 11.0)$$

$$= 5.825 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (253.0 + 0.6 \cdot 12.0)$$

$$= 6.364 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm
Corroded Small Hub,	g0Cor = go-ci	11.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	44.850 mm
Gasket Contact Width,	N = (Go - Gi)/2	14.800 mm
Basic Gasket Width,	bo = N/2	7.400 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.855 mm
Gasket Reaction Diameter,	G = Go - 2 * b	566.590 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 566.5901^2 \cdot 0.343$$

$$= 86532.648 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.8549 \cdot \pi \cdot 566.5901 \cdot 3.0 \cdot 0.34$$

$$= 25126.088 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 0.3432$$

$$= 69561.586 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 86533 - 69562$$

$$= 16971.061 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(86533 + 25126 + 0, 0)$$

$$= 111658.734 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.8549 \cdot \pi \cdot 566.59 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 841237.812 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((111659 + 0 + 4 \cdot 0 / 566.59) / 172, 841238 / 172)$$

$$= 4891.421 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 62.0 / (3.0 + 0.5)$$

$$= 169.786 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 635.7 \cdot \sin(\pi/16)$$

$$= 124.019 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{124.019}{2 \cdot 31.75 + 62.0}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4891.421	9589.658	
Radial Distance between Hub and Bolts:	44.450	44.850	
Radial Distance between Bolts and Edge:	31.750	35.500	
Circ. Spacing between the Bolts:	71.450	124.019	169.786

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9589.658 \cdot 172.0 / (68.95 \cdot \pi(580.3 + 550.7))$$

$$= 6.733 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 9589.6582$$

$$= 1649251.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 111659 - 86533$$

$$= 25126.08 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (635.7 - 566.5901) / 2$$

$$= 34.5549 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((635.7 - 508.0) / 2 + 34.5549)$$

$$= 49.2025 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (635.7 - 508.0 - 19.0) / 2$$

$$= 54.3500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	69562.	54.3500	1.0000	3782204.
Face Pressure, Mt	16971.	49.2025	1.0000	835356.
Gasket Load, Mg	25126.	34.5549	1.0000	868582.
Gasket Seating, Matm	1649251.	34.5549	1.0000	57012888.

Total Moment for Operation, Mo 5486142. N-mm
 Total Moment for Gasket seating, Mg 57012888. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 74.753 mm
 Hub Ratio, h/h0 = HL / H0 0.669
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.807



Factor V per 4.16.5 0.232
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B \\ = 706.7/508.0 \\ = 1.391$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ = (0.66845 + (5.7169(1.391^2 \cdot \log_{10}(1.391))) / (1.391^2 - 1)) / (1.391 - 1) \\ = 6.045$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ = (1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / ((1.0472 + 1.9448 \cdot 1.391^2) \cdot (1.391 - 1)) \\ = 1.758$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ = (1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / (1.36136(1.391^2 - 1) \cdot (1.391 - 1)) \\ = 6.643$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1) \\ = (1.391^2 + 1) / (1.391^2 - 1) \\ = 3.138$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V \\ = 6.643 \cdot 11.0^2 \cdot 74.753 / 0.232 \\ = 258517.531 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0 \\ = 0.807 / 74.753 \\ = 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / (T + t^3/d) \\ = (62.0 \cdot 0.011 + 1) / (1.758 + 62.0^3/258517.531) \\ = 1.871$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ = (1.0 \cdot 5486142 / 508.0) / (1.8712 \cdot 19.0^2) \\ = 15.98 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ = (1.0 \cdot 57012884 / 508.0) / (1.8712 \cdot 19.0^2) \\ = 166.09 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ = (1.8919 \cdot 5486142 / 508.0) / (1.8712 \cdot 62.0^2) \\ = 2.84 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ = (1.8919 \cdot 57012884 / 508.0) / (1.8712 \cdot 62.0^2) \\ = 29.51 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ = (6.045 \cdot 5486142 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 3$$



= 8.07 MPa

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (6.045 \cdot 57012884 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 30$$

= 83.82 MPa

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (16 + \max(3, 8)) / 2$$

= 12.02 MPa

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (166 + \max(30, 84)) / 2$$

= 124.95 MPa

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 111659 / 9589.6582$$

= 11.64 MPa

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (841238 / 9589.6582)$$

= 87.73 MPa

Flange Stress Analysis Results: MPa

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	15.98	207.00	166.09	207.00
Radial Flange	2.84	138.00	29.51	138.00
Tangential Flange	8.07	138.00	83.82	138.00
Maximum Average	12.02	138.00	124.95	138.00
Bolting	11.64	172.00	87.73	172.00

Minimum Required Flange Thickness

57.887 mm

Estimated M.A.W.P. (Operating)

4 MPa

Estimated Finished Weight of Flange at given Thk.

102.0 kgm

Estimated Unfinished Weight of Forging at given Thk

165.9 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 57627740 / 1.0108 \cdot 1.0 \cdot 0.232 / (1.674 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.750 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5545308 / 1.0108 \cdot 1.0 \cdot 0.232 / (1.674 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.076 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 57012884 / 1.0 \cdot 1.0 \cdot 0.232 / (1.871 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.671 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5486142 / 1.0 \cdot 1.0 \cdot 0.232 / (1.871 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.068 (should be ≤ 1)

**Minimum Design Metal Temperature Results:**

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	15.66	207.00	161.51	207.00
Radial Flange	3.04	138.00	31.33	138.00
Tangential Flange	7.55	138.00	77.87	138.00
Maximum Average	11.60	138.00	119.69	138.00
Bolting	11.64	172.00	87.73	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 55.905 mm

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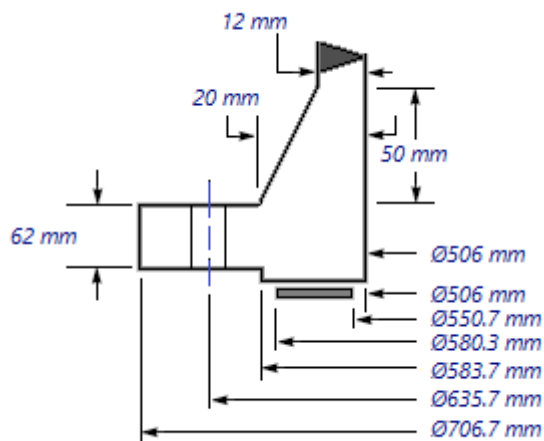


Flange Input Data Values

Description: FLANGE-2 :

Flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	0.34	MPa
Design Temperature		225	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm
Flange Outside Diameter	A	706.700	mm
Flange Thickness	t	62.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	50.0000	mm
Flange Material	SA-516 70 +IT		
Flange Material UNS number	K02700		
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	635.700	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	16		
Flange Face Outside Diameter	Fod	583.700	mm
Flange Face Inside Diameter	Fid	506.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	580.300	mm
Gasket Inside Diameter	Gi	550.700	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:



$$\begin{aligned}
 &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.34 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca \\
 &= 1.6327 \text{ mm}
 \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (254.0 + 0.6 \cdot 11.0) \\
 &= 5.825 \text{ MPa}
 \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned}
 &= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (253.0 + 0.6 \cdot 12.0) \\
 &= 6.364 \text{ MPa}
 \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm
Corroded Small Hub,	g0Cor = go-ci	11.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	44.850 mm
Gasket Contact Width,	N = (Go - Gi)/2	14.800 mm
Basic Gasket Width,	bo = N/2	7.400 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.855 mm
Gasket Reaction Diameter,	G = Go - 2 * b	566.590 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned}
 &= \pi/4 \cdot G^2 \cdot P \\
 &= \pi/4 \cdot 566.5901^2 \cdot 0.343 \\
 &= 86532.648 \text{ N}
 \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned}
 &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P \\
 &= 2 \cdot 6.8549 \cdot \pi \cdot 566.5901 \cdot 3.0 \cdot 0.34 \\
 &= 25126.088 \text{ N}
 \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned}
 &= \pi/4 \cdot Bcor^2 \cdot P \\
 &= \pi/4 \cdot 508.0^2 \cdot 0.3432 \\
 &= 69561.586 \text{ N}
 \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned}
 &= H - Hd \\
 &= 86533 - 69562 \\
 &= 16971.061 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned}
 &= \max(Wo,1 + Wo,2 + H'p, 0) \\
 &= \max(86533 + 25126 + 0, 0) \\
 &= 111658.734 \text{ N}
 \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned}
 &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\
 &= 68.95 \cdot 6.8549 \cdot \pi \cdot 566.59 + 0.0 \cdot 0.0 \cdot 0.0 \\
 &= 841237.812 \text{ N}
 \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned}
 &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\
 &= \max((111659 + 0 + 4 \cdot 0 / 566.59) / 172, 841238 / 172) \\
 &= 4891.421 \text{ mm}^2
 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned}
 &= 2a + 6t / (m + 0.5) \\
 &= 2 \cdot 31.75 + 6 \cdot 62.0 / (3.0 + 0.5) \\
 &= 169.786 \text{ mm}
 \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 635.7 \cdot \sin(\pi/16)$$

$$= 124.019 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a+t}}, 1)$$

$$= \max(\sqrt{\frac{124.019}{2 \cdot 31.75 + 62.0}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4891.421	9589.658	
Radial Distance between Hub and Bolts:	44.450	44.850	
Radial Distance between Bolts and Edge:	31.750	35.500	
Circ. Spacing between the Bolts:	71.450	124.019	169.786

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9589.658 \cdot 172.0 / (68.95 \cdot \pi(580.3 + 550.7))$$

$$= 6.733 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 9589.6582$$

$$= 1649251.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 111659 - 86533$$

$$= 25126.08 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (635.7 - 566.5901) / 2$$

$$= 34.5549 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((635.7 - 508.0) / 2 + 34.5549)$$

$$= 49.2025 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (635.7 - 508.0 - 19.0) / 2$$

$$= 54.3500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	69562.	54.3500	1.0000	3782204.
Face Pressure, Mt	16971.	49.2025	1.0000	835356.
Gasket Load, Mg	25126.	34.5549	1.0000	868582.
Gasket Seating, Matm	1649251.	34.5549	1.0000	57012888.

Total Moment for Operation, Mo 5486142. N-mm
 Total Moment for Gasket seating, Mg 57012888. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 74.753 mm
 Hub Ratio, h/h0 = HL / H0 0.669
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.807



Factor V per 4.16.5 0.232
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 706.7/508.0
= 1.391

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$
= $(0.66845 + (5.7169 \cdot 1.391^2 \cdot \log_{10}(1.391))) / (1.391^2 - 1) / (1.391 - 1)$
= 6.045

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / ((1.0472 + 1.9448 \cdot 1.391^2) \cdot (1.391 - 1))$
= 1.758

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / (1.36136(1.391^2 - 1) \cdot (1.391 - 1))$
= 6.643

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.391^2 + 1) / (1.391^2 - 1)$
= 3.138

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
= $6.643 \cdot 11.0^2 \cdot 74.753 / 0.232$
= 258517.531 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
= 0.807/74.753
= 0.011 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
= $(62.0 \cdot 0.011 + 1) / 1.758 + 62.0^3/258517.531$
= 1.871

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 5486142 / 508.0) / (1.8712 \cdot 19.0^2)$
= 15.98 MPa

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 57012884 / 508.0) / (1.8712 \cdot 19.0^2)$
= 166.09 MPa

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(1.8919 \cdot 5486142 / 508.0) / (1.8712 \cdot 62.0^2)$
= 2.84 MPa

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(1.8919 \cdot 57012884 / 508.0) / (1.8712 \cdot 62.0^2)$
= 29.51 MPa

Tangential Flange Stress, Operating [STo]:

= $(Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
= $(6.045 \cdot 5486142 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 3$



= 8.07 MPa

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (6.045 \cdot 57012884 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 30$$

= 83.82 MPa

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (16 + \max(3, 8)) / 2$$

= 12.02 MPa

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (166 + \max(30, 84)) / 2$$

= 124.95 MPa

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 111659 / 9589.6582$$

= 11.64 MPa

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (841238 / 9589.6582)$$

= 87.73 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	15.98	207.00	166.09	207.00
Radial Flange	2.84	138.00	29.51	138.00
Tangential Flange	8.07	138.00	83.82	138.00
Maximum Average	12.02	138.00	124.95	138.00
Bolting	11.64	172.00	87.73	172.00

Minimum Required Flange Thickness 57.887 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 102.0 kgm

Estimated Unfinished Weight of Forging at given Thk 165.9 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 57627740 / 1.0108 \cdot 1.0 \cdot 0.232 / (1.674 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.750 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5545308 / 1.0108 \cdot 1.0 \cdot 0.232 / (1.674 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.076 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 57012884 / 1.0 \cdot 1.0 \cdot 0.232 / (1.871 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.671 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5486142 / 1.0 \cdot 1.0 \cdot 0.232 / (1.871 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.068 (should be ≤ 1)

**Minimum Design Metal Temperature Results:**

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:**Flange Stress Analysis Results: MPa**

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	15.66	207.00	161.51	207.00
Radial Flange	3.04	138.00	31.33	138.00
Tangential Flange	7.55	138.00	77.87	138.00
Maximum Average	11.60	138.00	119.69	138.00
Bolting	11.64	172.00	87.73	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 55.905 mm

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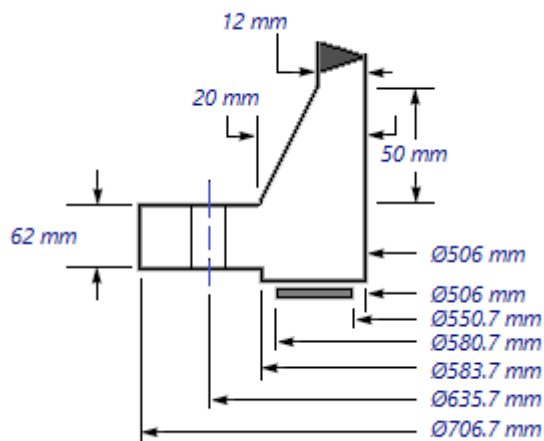


Flange Input Data Values

Description: FLANGE-3 :

Flange-3

Flange Type	Integral Weld Neck		
Design Pressure	P	0.34	MPa
Design Temperature		225	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm
Flange Outside Diameter	A	706.700	mm
Flange Thickness	t	62.0000	mm
Thickness of Hub at Small End	go	12.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-516 70 +IT	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	635.700	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	583.700	mm
Flange Face Inside Diameter	Fid	506.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	580.700	mm
Gasket Inside Diameter	Gi	550.700	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:



$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.34 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca$$

$$= 1.6327 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (254.0 + 0.6 \cdot 11.0)$$

$$= 5.825 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (253.0 + 0.6 \cdot 12.0)$$

$$= 6.364 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm
Corroded Small Hub,	g0Cor = go-ci	11.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	44.850 mm
Gasket Contact Width,	N = (Go - Gi)/2	15.000 mm
Basic Gasket Width,	bo = N/2	7.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.901 mm
Gasket Reaction Diameter,	G = Go - 2 * b	566.898 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 566.8978^2 \cdot 0.343$$

$$= 86626.664 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 566.8978 \cdot 3.0 \cdot 0.34$$

$$= 25309.037 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 0.3432$$

$$= 69561.586 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 86627 - 69562$$

$$= 17065.078 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(86627 + 25309 + 0, 0)$$

$$= 111935.711 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.9011 \cdot \pi \cdot 566.898 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 847363.125 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((111936 + 0 + 4 \cdot 0 / 566.9) / 172, 847363 / 172)$$

$$= 4927.038 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 62.0 / (3.0 + 0.5)$$

$$= 169.786 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 635.7 \cdot \sin(\pi/16)$$

$$= 124.019 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{2a + t}}, 1)$$

$$= \max(\sqrt{\frac{124.019}{2 \cdot 31.75 + 62.0}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4927.038	9589.658	
Radial Distance between Hub and Bolts:	44.450	44.850	
Radial Distance between Bolts and Edge:	31.750	35.500	
Circ. Spacing between the Bolts:	71.450	124.019	169.786

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9589.658 \cdot 172.0 / (68.95 \cdot \pi(580.7 + 550.7))$$

$$= 6.730 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 9589.6582$$

$$= 1649251.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 111936 - 86627$$

$$= 25309.04 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (635.7 - 566.8978) / 2$$

$$= 34.4011 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((635.7 - 508.0) / 2 + 34.4011)$$

$$= 49.1255 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (635.7 - 508.0 - 19.0) / 2$$

$$= 54.3500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	69562.	54.3500	1.0000	3782204.
Face Pressure, Mt	17065.	49.1255	1.0000	838671.
Gasket Load, Mg	25309.	34.4011	1.0000	871011.
Gasket Seating, Matm	1649251.	34.4011	1.0000	56759016.

Total Moment for Operation, Mo 5491886. N-mm
 Total Moment for Gasket seating, Mg 56759016. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 74.753 mm
 Hub Ratio, h/h0 = HL / H0 0.669
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.807



Factor V per 4.16.5 0.232
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 706.7/508.0
= 1.391

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$
= $(0.66845 + (5.7169 \cdot 1.391^2 \cdot \log_{10}(1.391))) / (1.391^2 - 1) / (1.391 - 1)$
= 6.045

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / ((1.0472 + 1.9448 \cdot 1.391^2) \cdot (1.391 - 1))$
= 1.758

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / (1.36136(1.391^2 - 1) \cdot (1.391 - 1))$
= 6.643

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.391^2 + 1) / (1.391^2 - 1)$
= 3.138

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
= $6.643 \cdot 11.0^2 \cdot 74.753 / 0.232$
= 258517.531 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
= 0.807/74.753
= 0.011 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
= $(62.0 \cdot 0.011 + 1) / 1.758 + 62.0^3/258517.531$
= 1.871

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 5491886 / 508.0) / (1.8712 \cdot 19.0^2)$
= 16.00 MPa

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 56759016 / 508.0) / (1.8712 \cdot 19.0^2)$
= 165.35 MPa

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(1.8919 \cdot 5491886 / 508.0) / (1.8712 \cdot 62.0^2)$
= 2.84 MPa

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(1.8919 \cdot 56759016 / 508.0) / (1.8712 \cdot 62.0^2)$
= 29.38 MPa

Tangential Flange Stress, Operating [STo]:

= $(Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
= $(6.045 \cdot 5491886 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 3$



= 8.07 MPa

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot M_{atm} / (t^2 \cdot B_{cor})) - Z \cdot S_{Ra}$$

$$= (6.045 \cdot 56759016 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 29$$

= 83.45 MPa

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (16 + \max(3, 8)) / 2$$

= 12.04 MPa

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (165 + \max(29, 83)) / 2$$

= 124.40 MPa

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 111936 / 9589.6582$$

= 11.67 MPa

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (847363 / 9589.6582)$$

= 88.37 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	16.00	207.00	165.35	207.00
Radial Flange	2.84	138.00	29.38	138.00
Tangential Flange	8.07	138.00	83.45	138.00
Maximum Average	12.04	138.00	124.40	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Required Flange Thickness 57.709 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 102.0 kgm

Estimated Unfinished Weight of Forging at given Thk 165.9 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 57413200 / 1.0115 \cdot 1.0 \cdot 0.232 / (1.666 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.750 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5555183 / 1.0115 \cdot 1.0 \cdot 0.232 / (1.666 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.077 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 56759016 / 1.0 \cdot 1.0 \cdot 0.232 / (1.871 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.668 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 5491886 / 1.0 \cdot 1.0 \cdot 0.232 / (1.871 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

= 0.068 (should be ≤ 1)

**Minimum Design Metal Temperature Results:**

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	15.67	207.00	160.79	207.00
Radial Flange	3.04	138.00	31.19	138.00
Tangential Flange	7.56	138.00	77.52	138.00
Maximum Average	11.61	138.00	119.16	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 55.728 mm

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Warnings for Flange : FLANGE-4 :

Warning: Flange small end hub thickness > attached shell thickness.

Review the Code definition of g0 and change the small end hub thickness in the flange input to be the attached shell thickness.

Flange Input Data Values

Description: FLANGE-4 :

Flange-4

Flange Type	Integral Weld Neck		
Design Pressure	P	0.34	MPa
Design Temperature		225	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm
Flange Outside Diameter	A	706.700	mm
Flange Thickness	t	62.0000	mm
Thickness of Hub at Small End	g0	12.0000	mm
Thickness of Hub at Large End	g1	20.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-516 70	+IT
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	635.700	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	583.700	mm
Flange Face Inside Diameter	Fid	506.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	580.700	mm
Gasket Inside Diameter	Gi	550.700	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm

$$\begin{aligned} &= H - H_d \\ &= 86627 - 69562 \\ &= 17065.078 \text{ N} \end{aligned}$$



Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(86627 + 25309 + 0, 0)$$

$$= 111935.711 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.9011 \cdot \pi \cdot 566.898 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 847363.125 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((111936 + 0 + 4 \cdot 0 / 566.9) / 172, 847363 / 172)$$

$$= 4927.038 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 62.0 / (3.0 + 0.5)$$

$$= 169.786 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 635.7 \cdot \sin(\pi / 16)$$

$$= 124.019 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 124.019 / (2 \cdot 31.75 + 62.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4927.038	9589.658	
Radial Distance between Hub and Bolts:	44.450	44.850	
Radial Distance between Bolts and Edge:	31.750	35.500	
Circ. Spacing between the Bolts:	71.450	124.019	169.786

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 9589.658 \cdot 172.0 / (68.95 \cdot \pi (580.7 + 550.7))$$

$$= 6.730 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 9589.6582$$

$$= 1649251.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 111936 - 86627$$

$$= 25309.04 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (635.7 - 566.8978) / 2$$

$$= 34.4011 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((635.7 - 508.0) / 2 + 34.4011)$$

$$= 49.1255 \text{ mm}$$



Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (635.7 - 508.0 - 19.0) / 2$$

$$= 54.3500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	69562.	54.3500	1.0000	3782204.
Face Pressure, Mt	17065.	49.1255	1.0000	838671.
Gasket Load, Mg	25309.	34.4011	1.0000	871011.
Gasket Seating, Matm	1649251.	34.4011	1.0000	56759016.

Total Moment for Operation, Mo 5491886. N-mm

Total Moment for Gasket seating, Mg 56759016. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 74.753 mm

Hub Ratio, h/h0 = HL / H0 0.669

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.727

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.807

Factor V per 4.16.5 0.232

Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 706.7/508.0$$

$$= 1.391$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169(1.391^2 \cdot \log_{10}(1.391))) / (1.391^2 - 1)) / (1.391 - 1)$$

$$= 6.045$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / ((1.0472 + 1.9448 \cdot 1.391^2) \cdot (1.391 - 1))$$

$$= 1.758$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / (1.36136(1.391^2 - 1) \cdot (1.391 - 1))$$

$$= 6.643$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.391^2 + 1) / (1.391^2 - 1)$$

$$= 3.138$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 6.643 \cdot 11.0^2 \cdot 74.753 / 0.232$$

$$= 258517.531 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.807 / 74.753$$

$$= 0.011 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (62.0 \cdot 0.011 + 1) / 1.758 + 62.0^3 / 258517.531$$

$$= 1.871$$

Flange Stresses



Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1.2})$$

$$= (1.0 \cdot 5491886 / 508.0) / (1.8712 \cdot 19.0^2)$$

$$= 16.00 \text{ MPa}$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1.2})$$

$$= (1.0 \cdot 56759016 / 508.0) / (1.8712 \cdot 19.0^2)$$

$$= 165.35 \text{ MPa}$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (1.8919 \cdot 5491886 / 508.0) / (1.8712 \cdot 62.0^2)$$

$$= 2.84 \text{ MPa}$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (1.8919 \cdot 56759016 / 508.0) / (1.8712 \cdot 62.0^2)$$

$$= 29.38 \text{ MPa}$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (6.045 \cdot 5491886 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 3$$

$$= 8.07 \text{ MPa}$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (6.045 \cdot 56759016 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 29$$

$$= 83.45 \text{ MPa}$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (16 + \max(3, 8)) / 2$$

$$= 12.04 \text{ MPa}$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (165 + \max(29, 83)) / 2$$

$$= 124.40 \text{ MPa}$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 111936 / 9589.6582$$

$$= 11.67 \text{ MPa}$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (847363 / 9589.6582)$$

$$= 88.37 \text{ MPa}$$

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	16.00	207.00	165.35	207.00
Radial Flange	2.84	138.00	29.38	138.00
Tangential Flange	8.07	138.00	83.45	138.00
Maximum Average	12.04	138.00	124.40	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Required Flange Thickness

57.709 mm

Estimated M.A.W.P. (Operating)

4 MPa

Estimated Finished Weight of Flange at given Thk.

102.0 kgm

Estimated Unfinished Weight of Forging at given Thk

165.9 kgm

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 57413200 / 1.0115 \cdot 1 \cdot 0.232 / (1.666 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

$$= 0.750 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5555183 / 1.0115 \cdot 1 \cdot 0.232 / (1.666 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

$$= 0.077 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 56759016 / 1.0 \cdot 1 \cdot 0.232 / (1.871 \cdot 202713 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

$$= 0.668 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5491886 / 1.0 \cdot 1 \cdot 0.232 / (1.871 \cdot 190841 \cdot 11.0^2 \cdot 74.753 \cdot 0.3)$$

$$= 0.068 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	15.67	207.00	160.79	207.00
Radial Flange	3.04	138.00	31.19	138.00
Tangential Flange	7.56	138.00	77.52	138.00
Maximum Average	11.61	138.00	119.16	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 55.728 mm

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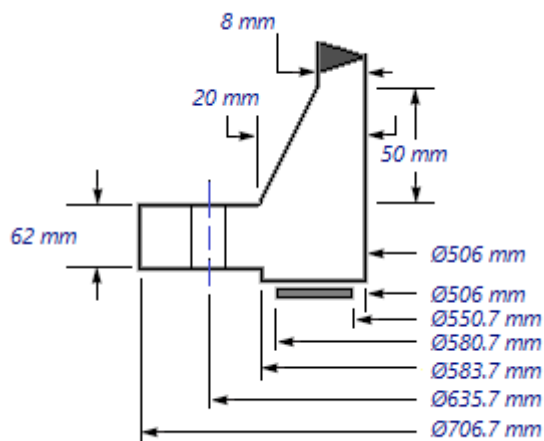


Flange Input Data Values

Description: FLANGE-5 :

Flange-5

Flange Type	Integral Weld Neck		
Design Pressure	P	0.34	MPa
Design Temperature		225	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm
Flange Outside Diameter	A	706.700	mm
Flange Thickness	t	62.0000	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-516 70 +IT	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	635.700	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	583.700	mm
Flange Face Inside Diameter	Fid	506.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	580.700	mm
Gasket Inside Diameter	Gi	550.700	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:



$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.34 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca$$

$$= 1.6327 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 7.0) / (254.0 + 0.6 \cdot 7.0)$$

$$= 3.741 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.0) / (253.0 + 0.6 \cdot 8.0)$$

$$= 4.282 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	44.850 mm
Gasket Contact Width,	N = (Go - Gi)/2	15.000 mm
Basic Gasket Width,	bo = N/2	7.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.901 mm
Gasket Reaction Diameter,	G = Go - 2 * b	566.898 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 566.8978^2 \cdot 0.343$$

$$= 86626.664 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 566.8978 \cdot 3.0 \cdot 0.34$$

$$= 25309.037 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 0.3432$$

$$= 69561.586 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 86627 - 69562$$

$$= 17065.078 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(86627 + 25309 + 0, 0)$$

$$= 111935.711 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.9011 \cdot \pi \cdot 566.898 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 847363.125 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((111936 + 0 + 4 \cdot 0 / 566.9) / 172, 847363 / 172)$$

$$= 4927.038 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 62.0 / (3.0 + 0.5)$$

$$= 169.786 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 635.7 \cdot \sin(\pi/16)$$

$$= 124.019 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a+t)}, 1)$$

$$= \max(\sqrt{124.019/(2 \cdot 31.75 + 62.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4927.038	9589.658	
Radial Distance between Hub and Bolts:	44.450	44.850	
Radial Distance between Bolts and Edge:	31.750	35.500	
Circ. Spacing between the Bolts:	71.450	124.019	169.786

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9589.658 \cdot 172.0 / (68.95 \cdot \pi(580.7 + 550.7))$$

$$= 6.730 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 9589.6582$$

$$= 1649251.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 111936 - 86627$$

$$= 25309.04 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (635.7 - 566.8978) / 2$$

$$= 34.4011 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((635.7 - 508.0) / 2 + 34.4011)$$

$$= 49.1255 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (635.7 - 508.0 - 19.0) / 2$$

$$= 54.3500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	69562.	54.3500	1.0000	3782204.
Face Pressure, Mt	17065.	49.1255	1.0000	838671.
Gasket Load, Mg	25309.	34.4011	1.0000	871011.
Gasket Seating, Matm	1649251.	34.4011	1.0000	56759016.

Total Moment for Operation, Mo 5491886. N-mm
 Total Moment for Gasket seating, Mg 56759016. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 59.632 mm
 Hub Ratio, h/h0 = HL / H0 0.838
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 2.714

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.734



Factor V per 4.16.5 0.109
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 706.7/508.0
= 1.391

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$
= $(0.66845 + (5.7169 \cdot 1.391^2 \cdot \log_{10}(1.391))) / (1.391^2 - 1) / (1.391 - 1)$
= 6.045

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / ((1.0472 + 1.9448 \cdot 1.391^2) \cdot (1.391 - 1))$
= 1.758

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / (1.36136(1.391^2 - 1) \cdot (1.391 - 1))$
= 6.643

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.391^2 + 1) / (1.391^2 - 1)$
= 3.138

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
= $6.643 \cdot 7.0^2 \cdot 59.632 / 0.109$
= 178098.031 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
= 0.734/59.632
= 0.012 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
= $(62.0 \cdot 0.012 + 1) / 1.758 + 62.0^3/178098.031$
= 2.341

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 5491886 / 508.0) / (2.341 \cdot 19.0^2)$
= 12.79 MPa

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 56759016 / 508.0) / (2.341 \cdot 19.0^2)$
= 132.17 MPa

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(2.0174 \cdot 5491886 / 508.0) / (2.341 \cdot 62.0^2)$
= 2.42 MPa

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(2.0174 \cdot 56759016 / 508.0) / (2.341 \cdot 62.0^2)$
= 25.04 MPa

Tangential Flange Stress, Operating [STo]:

= $(Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
= $(6.045 \cdot 5491886 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 2$



= 9.39 MPa

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (6.045 \cdot 56759016 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 25$$

= 97.06 MPa

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (13 + \max(2, 9)) / 2$$

= 11.09 MPa

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (132 + \max(25, 97)) / 2$$

= 114.62 MPa

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 111936 / 9589.6582$$

= 11.67 MPa

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (847363 / 9589.6582)$$

= 88.37 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	12.79	207.00	132.17	207.00
Radial Flange	2.42	138.00	25.04	138.00
Tangential Flange	9.39	138.00	97.06	138.00
Maximum Average	11.09	138.00	114.62	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Required Flange Thickness 55.093 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 100.7 kgm

Estimated Unfinished Weight of Forging at given Thk 165.9 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58043024 / 1.0226 \cdot 1.0 \cdot 109 / (1.892 \cdot 202713 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.958 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5616124 / 1.0226 \cdot 1.0 \cdot 109 / (1.892 \cdot 190841 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.098 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 56759016 / 1.0 \cdot 1.0 \cdot 109 / (2.341 \cdot 202713 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.775 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5491886 / 1.0 \cdot 1.0 \cdot 109 / (2.341 \cdot 190841 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.079 (should be ≤ 1)

**Minimum Design Metal Temperature Results:**

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	12.85	207.00	131.86	207.00
Radial Flange	2.64	138.00	27.10	138.00
Tangential Flange	8.79	138.00	90.23	138.00
Maximum Average	10.82	138.00	111.04	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 53.518 mm

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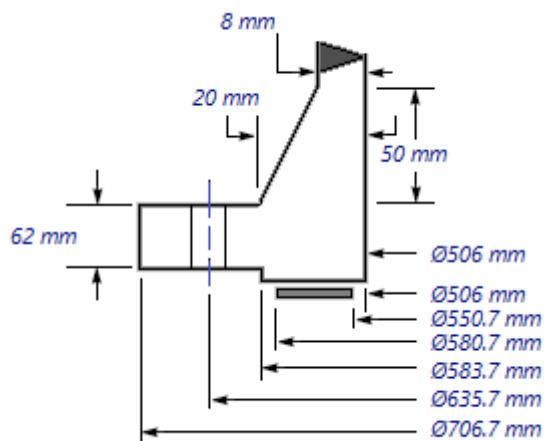


Flange Input Data Values

Description: FLANGE-6 :

Flange-6

Flange Type	Integral Weld Neck		
Design Pressure	P	0.34	MPa
Design Temperature		225	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	506.000	mm
Flange Outside Diameter	A	706.700	mm
Flange Thickness	t	62.0000	mm
Thickness of Hub at Small End	go	8.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-516 70 +IT	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	MPa
Flange Allowable Stress At Ambient	Sfa	138.00	MPa
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	MPa
Bolt Allowable Stress At Ambient	Sa	172.00	MPa
Diameter of Bolt Circle	C	635.700	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		16	
Flange Face Outside Diameter	Fod	583.700	mm
Flange Face Inside Diameter	Fid	506.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	580.700	mm
Gasket Inside Diameter	Gi	550.700	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	MPa
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Hub Small End Required Thickness due to Internal Pressure:



$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.34 \cdot (506.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 0.34) + Ca$$

$$= 1.6327 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 7.0) / (254.0 + 0.6 \cdot 7.0)$$

$$= 3.741 \text{ MPa}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.0) / (253.0 + 0.6 \cdot 8.0)$$

$$= 4.282 \text{ MPa}$$

Corroded Flange ID,	Bcor = B+2*ci	508.000 mm
Corroded Large Hub,	g1Cor = g1-ci	19.000 mm
Corroded Small Hub,	g0Cor = go-ci	7.000 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	44.850 mm
Gasket Contact Width,	N = (Go - Gi)/2	15.000 mm
Basic Gasket Width,	bo = N/2	7.500 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	6.901 mm
Gasket Reaction Diameter,	G = Go - 2 * b	566.898 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 566.8978^2 \cdot 0.343$$

$$= 86626.664 \text{ N}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.9011 \cdot \pi \cdot 566.8978 \cdot 3.0 \cdot 0.34$$

$$= 25309.037 \text{ N}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 508.0^2 \cdot 0.3432$$

$$= 69561.586 \text{ N}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 86627 - 69562$$

$$= 17065.078 \text{ N}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(86627 + 25309 + 0, 0)$$

$$= 111935.711 \text{ N}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.9011 \cdot \pi \cdot 566.898 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 847363.125 \text{ N}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((111936 + 0 + 4 \cdot 0 / 566.9) / 172, 847363 / 172)$$

$$= 4927.038 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 31.75 + 6 \cdot 62.0 / (3.0 + 0.5)$$

$$= 169.786 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$



$$= 635.7 \cdot \sin(\pi/16)$$

$$= 124.019 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{\frac{Bs}{(2a + t)}}, 1)$$

$$= \max(\sqrt{\frac{124.019}{(2 \cdot 31.75 + 62.0)}}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	4927.038	9589.658	
Radial Distance between Hub and Bolts:	44.450	44.850	
Radial Distance between Bolts and Edge:	31.750	35.500	
Circ. Spacing between the Bolts:	71.450	124.019	169.786

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 9589.658 \cdot 172.0 / (68.95 \cdot \pi(580.7 + 550.7))$$

$$= 6.730 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 9589.6582$$

$$= 1649251.25 \text{ N}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 111936 - 86627$$

$$= 25309.04 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (635.7 - 566.8978) / 2$$

$$= 34.4011 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((635.7 - 508.0) / 2 + 34.4011)$$

$$= 49.1255 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (635.7 - 508.0 - 19.0) / 2$$

$$= 54.3500 \text{ mm}$$

Summary of Moments for Internal Pressure: (N-mm)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	69562.	54.3500	1.0000	3782204.
Face Pressure, Mt	17065.	49.1255	1.0000	838671.
Gasket Load, Mg	25309.	34.4011	1.0000	871011.
Gasket Seating, Matm	1649251.	34.4011	1.0000	56759016.

Total Moment for Operation, Mo 5491886. N-mm
 Total Moment for Gasket seating, Mg 56759016. N-mm

Effective Hub Length, ho = sqrt(Bcor*goCor) 59.632 mm
 Hub Ratio, h/h0 = HL / H0 0.838
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 2.714

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.734



Factor V per 4.16.5 0.109
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
= 706.7/508.0
= 1.391

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169 \cdot K^2 \cdot \log_{10}(K))) / (K^2 - 1) / (K - 1)$
= $(0.66845 + (5.7169 \cdot 1.391^2 \cdot \log_{10}(1.391))) / (1.391^2 - 1) / (1.391 - 1)$
= 6.045

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / ((1.0472 + 1.9448 \cdot 1.391^2) \cdot (1.391 - 1))$
= 1.758

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
= $(1.391^2(1 + 8.55246 \cdot \log_{10}(1.391)) - 1) / (1.36136(1.391^2 - 1) \cdot (1.391 - 1))$
= 6.643

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
= $(1.391^2 + 1) / (1.391^2 - 1)$
= 3.138

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g_0^2 \cdot h_0 / V$
= $6.643 \cdot 7.0^2 \cdot 59.632 / 0.109$
= 178098.031 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h₀
= 0.734/59.632
= 0.012 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
= $(62.0 \cdot 0.012 + 1) / 1.758 + 62.0^3/178098.031$
= 2.341

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 5491886/508.0) / (2.341 \cdot 19.0^2)$
= 12.79 MPa

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g_1^2)$
= $(1.0 \cdot 56759016/508.0) / (2.341 \cdot 19.0^2)$
= 132.17 MPa

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
= $(2.0174 \cdot 5491886/508.0) / (2.341 \cdot 62.0^2)$
= 2.42 MPa

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
= $(2.0174 \cdot 56759016/508.0) / (2.341 \cdot 62.0^2)$
= 25.04 MPa

Tangential Flange Stress, Operating [STo]:

= $(Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$
= $(6.045 \cdot 5491886 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 2$



= 9.39 MPa

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (6.045 \cdot 56759016 / (62.0^2 \cdot 508.0)) - 3.1384 \cdot 25$$

= 97.06 MPa

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (13 + \max(2, 9)) / 2$$

= 11.09 MPa

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRA, STa)) / 2$$

$$= (132 + \max(25, 97)) / 2$$

= 114.62 MPa

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 111936 / 9589.6582$$

= 11.67 MPa

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (847363 / 9589.6582)$$

= 88.37 MPa

Flange Stress Analysis Results: MPa

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	12.79	207.00	132.17	207.00
Radial Flange	2.42	138.00	25.04	138.00
Tangential Flange	9.39	138.00	97.06	138.00
Maximum Average	11.09	138.00	114.62	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Required Flange Thickness 55.093 mm

Estimated M.A.W.P. (Operating) 4 MPa

Estimated Finished Weight of Flange at given Thk. 100.7 kgm

Estimated Unfinished Weight of Forging at given Thk 165.9 kgm

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 58043024 / 1.0226 \cdot 1.0 \cdot 109 / (1.892 \cdot 202713 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.958 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5616124 / 1.0226 \cdot 1.0 \cdot 109 / (1.892 \cdot 190841 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.098 (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 56759016 / 1.0 \cdot 1.0 \cdot 109 / (2.341 \cdot 202713 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.775 (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5491886 / 1.0 \cdot 1.0 \cdot 109 / (2.341 \cdot 190841 \cdot 7.0^2 \cdot 59.632 \cdot 0.3)$$

= 0.079 (should be ≤ 1)

**Minimum Design Metal Temperature Results:**

Note: This Element/Detail was specified as being Impact Tested.

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:**Flange Stress Analysis Results: MPa**

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	12.85	207.00	131.86	207.00
Radial Flange	2.64	138.00	27.10	138.00
Tangential Flange	8.79	138.00	90.23	138.00
Maximum Average	10.82	138.00	111.04	138.00
Bolting	11.67	172.00	88.37	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 53.518 mm

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Bottom head		0.3432	10	1	506	138	1
Shell-1		0.3432	12	1	506	138	2
Shell-2		0.3432	12	1	506	138	3
Flange-1		0.3432	64	1	506	138	4
Flange-2		0.3432	64	1	506	138	5
Shell-3		0.3432	14	1	506	138	6
Shell-4		0.3432	12	1	506	138	7
Flange-3		0.3432	64	1	506	138	8
Flange-4		0.3432	64	1	506	138	9
Shell-5		0.3432	8	1	506	138	10
Flange-5		0.3432	64	1	506	138	11
Flange-6		0.3432	64	1	506	138	12
Shell-6		0.3432	8	1	506	138	13
Top head		0.3432	10	1	506	138	14

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Bottom head		0.34323	3.81247	4.3498	8	2.5	1
Shell-1		0.34323	5.82489	6.3642	12	2.5	2
Shell-2		0.34323	5.82489	6.3642	12	2.5	3
Flange-1		0.34323	3.93926	4.08239	62	57.8866	4
Flange-2		0.34323	3.93926	4.08239	62	57.8866	5
Shell-3		0.34323	6.85241	7.39081	14	2.5	6
Shell-4		0.34323	5.82489	6.3642	12	2.5	7
Flange-3		0.34323	3.93514	4.07815	62	57.7088	8
Flange-4		0.34323	3.93514	4.07815	62	57.7088	9
Shell-5		0.34323	3.7412	4.2823	8	2.5	10
Flange-5		0.34323	3.7412	4.2823	62	55.0926	11
Flange-6		0.34323	3.7412	4.2823	62	55.0926	12
Shell-6		0.34323	3.7412	4.2823	8	2.5	13
Top head		0.34323	3.81247	4.3498	8	2.5	14

Minimum

4

4

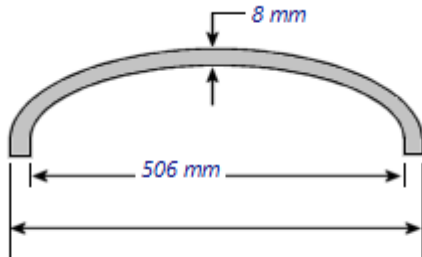
MAWP: 4 MPa, limited by: Shell-6.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Bottom head

Elliptical Head Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature		225 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress		12.4 MPa
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		382536.6 mm²	SG of Contents		0.000
Element Volume		27.0 ltr	Weight of Contents		0.0 N
Empty Weight		303.8 N	Operating Weight		303.8 N
			Comp. All. Ope. Stress		95.2 MPa
			Aspect Ratio		2.00
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4 (c)}$$

$$= (0.343 \cdot 508.0 \cdot 0.995) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.343)$$

$$= 0.6286 + 1.0000 = 1.6286 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 7.0) / (0.995 \cdot 508.0 + 0.2 \cdot 7.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.0) / (1.0 \cdot 506.0 + 0.2 \cdot 8.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:



$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.343 \cdot (0.995 \cdot 508.0 + 0.2 \cdot 7.0)) / (2 \cdot 1.0 \cdot 7.0)$$

$$= 12.424 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) + 1.0$$

$$= 1.633 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (254.0 + 0.6 \cdot 9.0)$$

$$= 5 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (508.0 / (2 \cdot 127.5))^2) / 6$$

$$= 0.994782$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.240 %**Note: Please Check Requirements of UCS-79 as Elongation is > 5%.****MDMT Calculations in the Knuckle Portion:**Govrn. thk, $t_g = 8.0$, $t_r = 6.869$, $c = 1.0$ mm, $E^* = 1.0$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

MDMT Calculations in the Head Straight Flange:Govrn. thk, $t_g = 10.0$, $t_r = 7.0$, $c = 1.0$ mm, $E^* = 1.0$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.741 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.778	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT**Tolerance for Formed Heads per UG-81(a):**

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm

EXTERNAL PRESSURE RESULTS for: Bottom head**Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 150 °C**

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.000	522.00	74.57	0.0018625	101.88



$$MAEP = B/(K0*Do/t) = 101.8843/(0.9 * 74.5714) = 1.5181 \text{ MPa}$$

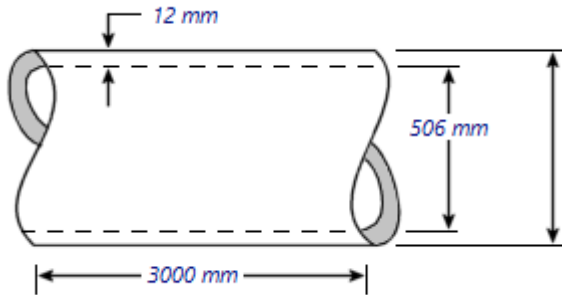
Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.352	522.00	386.10	0.0003597	35.94

$$MAEP = B/(K0*Do/t) = 35.9398/(0.9 * 386.0981) = 0.1034 \text{ MPa}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature	225 °C	
Material		SA-516 70	Ext. Design Temperature	150 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress	8.1 MPa	
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.	12.000 mm	
Surface Area		4995132.0 mm²	SG of Contents	0.000	
Element Volume		603.4 ltr	Weight of Contents	0.0 N	
Empty Weight		4452.5 N	Operating Weight	4452.5 N	
			Comp. All. Ope. Stress	101.4 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) \\
 &= 0.6327 + 1.0000 = 1.6327 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (254.0 + 0.6 \cdot 11.0) \\
 &= 6 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (253.0 + 0.6 \cdot 12.0) \\
 &= 6 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.343 \cdot (254.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0) \\
 &= 8.132 \text{ MPa}
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.317 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	530.00	5092.17	48.18	9.6079	0.0005015	50.11

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 50.1069) / (3 \cdot 48.1818) = 1.3866 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.412	530.00	5092.17	120.12	9.6079	0.0000933	9.32

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.3176) / (3 \cdot 120.1212) = 0.1034 \text{ MPa}$$

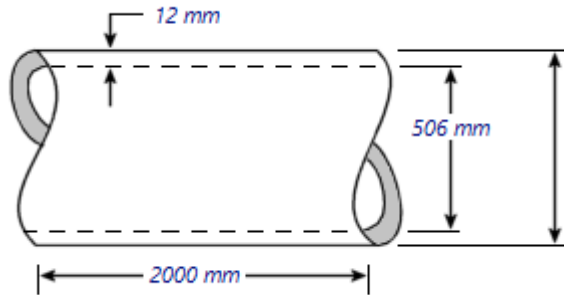
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	530.00	239145.73	48.18	50.0000	0.0004795	47.91

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 47.9057) / (3 \cdot 48.1818) = 1.3257 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature		225 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		2000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress		8.1 MPa
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.		12.000 mm
Surface Area		3330088.0 mm²	SG of Contents		0.000
Element Volume		402.3 ltr	Weight of Contents		0.0 N
Empty Weight		2968.3 N	Operating Weight		2968.3 N
			Comp. All. Ope. Stress		101.4 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) \\
 &= 0.6327 + 1.0000 = 1.6327 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (254.0 + 0.6 \cdot 11.0) \\
 &= 6 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (253.0 + 0.6 \cdot 12.0) \\
 &= 6 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.343 \cdot (254.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0) \\
 &= 8.132 \text{ MPa}
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.317 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	530.00	5092.17	48.18	9.6079	0.0005015	50.11

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 50.1069) / (3 \cdot 48.1818) = 1.3866 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.412	530.00	5092.17	120.12	9.6079	0.0000933	9.32

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.3176) / (3 \cdot 120.1212) = 0.1034 \text{ MPa}$$

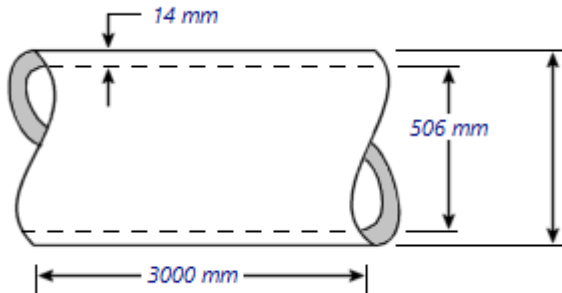
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	530.00	239145.73	48.18	50.0000	0.0004795	47.91

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 47.9057) / (3 \cdot 48.1818) = 1.3257 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature	225 °C	
Material		SA-516 70	Ext. Design Temperature	150 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress	6.9 MPa	
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		5032831.5 mm²	SG of Contents	0.000	
Element Volume		603.4 ltr	Weight of Contents	0.0 N	
Empty Weight		5214.6 N	Operating Weight	5214.6 N	
			Comp. All. Ope. Stress	103.5 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) \\
 &= 0.6327 + 1.0000 = 1.6327 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 13.0) / (254.0 + 0.6 \cdot 13.0) \\
 &= 7 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (253.0 + 0.6 \cdot 14.0) \\
 &= 7 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.343 \cdot (254.0 + 0.6 \cdot 13.0)) / (1.0 \cdot 13.0) \\
 &= 6.912 \text{ MPa}
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.692 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 60 to 70 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	534.00	4707.65	41.08	8.8158	0.0006894	68.88

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 68.8754) / (3 \cdot 41.0769) = 2.2357 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.343	534.00	4707.65	122.95	8.8158	0.0000955	9.54

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.5369) / (3 \cdot 122.9498) = 0.1034 \text{ MPa}$$

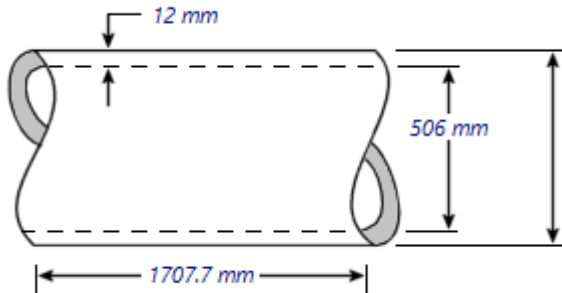
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	534.00	99333.07	41.08	50.0000	0.0006568	65.62

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 65.6222) / (3 \cdot 41.0769) = 2.1301 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature	225 °C	
Material		SA-516 70	Ext. Design Temperature	150 °C	
Length		1707.650 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress	8.1 MPa	
Spec. Min. Thk t,e, or sa		12.0000 mm	Spec. Nominal Thk.	12.000 mm	
Surface Area		2843312.5 mm²	SG of Contents	0.000	
Element Volume		343.5 ltr	Weight of Contents	0.0 N	
Empty Weight		2534.4 N	Operating Weight	2534.4 N	
			Comp. All. Ope. Stress	101.4 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) \\
 &= 0.6327 + 1.0000 = 1.6327 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (254.0 + 0.6 \cdot 11.0) \\
 &= 6 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (253.0 + 0.6 \cdot 12.0) \\
 &= 6 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.343 \cdot (254.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0) \\
 &= 8.132 \text{ MPa}
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.317 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	530.00	4707.65	48.18	8.8824	0.0005087	50.83

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 50.8255) / (3 \cdot 48.1818) = 1.4065 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.322	530.00	4707.65	122.62	8.8824	0.0000952	9.51

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.5111) / (3 \cdot 122.6155) = 0.1034 \text{ MPa}$$

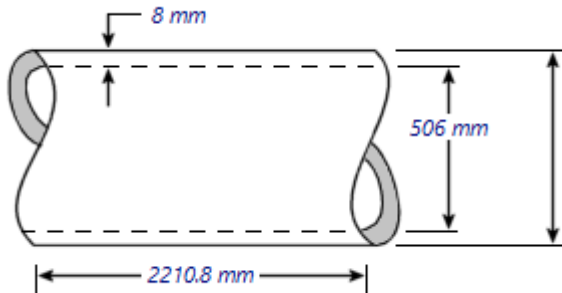
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	530.00	222739.77	48.18	50.0000	0.0004795	47.91

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 47.9057) / (3 \cdot 48.1818) = 1.3257 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature	225 °C	
Material		SA-516 70	Ext. Design Temperature	150 °C	
Length		2210.820 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress	12.7 MPa	
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.	8.000 mm	
Surface Area		3625549.2 mm²	SG of Contents	0.000	
Element Volume		444.7 ltr	Weight of Contents	0.0 N	
Empty Weight		2170.6 N	Operating Weight	2170.6 N	
			Comp. All. Ope. Stress	95.2 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) \\
 &= 0.6327 + 1.0000 = 1.6327 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 7.0) / (254.0 + 0.6 \cdot 7.0) \\
 &= 4 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 8.0) / (253.0 + 0.6 \cdot 8.0) \\
 &= 4 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.343 \cdot (254.0 + 0.6 \cdot 7.0)) / (1.0 \cdot 7.0) \\
 &= 12.660 \text{ MPa}
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.556 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 100 to 110 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	522.00	2210.82	74.57	4.2353	0.0004471	44.67

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 44.6721) / (3 \cdot 74.5714) = 0.7987 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
3.054	522.00	2210.82	170.91	4.2353	0.0001327	13.26

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 13.258) / (3 \cdot 170.9138) = 0.1034 \text{ MPa}$$

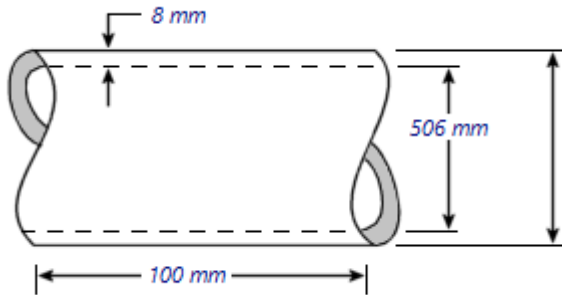
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	522.00	77724.60	74.57	50.0000	0.0002032	20.30

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 20.2996) / (3 \cdot 74.5714) = 0.363 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:			
Design Pressure P	0.343 MPa	Int. Design Temperature	225 °C
Material	SA-516 70	Ext. Design Temperature	150 °C
Length	100.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	506.000 mm	Actual Stress	12.7 MPa
Spec. Min. Thk t,e, or sa	8.0000 mm	Spec. Nominal Thk.	8.000 mm
Surface Area	163991.1 mm ²	SG of Contents	0.000
Element Volume	20.1 ltr	Weight of Contents	0.0 N
Empty Weight	98.2 N	Operating Weight	98.2 N
		Comp. All. Ope. Stress	95.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) \\
 &= 0.6327 + 1.0000 = 1.6327 \text{ mm}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 7.0) / (254.0 + 0.6 \cdot 7.0) \\
 &= 4 \text{ MPa}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 8.0) / (253.0 + 0.6 \cdot 8.0) \\
 &= 4 \text{ MPa}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (0.343 \cdot (254.0 + 0.6 \cdot 7.0)) / (1.0 \cdot 7.0) \\
 &= 12.660 \text{ MPa}
 \end{aligned}$$



% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.556 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 130 to 140 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	522.00	192.17	74.57	0.3681	0.0064843	121.76

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 121.7558) / (3 \cdot 74.5714) = 2.177 \text{ MPa}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
1.104	522.00	192.17	472.79	0.3681	0.0003671	36.67

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 36.674) / (3 \cdot 472.7882) = 0.1034 \text{ MPa}$$

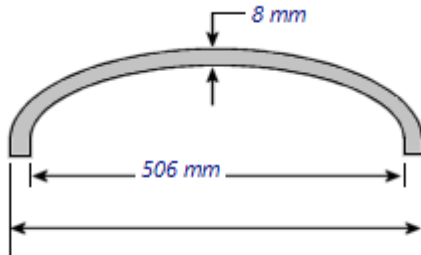
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.000	522.00	7558.76	74.57	14.4804	0.0002071	20.69

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 20.6877) / (3 \cdot 74.5714) = 0.3699 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature		225 °C
Material		SA-516 70	Ext. Design Temperature		150 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	506.000 mm	Actual Stress		12.4 MPa
Spec. Min. Thk t,e, or sa		8.0000 mm	Spec. Nominal Thk.		10.000 mm
Surface Area		382536.6 mm²	SG of Contents		0.000
Element Volume		27.0 ltr	Weight of Contents		0.0 N
Empty Weight		303.8 N	Operating Weight		303.8 N
			Comp. All. Ope. Stress		95.2 MPa
			Aspect Ratio		2.00
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= (0.343 \cdot 508.0 \cdot 0.995) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 0.343)$$

$$= 0.6286 + 1.0000 = 1.6286 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 7.0) / (0.995 \cdot 508.0 + 0.2 \cdot 7.0)$$

$$= 4 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.0) / (1.0 \cdot 506.0 + 0.2 \cdot 8.0)$$

$$= 4 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:



$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.343 \cdot (0.995 \cdot 508.0 + 0.2 \cdot 7.0)) / (2 \cdot 1.0 \cdot 7.0)$$

$$= 12.424 \text{ MPa}$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (0.343 \cdot 254.0) / (138.0 \cdot 1.0 - 0.6 \cdot 0.343) + 1.0$$

$$= 1.633 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.0) / (254.0 + 0.6 \cdot 9.0)$$

$$= 5 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (508.0 / (2 \cdot 127.5))^2) / 6$$

$$= 0.994782$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.240 %**Note:** Please Check Requirements of UCS-79 as Elongation is > 5%.**MDMT Calculations in the Knuckle Portion:**

Govrn. thk, tg = 8.0, tr = 6.869, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 10.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	3.741 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	10.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.778	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT**Tolerance for Formed Heads per UG-81(a):**

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.325 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.163 mm

EXTERNAL PRESSURE RESULTS for: Top head**Elliptical Head From 140 to 150 Ext. Chart: CS-2 at 150 °C**

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.000	522.00	74.57	0.0018625	101.88



$$MAEP = B/(K0*Do/t) = 101.8843/(0.9 * 74.5714) = 1.5181 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.352	522.00	386.10	0.0003597	35.94

$$MAEP = B/(K0*Do/t) = 35.9398/(0.9 * 386.0981) = 0.1034 \text{ MPa}$$

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4.864 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.446 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	5.297 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.115 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.491 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.491 MPa
App 27-4	= M.A.W.P.	3.741 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 3.741 • 1.0

= 4.864 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	154.5	234.0	0.660	4.87
Shell-1	105.6	234.0	0.451	4.87
Shell-2	105.6	234.0	0.451	4.87
Shell-3	90.9	234.0	0.388	4.87
Shell-4	105.6	234.0	0.451	4.87
Shell-5	156.9	234.0	0.670	4.87
Shell-6	156.9	234.0	0.670	4.87
Top head	154.5	234.0	0.660	4.87

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 2"	Nozzle	138.00	137.00	1.007
N7 2"	Nozzle	138.00	137.00	1.007
N10 4"	Nozzle	138.00	137.00	1.007
N2 2"	Nozzle	138.00	137.00	1.007
N8 2"	Nozzle	138.00	137.00	1.007
N3 2"	Nozzle	138.00	137.00	1.007
K1A 2"	Nozzle	138.00	137.00	1.007
K2A 2"	Nozzle	138.00	137.00	1.007
K3A 2"	Nozzle	138.00	137.00	1.007
K4A 2"	Nozzle	138.00	137.00	1.007
K1B 2"	Nozzle	138.00	137.00	1.007
K2B 2"	Nozzle	138.00	137.00	1.007
K3B 2"	Nozzle	138.00	137.00	1.007
K4B 2"	Nozzle	138.00	137.00	1.007
N11 8"	Nozzle	138.00	137.00	1.007
A3 8"	Nozzle	138.00	137.00	1.007
K6 2"	Nozzle	138.00	137.00	1.007
P1A 2"	Nozzle	138.00	137.00	1.007
P1B 2"	Nozzle	138.00	137.00	1.007
N4 6"	Nozzle	138.00	137.00	1.007
A2 8"	Nozzle	138.00	137.00	1.007
N5 2"	Nozzle	118.00	118.00	1.000
K5A 2"	Nozzle	138.00	137.00	1.007
K5B 2"	Nozzle	138.00	137.00	1.007
K5C 2"	Nozzle	138.00	137.00	1.007
P1C 2"	Nozzle	138.00	137.00	1.007
P1D 2"	Nozzle	138.00	137.00	1.007
A1 8"	Nozzle	138.00	137.00	1.007



N6 12"	Nozzle	138.00	137.00	1.007
N9 2"	Nozzle	138.00	137.00	1.007
K7 2"	Nozzle	138.00	137.00	1.007

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 2"	10.30	225.00	0.046
N7 2"	12.11	225.00	0.054
N10_4"	33.84	225.00	0.150
N2 2"	12.11	225.00	0.054
N8 2"	12.11	225.00	0.054
N3 2"	12.11	225.00	0.054
K1A 2"	10.30	225.00	0.046
K2A 2"	10.30	225.00	0.046
K3A 2"	10.30	225.00	0.046
K4A 2"	10.30	225.00	0.046
K1B 2"	10.30	225.00	0.046
K2B 2"	10.30	225.00	0.046
K3B_2"	10.30	225.00	0.046
K4B 2"	10.30	225.00	0.046
N11_8"	52.39	225.00	0.233
A3 8"	52.39	225.00	0.233
K6 2"	10.30	225.00	0.046
P1A 2"	10.30	225.00	0.046
P1B 2"	10.30	225.00	0.046
N4 6"	35.38	225.00	0.157
A2 8"	52.39	225.00	0.233
N5 2"	12.11	216.00	0.056
K5A 2"	10.30	225.00	0.046
K5B 2"	10.30	225.00	0.046
K5C 2"	10.30	225.00	0.046
P1C 2"	10.30	225.00	0.046
P1D 2"	10.30	225.00	0.046
A1 8"	52.39	225.00	0.233
N6 12"	74.50	225.00	0.331
N9 2"	12.11	225.00	0.054
K7 2"	10.30	225.00	0.046

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	4.864 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.446 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	5.274 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	4.115 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.491 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.491 MPa
App 27-4	= M.A.W.P.	3.741 MPa

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 3.741 • 1.0

= 4.864 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	180.7	234.0	0.772	4.99
Shell-1	118.2	234.0	0.505	4.99
Shell-2	117.5	234.0	0.502	4.96
Shell-3	99.5	234.0	0.425	4.94
Shell-4	116.3	234.0	0.497	4.91
Shell-5	180.4	234.0	0.771	4.89
Shell-6	179.5	234.0	0.767	4.87
Top head	176.1	234.0	0.753	4.87

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 2"	Nozzle	138.00	137.00	1.007
N7 2"	Nozzle	138.00	137.00	1.007
N10 4"	Nozzle	138.00	137.00	1.007
N2 2"	Nozzle	138.00	137.00	1.007
N8 2"	Nozzle	138.00	137.00	1.007
N3 2"	Nozzle	138.00	137.00	1.007
K1A 2"	Nozzle	138.00	137.00	1.007
K2A 2"	Nozzle	138.00	137.00	1.007
K3A 2"	Nozzle	138.00	137.00	1.007
K4A 2"	Nozzle	138.00	137.00	1.007
K1B 2"	Nozzle	138.00	137.00	1.007
K2B 2"	Nozzle	138.00	137.00	1.007
K3B 2"	Nozzle	138.00	137.00	1.007
K4B 2"	Nozzle	138.00	137.00	1.007
N11 8"	Nozzle	138.00	137.00	1.007
A3 8"	Nozzle	138.00	137.00	1.007
K6 2"	Nozzle	138.00	137.00	1.007
P1A 2"	Nozzle	138.00	137.00	1.007
P1B 2"	Nozzle	138.00	137.00	1.007
N4 6"	Nozzle	138.00	137.00	1.007
A2 8"	Nozzle	138.00	137.00	1.007
N5 2"	Nozzle	118.00	118.00	1.000
K5A 2"	Nozzle	138.00	137.00	1.007
K5B 2"	Nozzle	138.00	137.00	1.007
K5C 2"	Nozzle	138.00	137.00	1.007
P1C 2"	Nozzle	138.00	137.00	1.007
P1D 2"	Nozzle	138.00	137.00	1.007
A1 8"	Nozzle	138.00	137.00	1.007



N6 12"	Nozzle	138.00	137.00	1.007
N9 2"	Nozzle	138.00	137.00	1.007
K7 2"	Nozzle	138.00	137.00	1.007

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	138.00	1.000
Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Flange-1	138.00	138.00	1.000
Flange-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Flange-3	138.00	138.00	1.000
Flange-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Flange-5	138.00	138.00	1.000
Flange-6	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Top head	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 2"	10.56	225.00	0.047
N7 2"	12.41	225.00	0.055
N10_4"	34.68	225.00	0.154
N2 2"	12.41	225.00	0.055
N8 2"	12.41	225.00	0.055
N3 2"	12.41	225.00	0.055
K1A 2"	10.55	225.00	0.047
K2A 2"	10.55	225.00	0.047
K3A 2"	10.55	225.00	0.047
K4A 2"	10.55	225.00	0.047
K1B 2"	10.55	225.00	0.047
K2B 2"	10.55	225.00	0.047
K3B_2"	10.55	225.00	0.047
K4B 2"	10.55	225.00	0.047
N11_8"	53.69	225.00	0.239
A3 8"	53.69	225.00	0.239
K6 2"	10.45	225.00	0.046
P1A 2"	10.45	225.00	0.046
P1B 2"	10.38	225.00	0.046
N4 6"	35.68	225.00	0.159
A2 8"	52.82	225.00	0.235
N5 2"	12.16	216.00	0.056
K5A 2"	10.34	225.00	0.046
K5B 2"	10.34	225.00	0.046
K5C 2"	10.34	225.00	0.046
P1C 2"	10.34	225.00	0.046
P1D 2"	10.34	225.00	0.046
A1 8"	52.62	225.00	0.234
N6 12"	74.45	225.00	0.331
N9 2"	12.10	225.00	0.054
K7 2"	10.29	225.00	0.046

Elements Suitable for Test Pressure.

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**External Pressure Calculation Results Summary:****ASME Code, Section VIII Division 1, 2025****External Pressure Calculations:**

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	No Calc	522	7	0.0018625	101.884
20	30	5092.17	530	11	0.00050152	50.1069
30	40	5092.17	530	11	0.00050152	50.1069
40	50	No Calc	...	61	No Calc	No Calc
50	60	No Calc	...	61	No Calc	No Calc
60	70	4707.65	534	13	0.00068938	68.8754
70	80	4707.65	530	11	0.00050872	50.8255
80	90	No Calc	...	61	No Calc	No Calc
90	100	No Calc	...	61	No Calc	No Calc
100	110	2210.82	522	7	0.00044713	44.6721
110	120	No Calc	...	61	No Calc	No Calc
120	130	No Calc	...	61	No Calc	No Calc
130	140	192.167	522	7	0.0064843	121.756
140	150	No Calc	522	7	0.0018625	101.884

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	8	2.35199	0.10342	1.51807
20	30	12	5.41221	0.10342	1.38661
30	40	12	5.41221	0.10342	1.38661
40	50	62	57.8866	0.10342	No Calc
50	60	62	57.8866	0.10342	No Calc
60	70	14	5.34324	0.10342	2.23566
70	80	12	5.32246	0.10342	1.40649
80	90	62	57.7088	0.10342	No Calc
90	100	62	57.7088	0.10342	No Calc
100	110	8	4.05417	0.10342	0.79874
110	120	62	55.0926	0.10342	No Calc
120	130	62	55.0926	0.10342	No Calc
130	140	8	2.10409	0.10342	2.17699
140	150	8	2.35199	0.10342	1.51807

Minimum

1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	5092.17	239146	No Calc	No Calc
30	40	5092.17	239146	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	4707.65	99333.1	No Calc	No Calc
70	80	4707.65	222740	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	2210.82	77724.6	No Calc	No Calc
110	120	No Calc	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc



130	140	192.167	7558.76	No Calc	No Calc
140	150	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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**Element and Detail Weights:**

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	30.9805	27.018	27.8824	27.2995	2.16863
20	30	454.059	603.379	417.024	608.158	31.7841
30	40	302.706	402.253	278.016	405.439	21.1894
40	50	104.988	23.0303	103.581	23.11	7.34918
50	60	104.988	23.0303	103.581	23.11	7.34918
60	70	531.78	603.379	494.745	608.158	37.2246
70	80	258.458	343.453	237.377	346.174	18.092
80	90	104.988	23.0303	103.581	23.11	7.34918
90	100	104.988	23.0303	103.581	23.11	7.34918
100	110	221.353	444.654	194.061	448.176	15.4947
110	120	103.683	23.0303	102.276	23.11	7.25782
120	130	103.683	23.0303	102.276	23.11	7.25782
130	140	10.0123	20.1126	8.77777	20.2719	0.70086
140	150	30.9805	27.018	27.8824	27.2995	2.16863
Total		2467	2609.45	2304	2629.64	172

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Lini	9.20551	...	-38.25	...	Clad
10	Nozl	38.5022	...	-730.216	...	N1 2"
10	Nozl	9.20749	150	-272.875	...	N7 2"
20	Lini	114.294	...	1500	...	Clad
20	Nozl	24.6683	403	1500	...	N10_4"
20	Nozl	18.1448	357.775	100	...	N2 2"
20	Nozl	17.4594	-357.775	100	...	N8 2"
20	Nozl	10.2568	357.775	2000	...	N3 2"
20	Nozl	10.9038	-356.188	500	...	K1A 2"
20	Nozl	10.9038	-356.188	1500	...	K2A 2"
20	Nozl	10.9038	-356.188	1000	...	K3A 2"
20	Nozl	10.9038	...	1000	-356.188	K4A 2"
20	Nozl	10.9038	-178.094	500	308.467	K1B 2
20	Nozl	11.4195	-178.094	1500	308.467	K2B 2"
20	Nozl	10.9038	-178.094	1000	308.467	K3B_2"
20	Nozl	11.4195	...	500	-356.188	K4B 2"
20	Nozl	68.2798	356.188	1000	...	N11_8"
20	Nozl	117.053	...	2500	378.006	A3 8"
20	Wght	5942	-1299.04	1250	-750	REBOILER
20	Wght	566	1367	1000	...	N11 Fy wt
20	Wght	10	...	150	...	N8 dist wt
20	Wght	10	...	100	...	N8 dist sup wt
20	Wght	30	...	2500	300	A3 davit wt
30	Lini	76.1957	...	1000	...	Clad
30	Wght	60	...	1450	...	CLEAT WEIGHT
40	Lini	4.26696	...	56	...	Clad
50	Lini	4.26696	...	56	...	Clad
60	Pack	63.6917	...	2100	...	Pack: [1 of 1]
60	Lini	114.294	...	1500	...	Clad
60	Nozl	11.0224	356.188	1500	...	K6 2"
60	Nozl	11.0224	356.188	1100	...	P1A 2"
60	Lugs	319.775	...	1178	...	Support Lugs
60	Wght	30	...	1500	...	Noz clad wt
60	Wght	20	...	1150	...	Bed2 sup ring w
60	Wght	20	...	1200	...	Bed2 sup grid w
70	Pack	42.4612	...	600	...	Bed2
70	Lini	65.0578	...	853.825	...	Clad
70	Nozl	11.0224	356.188	1099.99	...	P1B 2"



70	Nozl	147.478	-378.006	1400	...
70	Nozl	85.5104	-378.006	799.995	...
70	Wght	783	-1339	1400	...
70	Wght	60	...	1400	...
70	Wght	20	...	1350	...
70	Wght	20	...	1250	...
70	Wght	30	-300	799.995	...
80	Lini	4.26696	...	56	...
90	Lini	4.26696	...	56	...
100	Pack	53.0765	...	950	...
100	Lini	84.2275	...	1105.41	...
100	Nozl	8.29628	357.775	1800	...
100	Nozl	10.9044	-356.188	99.9997	...
100	Nozl	10.9044	...	99.9997	-356.188
100	Nozl	10.9044	...	99.9997	356.188
100	Nozl	10.9038	356.188	99.9997	...
100	Nozl	10.9038	356.188	1600	...
100	Nozl	115.733	-378.006	1800	...
100	Wght	200	280	1500	...
100	Wght	200	-280	1500	...
100	Wght	60	...	1800	...
100	Wght	20	...	1750	...
100	Wght	10	...	1700	...
100	Wght	10	...	149.999	...
100	Wght	20	...	179.999	...
100	Wght	30	-300	1800	...
110	Lini	4.26696	...	56	...
120	Lini	4.26696	...	56	...
130	Lini	3.80979	...	50	...
140	Lini	9.20551	...	88.25	...
140	Nozl	94.1719	...	297.506	...
140	Nozl	13.1337	200	228.249	...
140	Nozl	14.557	-200	226.662	...
140	Wght	1235	1415	...	...
140	Wght	500	280	0.0046512	...
140	Wght	30	...	125	...
140	Forc	...	...	125	...
60	Pliq	40.12	...	2100.000	...
70	Pliq	26.75	...	600.000	...
100	Pliq	33.44	...	950.000	...

N4 6"
 A2 8"
 N4 Fy wt
 N4 dist wt
 N4 dist sup wt
 Bed2 limiter
 A2 davit wt
 Clad
 Clad
 Bed 1
 Clad
 N5 2"
 K5A 2"
 K5B 2"
 K5C 2"
 P1C 2"
 P1D 2"
 A1 8"
 Lifting lug1
 Lifting lug 2
 N5 dist wt
 N5 dist sup wt
 Bed 1 limiter
 Bed1 sup ring
 Bed1 sup grid
 A1 davit wt
 Clad
 Clad
 Clad
 Clad
 N6 12"
 N9 2"
 K7 2"
 N6 Fy wt
 PIPE DAVIT
 Insert plate wt
 N6 F/M

Total Weight of Each Detail Type:

Packing	159.2
Packing Liquid	100.3
Lining	501.9
Nozzles	958.3
Lugs	319.8
Weights	9916.0

Sum of the Detail Weights 11955.5 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	2640.4	2640.4	2640.4	2640.4	2640.4	2640.4
Nozzles	958.3	958.3	958.3	958.3	958.3	958.3
Lugs	319.8	319.8	319.8	319.8	319.8	319.8
Wld Weights	1756.0	1756.0	1756.0	1756.0	1756.0	1756.0
Packing	...	...	...	...	159.2	159.2
Lining	501.9	501.9	501.9	501.9	501.9	501.9
Ope Weights	...	...	...	...	...	8160.0
Packing Liq	...	...	...	...	...	100.3
Test Liquid	...	2545.4	...	...	...	...
Totals	6176.4	8721.8	6176.4	6176.4	6335.6	14595.9

**Field Installation Options:**

* Packing installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	6176.4 kgm
Shop Test Wt.	- Fabricated weight + water (full)	8721.8 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	6176.4 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	6176.4 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	6335.6 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	14595.9 kgm
Field Test Wt.	- Empty weight + test medium (full)	8721.8 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	4988.4 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	3447.5 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	2804.9 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	382537	357022
20	30	4995132	4768938
30	40	3330088	3179292
40	50	419013	181220
50	60	419013	181220
60	70	5032831	4768938
70	80	2843312	2714558
80	90	419013	181220
90	100	419013	181220
100	110	3625549	3514421
110	120	419013	181220
120	130	419013	181220
130	140	163991	158965
140	150	382537	357022

Total	23270060.0	20906472.0 mm ²
Total	23.3	20.9 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	90.0644	90.0644	117.066	13548.8	90.0644
20	30	1550.26	7502.26	2153.27	67929696	7502.26
30	40	460.091	460.091	862.098	...	460.091
40	50	116.604	116.604	139.621	...	116.604
50	60	116.604	116.604	139.621	...	116.604
60	Lugs	251.781	293.053	452.784	77028.7	279.678
Lugs	70	503.562	586.106	905.569	...	559.356
70	80	635.619	1567.83	978.862	11198890	1541.08
80	90	116.604	116.604	139.621	...	116.604
90	100	116.604	116.604	139.621	...	116.604
100	110	959.626	1136.14	1404.01	450239	1102.7
110	120	115.208	115.208	138.224	...	115.208

120	130	115.208	115.208	138.224	...	115.208
130	140	14.5229	14.5229	34.6233	...	14.5229
140	150	694.217	1929.22	721.219	18513778	1929.22



Empty Support Force + the Sum of the Y forces 31695.92 N
 Operating Support Force + the Sum of the Y forces 115241.56 N
 Hydro Support Force + the Sum of the Y forces 58252.18 N

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	...	...	...
20	30	-90.0644	-90.0644	-117.066
30	40	-7592.32	-7592.32	-2270.34
40	50	-8052.42	-8052.42	-3132.43
50	60	-8169.02	-8169.02	-3272.05
60	Lugs	-8285.62	-8285.62	-3411.68
Lugs	70	5610.5	5697.44	4599.97
70	80	5051.15	5111.33	3694.4
80	90	3510.07	3543.5	2715.54
90	100	3393.46	3426.9	2575.92
100	110	3276.86	3310.3	2436.3
110	120	2174.16	2174.16	1032.29
120	130	2058.95	2058.95	894.066
130	140	1943.74	1943.74	755.842
140	150	1929.22	1929.22	721.219

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	13548.8	13548.8	13548.8
20	30	67943248	67943248	67943248
30	40	67943248	67943248	67943248
40	50	67943248	67943248	67943248
50	60	67943248	67943248	67943248
60	Lugs	68020272	68020272	68020272
Lugs	70	30162908	30162908	30162908
70	80	30162908	30162908	30162908
80	90	18964018	18964018	18964018
90	100	18964018	18964018	18964018
100	110	18964018	18964018	18964018
110	120	18513778	18513778	18513778
120	130	18513778	18513778	18513778
130	140	18513778	18513778	18513778
140	150	18513778	18513778	18513778



Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1 2"	8.58	10.20	225	600	GR 1.1	6.129	6.735	8.576	6.735
N7 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
N10_4"	8.58	10.20	225	600	GR 1.1	4.621	8.242	8.576	8.242
N2 2"	8.58	10.20	225	600	GR 1.1	6.129	6.735	8.576	6.735
N8 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
N3 2"	9.50	10.34	225	600	GR 1.11	6.129	8.115	9.496	8.115
K1A 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K2A 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K3A 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K4A 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K1B 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K2B 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K3B 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K4B 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
N11_8"	8.58	10.20	225	600	GR 1.1	3.074	8.576	8.576	8.576
A3 8"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K6 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
P1A 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
P1B 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
N4 6"	8.58	10.20	225	600	GR 1.1	3.585	8.576	8.576	8.576
A2 8"	4.29	5.10	225	300	GR 1.1	...	...	...	...
N5 2"	8.58	10.20	225	600	GR 1.1	6.129	6.735	8.576	6.735
K5A 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K5B 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K5C 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
P1C 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
P1D 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
A1 8"	4.29	5.10	225	300	GR 1.1	...	...	...	...
N6 12"	4.29	5.10	225	300	GR 1.1	2.214	4.221	4.290	4.221
N9 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
K7 2"	4.29	5.10	225	300	GR 1.1	...	...	...	...
Min Rating	4.221	5.100, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N1 2"	77.851	0.50	494975	3750	3.741	6.735	Pass
N10_4"	135.001	0.50	1979899	7500	3.741	8.242	Pass
N2 2"	77.851	0.50	494975	3750	3.741	6.735	Pass
N3 2"	77.851	0.50	494975	3750	3.741	8.115	Pass
N11_8"	244.602	0.50	7919597	15000	3.741	8.576	Pass
N4 6"	192.151	0.50	4454763	11250	3.741	8.576	Pass
N5 2"	77.851	0.50	494975	3750	3.741	6.735	Pass
N6 12"	357.251	0.50	17819094	22500	3.741	4.221	Pass

Evaluating Flange on Nozzle [N1 2"]:



$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(494975) + 4 \cdot \text{abs}(3750) 77.851 \leq \pi \cdot 77.851^3 [(8.576 - 3.741) + .500 \cdot 8.576]$$

$$= 0.90878E+07 \leq 0.13526E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1 2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.576, 0.500 \cdot 8.576 + 8.576 - 9087832.0 / (\pi \cdot 77.851^3))$$

$$= \min(8.576, 6.735)$$

$$= 6.735 \text{ MPa}$$

Evaluating Flange on Nozzle [N104"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(1979899) + 4 \cdot \text{abs}(7500) 135.001 \leq \pi \cdot 135.001^3 [(8.576 - 3.741) + .500 \cdot 8.576]$$

$$= 0.35730E+08 \leq 0.70534E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N104" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.576, 0.500 \cdot 8.576 + 8.576 - 35730028.0 / (\pi \cdot 135.001^3))$$

$$= \min(8.576, 8.242)$$

$$= 8.242 \text{ MPa}$$

Evaluating Flange on Nozzle [N2 2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(494975) + 4 \cdot \text{abs}(3750) 77.851 \leq \pi \cdot 77.851^3 [(8.576 - 3.741) + .500 \cdot 8.576]$$

$$= 0.90878E+07 \leq 0.13526E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2 2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.576, 0.500 \cdot 8.576 + 8.576 - 9087832.0 / (\pi \cdot 77.851^3))$$

$$= \min(8.576, 6.735)$$

$$= 6.735 \text{ MPa}$$

Evaluating Flange on Nozzle [N3 2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(494975) + 4 \cdot \text{abs}(3750) 77.851 \leq \pi \cdot 77.851^3 [(9.496 - 3.741) + .500 \cdot 9.496]$$

$$= 0.90878E+07 \leq 0.15573E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3 2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(9.496, 0.500 \cdot 9.496 + 9.496 - 9087832.0 / (\pi \cdot 77.851^3))$$

$$= \min(9.496, 8.115)$$

$$= 8.115 \text{ MPa}$$

Evaluating Flange on Nozzle [N118"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(7919597) + 4 \cdot \text{abs}(15000) 244.602 \leq \pi \cdot 244.602^3 [(8.576 - 3.741) + .500 \cdot 8.576]$$

$$= 0.14140E+09 \leq 0.41953E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N118" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.576, 0.500 \cdot 8.576 + 8.576 - 141395536.0 / (\pi \cdot 244.602^3))$$

$$= \min(8.576, 9.789)$$

$$= 8.576 \text{ MPa}$$

Evaluating Flange on Nozzle [N4 6"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(4454763) + 4 \cdot \text{abs}(11250) 192.151 \leq \pi \cdot 192.151^3 [(8.576 - 3.741) + .500 \cdot 8.576]$$

$$= 0.79927E+08 \leq 0.20338E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4 6" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.576, 0.500 \cdot 8.576 + 8.576 - 79926536.0 / (\pi \cdot 192.151^3))$$

$$= \min(8.576, 9.278)$$

$$= 8.576 \text{ MPa}$$

Evaluating Flange on Nozzle [N5 2"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(494975) + 4 \cdot \text{abs}(3750) 77.851 \leq \pi \cdot 77.851^3 [(8.576 - 3.741) + .500 \cdot 8.576]$$



= 0.90878E+07 ≤ 0.13526E+08 Check Passes

Maximum Pressure for Flange on Nozzle N5 2" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(8.576, 0.500 \cdot 8.576 + 8.576 - 9087832.0 / (\pi \cdot 77.851^3))$$

$$= \min(8.576, 6.735)$$

$$= 6.735 \text{ MPa}$$

Evaluating Flange on Nozzle [N6 12"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(17819094) + 4 \cdot \text{abs}(22500) 357.251 \leq \pi \cdot 357.251^3 [(4.290 - 3.741) + .500 \cdot 4.290]$$

$$= 0.31727E+09 \leq 0.38605E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N6 12" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.290, 0.500 \cdot 4.290 + 4.290 - 317271040.0 / (\pi \cdot 357.251^3))$$

$$= \min(4.290, 4.221)$$

$$= 4.221 \text{ MPa}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 3.10821 Hz.

The Natural Frequency for the Vessel (Ope...) is 3.1082 Hz.

The Natural Frequency for the Vessel (Filled) is 4.44639 Hz.

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**Forces/Moments Applied to Vessel (Combined w/Wind Loads)**

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	23334.5	22518996

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	70	...	...
70	80	...	...
80	90	...	...
90	100	...	...
100	110	...	...
110	120	...	...
120	130	...	...
130	140	...	...
140	150	23334.5	22518996

User Input Forces and Moments:

From Node	Distance From	- - - - Fx	Forces - - - - Fy	- - - - Fz	- - - - Mx	Moments - - - - My	- - - - Mz
140	125.00	16500.	24750.	16500.	0.139E+08	0.139E+08	

**Wind Load Calculation Results:****Input Values:**

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	3500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.209 \cdot 0.936)/(1 + 1.7 \cdot 3.4 \cdot 0.209)))$$

$$= \min(0.85, 0.893)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	3.108 Hz
Natural Frequency of Vessel (Empty)	3.108 Hz
Natural Frequency of Vessel (Test)	4.446 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.698
Structure Height to Diameter ratio	24.565
Height to top of Structure	12934.496 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 103.787 + 3500.0 = 3603.787 \text{ mm}$$

$$= 11.823 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (11.823 \text{ ft.}) < 15 \text{ ft.}$$

$$= 2.01(15 / Z_g)^{2/\alpha}$$

$$= 2.01(15/900.0)^{2/9.5}$$

$$= 0.849$$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:

$$= 1, \text{ per Table 26.6-1}$$



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$

$= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$

$= 16.0 \text{ psf } [0.766] \text{ kPa}$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$

$= 16.0 \cdot 0.85 \cdot 0.698 \cdot 1.871$

$= 17.8 \text{ lbs. } [79.0] \text{ N}$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Bottom head	3603.8	0.000	0.000	0.000	0.849	1.000	0.766
Shell-1	5176.5	0.000	0.000	0.000	0.871	1.000	0.766
Shell-2	7676.5	0.000	0.000	0.000	0.947	1.000	0.766
Flange-1	8735.7	0.000	0.000	0.000	0.973	1.000	0.766
Flange-2	8847.7	0.000	0.000	0.000	0.975	1.000	0.766
Shell-3	10403.7	0.000	0.000	0.000	1.009	1.000	0.766
Shell-4	12757.5	0.000	0.000	0.000	1.054	1.000	0.766
Flange-3	13670.5	0.000	0.000	0.000	1.069	1.000	0.766
Flange-4	13782.5	0.000	0.000	0.000	1.071	1.000	0.766
Shell-5	14943.9	0.000	0.000	0.000	1.089	1.000	0.766
Flange-5	16108.5	0.000	0.000	0.000	1.107	1.000	0.766
Flange-6	16220.5	0.000	0.000	0.000	1.108	1.000	0.766
Shell-6	16326.5	0.000	0.000	0.000	1.110	1.000	0.766
Top head	16457.2	0.000	0.000	0.000	1.112	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- D1 - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, ≤ 0.000003 (High); $0.000003 < 0.000004$ (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/12799.996 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)



:: [1.73/41.99²] • 10⁴ = 9.819 [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 141417 / (12800.0 \cdot 535.227^2)$$

$$= 0.38567E-04$$

Since Vp is > 0.000004 no further vibration analysis is required !

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
REBOILER	0.00	4800.00	0.77	0.00
N11 Fy wt	0.00	4550.00	0.77	0.00
N8 dist wt	0.00	3700.00	0.77	0.00
N8 dist sup w	0.00	3650.00	0.77	0.00
A3 davit wt	0.00	6050.00	0.77	0.00
CLEAT WEIGHT	0.00	8000.00	0.77	0.00
Noz clad wt	0.00	10277.18	0.77	0.00
Bed2 sup ring	0.00	9927.18	0.77	0.00
Bed2 sup grid	0.00	9977.18	0.77	0.00
N4 Fy wt	0.00	13177.17	0.77	0.00
N4 dist wt	0.00	13177.17	0.77	0.00
N4 dist sup w	0.00	13127.17	0.77	0.00
Bed2 limiter	0.00	13027.17	0.77	0.00
A2 davit wt	0.00	12577.17	0.77	0.00
Lifting lug1	0.00	15212.00	0.77	0.00
Lifting lug 2	0.00	15212.00	0.77	0.00
N5 dist wt	0.00	15512.00	0.77	0.00
N5 dist sup w	0.00	15462.00	0.77	0.00
Bed 1 limiter	0.00	15412.00	0.77	0.00
Bed1 sup ring	0.00	13862.00	0.77	0.00
Bed1 sup grid	0.00	13892.00	0.77	0.00
A1 davit wt	0.00	15512.00	0.77	0.00
N6 Fy wt	0.00	16250.00	0.77	0.00
PIPE DAVIT	0.00	16250.00	0.77	0.00
Insert plate	0.00	16375.00	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 3.1082 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	3603.79	1117.08	173858	0.76606	86.8671
20	30	5176.5	1134.2	3402599	0.76606	1700.09
30	40	7676.5	1134.2	2268400	0.76606	1133.39
40	50	8735.67	1082.84	121278	0.76606	60.5959
50	60	8847.67	1082.84	121278	0.76606	60.5959
60	70	10403.7	1142.76	3428280	0.76606	1712.92
70	80	12757.5	1134.2	1936816	0.76606	967.72
80	90	13670.5	1082.84	121278	0.76606	60.5959
90	100	13782.5	1082.84	121278	0.76606	60.5959
100	110	14943.9	1117.08	2469663	0.76606	1233.95
110	120	16108.5	1082.84	121278	0.76606	60.5959
120	130	16220.5	1082.84	121278	0.76606	60.5959
130	140	16326.5	1117.08	111708	0.76606	55.8143
140	150	16457.2	1117.08	173858	0.76606	86.8671

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



Be sure the wind speed is in accordance with the specified wind design code.

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**Earthquake Load Calculation Results:****Earthquake Load Calculation:****Input Values:**

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.450
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	No
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(42.40980.75)$
 $= 0.332$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/3.108)$
 $= 0.322$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.332, 0.322) \text{ per 12.8.2}$
 $= 0.322$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R / I)$
 $= 0.157 / (2.0/1.25)$
 $= 0.098$

Maximum value of Cs per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R / I_e))$
 $= 0.104 / (0.322(2.0/1.25))$
 $= 0.202$

Minimum value of Cs per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot I_e, 0.030)$
 $= \max(0.044 \cdot 0.1568 \cdot 1.25, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= Cs \cdot W$ (Equation 12.8-1):
 $= 0.098 \cdot 139991.6$
 $= 13719.173$ N

Distribute the Base shear force to each element according to the equations



$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_{xk} / (\text{Sum of } W_i * h_{ik})$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 3.1082 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	25	883.171	0.39576	0.30404	...
20	30	1550	73567.1	2043.94	324.468	...
30	40	4050	4511.65	327.524	251.614	...
40	50	5109.17	1143.42	104.715	80.4454	...
50	60	5221.17	1143.42	107.011	82.2089	...
60	Lugs	6277.17	2873.68	323.336	213.414	-24750
Lugs	70	7277.17	5747.35	749.692	494.825	24750
70	80	9131	15374.1	2516.29	783.702	24750
80	90	10044	1143.42	205.857	158.146	24750
90	100	10156	1143.42	208.152	159.909	24750
100	110	11317.4	11141	2260.07	1466.51	24750
110	120	12482	1129.73	252.761	194.179	24750
120	130	12594	1129.73	255.029	195.921	24750
130	140	12700	142.412	32.4191	24.9054	24750
140	150	12775	18917.9	4331.98	1197.55	24750



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	6307.89	...	...	...	...
20	30	4727.17	...	...	...	...
30	40	2227.18	...	...	...	...
40	50	1168	...	...	...	...
50	60	1056	...	...	...	...
60	Lugs	500	...	...	...	...
Lugs	70	1000	23334.5	23334.5	173620432	173620432
70	80	2853.83	23334.5	23334.5	126932464	126932464
80	90	3766.83	23334.5	23334.5	86994968	86994968
90	100	3878.83	23334.5	23334.5	84380448	84380448
100	110	5040.24	23334.5	23334.5	81765912	81765912
110	120	6204.82	23334.5	23334.5	30082448	30082448
120	130	6316.82	23334.5	23334.5	27467918	27467918
130	140	6422.82	23334.5	23334.5	24853390	24853390
140	150	6553.53	23334.5	23334.5	22518996	22518996

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	6307.89	...	...	...	...
20	30	4727.17	86.8671	0.39576	7014.17	31.9564
30	40	2227.18	1786.96	2044.33	2818890	3068370
40	50	1168	2920.35	2371.86	7528106	7486352
50	60	1056	2980.95	2476.57	7867988	7765508
60	Lugs	500	3041.54	2583.58	8205385	8048992
Lugs	70	1000	7341.2	13719.2	744105	41908164
70	80	2853.83	2586.74	10062.6	5959885	31820240
80	90	3766.83	1619.02	7546.27	2362305	16755310
90	100	3878.83	1558.42	7340.41	2184295	15921318
100	110	5040.24	1497.83	7132.26	2013076	15110518
110	120	6204.82	263.873	4872.19	64048.9	1819829
120	130	6316.82	203.277	4619.42	37877.8	1288083
130	140	6422.82	142.681	4364.39	18496.2	784785
140	150	6553.53	86.8671	4331.98	7014.21	349791

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

Note:

Review the Vessel Design Summary for the cumulative shear force and bending moment on the support.

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**Wind Deflection Calculations:**

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	6.53073	6.56183	...
20	30	...	4.63334	6.49962	...
30	40	...	1.52288	2.76707	...
40	50	...	0.20902	0.2787	...
50	60	...	0.069674	0.13935	...
60	Lugs	...	-0.47031	...	...
Lugs	70	23334.5	1.18415	4.02777	0.00081722
70	80	23334.5	7.20954	11.0306	0.0023147
80	90	23334.5	11.2997	11.5692	0.0023254
90	100	23334.5	11.8389	12.1089	0.0023357
100	110	23334.5	18.0306	24.8534	0.0039659
110	120	23334.5	25.2149	25.5765	0.0039695
120	130	23334.5	25.9382	26.3	0.0039727
130	140	23334.5	26.6234	26.9476	0.004004
140	150	23334.5	27.1099	27.2722	0.0040041

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)

Allowable deflection : 64.000 Actual deflection : 27.272 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	...	1.07117	1.07928	0.00029777
20	30	86.8671	0.61138	1.06306	0.00029776
30	40	1786.96	0.085497	0.25247	0.00017681
40	50	2920.35	0.0046451	0.0062534	0.21567E-05
50	60	2980.95	0.0015178	0.0030663	0.11007E-05
60	Lugs	3041.54	-0.011707	...	...
Lugs	70	7341.2	0.059735	0.20808	0.00012013
70	80	2586.74	0.37305	0.56323	0.00017966
80	90	1619.02	0.5763	0.58937	0.00017996
90	100	1558.42	0.60246	0.61555	0.00018023
100	110	1497.83	0.88702	1.17134	0.00020523
110	120	263.873	1.18584	1.20034	0.00020524
120	130	203.277	1.21483	1.22933	0.00020525
130	140	142.681	1.24228	1.25522	0.00020526
140	150	86.8671	1.26169	1.26817	0.00020526

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	62.3297	389.56
20	30	63.2849	395.531
30	40	63.2849	395.531
40	50	60.4192	377.62
50	60	60.4192	377.62
60	70	63.7625	398.516



70	80	63.2849	395.531
80	90	60.4192	377.62
90	100	60.4192	377.62
100	110	62.3297	389.56
110	120	60.4192	377.62
120	130	60.4192	377.62
130	140	62.3297	389.56
140	150	62.3297	389.56

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)

Allowable deflection : 64.000 Actual deflection : 1.268 mm

Total Deflection in the Operating Condition + Applied Forces:

Allowable deflection : 64.000 Actual deflection : 28.540 mm

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**Longitudinal Stress Constants:**

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	12918.2	11325.4	1634946	1438859
20	30	19528.1	17935.3	2472961	2279833
30	40	19528.1	17935.3	2472961	2279833
40	50	19528.1	17935.3	2472961	2279833
50	60	22870.8	21278	2897351	2705670
60	70	22870.8	21278	2897351	2705670
70	80	19528.1	17935.3	2472961	2279833
80	90	19528.1	17935.3	2472961	2279833
90	100	12918.2	11325.4	1634946	1438859
100	110	12918.2	11325.4	1634946	1438859
110	120	12918.2	11325.4	1634946	1438859
120	130	12918.2	11325.4	1634946	1438859
130	140	12918.2	11325.4	1634946	1438859
140	150	12918.2	11325.4	1634946	1438859

**Longitudinal Allowable Stresses:**

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	137.997	233.996	-95.1829	-113.798
20	30	137.997	233.996	-101.412	-119.814
30	40	137.997	233.996	-101.412	-119.814
40	50	137.997	233.996	-101.412	-119.814
50	60	137.997	233.996	-103.495	-121.447
60	Lugs	137.997	233.996	-103.495	-121.447
Lugs	70	137.997	233.996	-103.495	-121.447
70	80	137.997	233.996	-101.412	-119.814
80	90	137.997	233.996	-101.412	-119.814
90	100	137.997	233.996	-95.1829	-113.798
100	110	137.997	233.996	-95.1829	-113.798
110	120	137.997	233.996	-95.1829	-113.798
120	130	137.997	233.996	-95.1829	-113.798
130	140	137.997	233.996	-95.1829	-113.798
140	150	137.997	233.996	-95.1829	-113.798



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	6.15858	-1.95428	87.2663
20	30	3.89413	-1.27216	55.1794
30	40	3.89413	-1.27216	55.1794
40	50	...	...	...
50	60	...	...	...
60	70	3.28448	-1.08856	46.5406
70	80	3.89413	-1.27216	55.1794
80	90	...	...	...
90	100	...	...	...
100	110	6.15858	-1.95428	87.2663
110	120	...	...	...
120	130	...	...	...
130	140	6.15858	-1.95428	87.2663
140	150	6.15858	-1.95428	87.2663

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	...	...	...	0.0094133	0.0094133
20	30	0.049246	0.049246	0.06401	29.7923	29.7923
30	40	4.1514	4.1514	1.24139	29.7923	29.7923
40	50	4.40297	4.40297	1.71278	29.7923	29.7923
50	60	3.76503	3.76503	1.50806	25.1034	25.1034
60	Lugs	3.81877	3.81877	1.57241	25.1318	25.1318
Lugs	70	-2.58583	-2.6259	-2.58583	11.1445	11.1445
70	80	-2.76191	-2.79482	-2.76191	13.2261	13.2261
80	90	-1.91926	-1.93755	-1.91926	8.3155	8.3155
90	100	-2.93845	-2.9674	-2.93845	13.1757	13.1757
100	110	-2.83748	-2.86643	-2.83748	13.1757	13.1757
110	120	-1.88263	-1.88263	-1.88263	12.8629	12.8629
120	130	-1.78287	-1.78287	-1.78287	12.8629	12.8629
130	140	-1.68311	-1.68311	-1.68311	12.8629	12.8629
140	150	-1.67054	-1.67054	-1.67054	12.8629	12.8629

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.0094133	...	...	...	...
20	30	29.7923	0.0030756	0.14013E-04	0.0015378	...
30	40	29.7923	1.23605	1.34544	0.61802	...
40	50	29.7923	3.30099	3.28268	1.65049	...
50	60	25.1034	2.90703	2.86917	1.45352	...
60	Lugs	25.1318	3.03169	2.97391	1.51585	...
Lugs	70	11.1445	0.27493	15.4841	0.13746	...
70	80	13.2261	2.61334	13.9528	1.30667	...



80	90	8.3155	1.03584	7.347	0.51792	...
90	100	13.1757	1.51759	11.0617	0.75879	...
100	110	13.1757	1.39863	10.4984	0.69932	...
110	120	12.8629	0.044499	1.26437	0.02225	...
120	130	12.8629	0.026316	0.89492	0.013158	...
130	140	12.8629	0.012851	0.54525	0.0064253	...
140	150	12.8629	0.0048733	0.24303	0.0024366	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	...
20	30	...	...	...	0.10765E-04
30	40	...	...	...	0.21391
40	50	...	...	...	0.60922
50	60	...	...	...	0.53954
60	Lugs	...	...	...	0.56843
Lugs	70	...	...	...	7.1729
70	80	...	...	...	5.67327
80	90	...	...	...	2.83026
90	100	...	...	...	4.22618
100	110	...	...	...	3.98026
110	120	...	...	...	0.37205
120	130	...	...	...	0.25407
130	140	...	...	...	0.15128
140	150	...	...	...	0.067183

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	...	...
60	Lugs	-1.16327	-1.16327
Lugs	70	1.16327	1.16327
70	80	1.38008	1.38008
80	90	1.38008	1.38008
90	100	2.18554	2.18554
100	110	2.18554	2.18554
110	120	2.18554	2.18554
120	130	2.18554	2.18554
130	140	2.18554	2.18554
140	150	2.18554	2.18554

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	Lugs	...	...	...	...
Lugs	70	64.1486	64.1486	59.9047	59.9047



70	80	55.6584	55.6584	51.3117	51.3117
80	90	38.1463	38.1463	35.1672	35.1672
90	100	58.6252	58.6252	51.594	51.594
100	110	56.8087	56.8087	49.9953	49.9953
110	120	20.9005	20.9005	18.3938	18.3938
120	130	19.084	19.084	16.7951	16.7951
130	140	17.2675	17.2675	15.1965	15.1965
140	150	15.6456	15.6456	13.7691	13.7691

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	Lugs	...
Lugs	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	138.00	-0.01	95.18	0.0001	0.0001
20	29.84	138.00	-29.75	101.41	0.2163	0.2933
30	35.18	138.00	-26.88	101.41	0.2549	0.2650
60	30.82	138.00	-25.51	103.50	0.2233	0.2465
60	74.15	138.00	-76.99	103.50	0.5373	0.7439
70	70.12	138.00	-72.88	101.41	0.5081	0.7187
100	70.73	138.00	-72.03	95.18	0.5126	0.7568
130	30.65	138.00	-29.64	95.18	0.2221	0.3114
140	29.03	138.00	-28.00	95.18	0.2104	0.2942

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	138.00	-0.01	95.18	0.0001	0.0001
20	29.84	138.00	-29.74	101.41	0.2162	0.2933
30	34.16	138.00	-25.85	101.41	0.2475	0.2549
60	28.36	138.00	-23.04	103.50	0.2055	0.2227
60	81.04	138.00	-83.89	103.50	0.5873	0.8106
70	73.18	138.00	-75.94	101.41	0.5303	0.7488
100	73.31	138.00	-74.62	95.18	0.5313	0.7839
130	30.78	138.00	-29.78	95.18	0.2231	0.3129



140 29.09 138.00 -28.06 95.18 0.2108 0.2948

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	138.00	-0.01	95.18	0.0001	0.0001
20	29.84	138.00	-29.75	101.41	0.2163	0.2933
30	35.18	138.00	-26.88	101.41	0.2549	0.2650
60	30.82	138.00	-25.51	103.50	0.2233	0.2465
60	74.11	138.00	-77.03	103.50	0.5370	0.7443
70	70.08	138.00	-72.91	101.41	0.5079	0.7190
100	70.70	138.00	-72.06	95.18	0.5123	0.7571
130	30.65	138.00	-29.64	95.18	0.2221	0.3114
140	29.03	138.00	-28.00	95.18	0.2104	0.2942

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	138.00	-0.01	95.18	0.0001	0.0001
20	29.84	138.00	-29.74	101.41	0.2162	0.2933
30	35.29	138.00	-26.99	101.41	0.2557	0.2661
60	30.76	138.00	-25.45	103.50	0.2229	0.2459
60	89.31	138.00	-92.24	103.50	0.6472	0.8912
70	81.42	138.00	-84.25	101.41	0.5900	0.8308
100	79.80	138.00	-81.16	95.18	0.5783	0.8527
130	31.18	138.00	-30.17	95.18	0.2259	0.3170
140	29.27	138.00	-28.24	95.18	0.2121	0.2967

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	234.00	-0.01	113.80	0.0000	0.0001
20	29.86	234.00	-29.73	119.81	0.1276	0.2481
30	31.65	234.00	-29.17	119.81	0.1353	0.2435
60	28.22	234.00	-25.08	121.45	0.1206	0.2065
60	8.70	234.00	-13.87	121.45	0.0372	0.1142
70	11.77	234.00	-17.29	119.81	0.0503	0.1443
100	11.04	234.00	-16.71	113.80	0.0472	0.1469
130	11.19	234.00	-14.55	113.80	0.0478	0.1279
140	11.19	234.00	-14.54	113.80	0.0478	0.1277

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	234.00	-0.01	113.80	0.0000	0.0001
20	29.86	234.00	-29.73	119.81	0.1276	0.2481
30	31.03	234.00	-28.55	119.81	0.1326	0.2383
60	26.70	234.00	-23.56	121.45	0.1141	0.1940
60	8.56	234.00	-13.73	121.45	0.0366	0.1131
70	10.46	234.00	-15.99	119.81	0.0447	0.1334
100	10.34	234.00	-16.01	113.80	0.0442	0.1407
130	11.18	234.00	-14.55	113.80	0.0478	0.1278
140	11.19	234.00	-14.53	113.80	0.0478	0.1277

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	39.07	138.00	-22.98	101.41	0.2832	0.2266
60	34.10	138.00	-22.22	103.50	0.2471	0.2147
60	77.39	138.00	-73.75	103.50	0.5608	0.7126
70	73.98	138.00	-69.02	101.41	0.5361	0.6806



100	76.86	138.00	-65.91	95.18	0.5570	0.6924
130	36.80	138.00	-23.48	95.18	0.2667	0.2467
140	35.19	138.00	-21.84	95.18	0.2550	0.2295

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	39.18	138.00	-23.09	101.41	0.2839	0.2277
60	34.05	138.00	-22.17	103.50	0.2467	0.2142
60	92.60	138.00	-88.96	103.50	0.6710	0.8595
70	85.32	138.00	-80.36	101.41	0.6183	0.7924
100	85.96	138.00	-75.01	95.18	0.6229	0.7880
130	37.34	138.00	-24.01	95.18	0.2706	0.2523
140	35.43	138.00	-22.08	95.18	0.2567	0.2320

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-1.96	95.18		0.0206
20	28.57	138.00	-31.02	101.41	0.2071	0.3059
30	33.91	138.00	-28.15	101.41	0.2457	0.2776
60	29.73	138.00	-26.60	103.50	0.2154	0.2570
60	73.02	138.00	-78.12	103.50	0.5291	0.7548
70	68.81	138.00	-74.18	101.41	0.4986	0.7315
100	68.75	138.00	-74.02	95.18	0.4982	0.7776
130	28.69	138.00	-31.60	95.18	0.2079	0.3319
140	27.07	138.00	-29.95	95.18	0.1962	0.3147

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-1.96	95.18		0.0206
20	28.57	138.00	-31.02	101.41	0.2070	0.3058
30	34.02	138.00	-28.26	101.41	0.2465	0.2787
60	29.67	138.00	-26.54	103.50	0.2150	0.2564
60	88.23	138.00	-93.33	103.50	0.6393	0.9018
70	80.15	138.00	-85.52	101.41	0.5808	0.8433
100	77.85	138.00	-83.12	95.18	0.5641	0.8732
130	29.22	138.00	-32.13	95.18	0.2118	0.3375
140	27.31	138.00	-30.19	95.18	0.1979	0.3172

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	87.28	234.00		113.80	0.3730	
20	85.04	234.00		119.81	0.3634	
30	86.83	234.00		119.81	0.3711	
60	74.76	234.00		121.45	0.3195	
60	55.24	234.00		121.45	0.2361	
70	66.95	234.00		119.81	0.2861	
100	98.30	234.00		113.80	0.4201	
130	98.45	234.00		113.80	0.4207	
140	98.46	234.00		113.80	0.4208	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	87.28	234.00		113.80	0.3730	
20	85.04	234.00		119.81	0.3634	
30	86.21	234.00		119.81	0.3684	
60	73.24	234.00		121.45	0.3130	



Stress due to Combined Loads: Step: 33 1:48pm Aug 11,2026

60	55.10	234.00	121.45	0.2355
70	65.64	234.00	119.81	0.2805
100	97.60	234.00	113.80	0.4171
130	98.45	234.00	113.80	0.4207
140	98.46	234.00	113.80	0.4208

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	37.84	138.00	-21.75	101.41	0.2742	0.2144
60	32.24	138.00	-18.03	103.50	0.2336	0.1742
60	11.84	138.00	-10.45	103.50	0.0858	0.1009
70	14.36	138.00	-12.09	101.41	0.1040	0.1193
100	16.50	138.00	-9.85	95.18	0.1195	0.1035
130	17.34	138.00	-8.39	95.18	0.1256	0.0881
140	17.35	138.00	-8.37	95.18	0.1257	0.0880

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.16	138.00		95.18	0.0446	
20	3.94	138.00		101.41	0.0285	
30	7.71	138.00		101.41	0.0558	
60	6.84	138.00		103.50	0.0495	
60	0.84	138.00		103.50	0.0061	
70	1.33	138.00		101.41	0.0096	
100	3.65	138.00		95.18	0.0264	
130	4.68	138.00		95.18	0.0339	
140	4.69	138.00		95.18	0.0340	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	37.84	138.00	-21.75	101.41	0.2742	0.2144
60	32.24	138.00	-18.03	103.50	0.2336	0.1742
60	11.80	138.00	-10.49	103.50	0.0855	0.1013
70	14.33	138.00	-12.13	101.41	0.1038	0.1196
100	16.47	138.00	-9.88	95.18	0.1193	0.1038
130	17.34	138.00	-8.39	95.18	0.1256	0.0881
140	17.35	138.00	-8.37	95.18	0.1257	0.0880

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	37.84	138.00	-21.75	101.41	0.2742	0.2144
60	32.24	138.00	-18.03	103.50	0.2336	0.1742
60	11.84	138.00	-10.45	103.50	0.0858	0.1009
70	14.36	138.00	-12.09	101.41	0.1040	0.1193
100	16.50	138.00	-9.85	95.18	0.1195	0.1035
130	17.34	138.00	-8.39	95.18	0.1256	0.0881
140	17.35	138.00	-8.37	95.18	0.1257	0.0880

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.01	138.00	-0.01	95.18	0.0001	0.0001
20	29.84	138.00	-29.74	101.41	0.2162	0.2933



Stress due to Combined Loads: Step: 33 1:48pm Aug 11,2026

30	33.94	138.00	-25.64	101.41	0.2460	0.2528
60	28.95	138.00	-21.31	103.50	0.2098	0.2059
60	8.52	138.00	-13.77	103.50	0.0617	0.1331
70	10.43	138.00	-16.02	101.41	0.0756	0.1580
100	10.31	138.00	-16.04	95.18	0.0747	0.1685
130	11.18	138.00	-14.55	95.18	0.0810	0.1528
140	11.19	138.00	-14.53	95.18	0.0811	0.1527

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	37.84	138.00	-21.75	101.41	0.2742	0.2144
60	31.07	138.00	-19.19	103.50	0.2252	0.1854
60	77.11	138.00	-73.47	103.50	0.5588	0.7099
70	71.36	138.00	-66.41	101.41	0.5171	0.6548
100	75.46	138.00	-64.51	95.18	0.5468	0.6777
130	36.79	138.00	-23.47	95.18	0.2666	0.2466
140	35.18	138.00	-21.83	95.18	0.2549	0.2294

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		138.00	-1.96	95.18		0.0206
20	28.57	138.00	-31.02	101.41	0.2070	0.3058
30	32.67	138.00	-26.91	101.41	0.2368	0.2654
60	26.70	138.00	-23.56	103.50	0.1935	0.2277
60	72.74	138.00	-77.84	103.50	0.5271	0.7522
70	66.20	138.00	-71.57	101.41	0.4797	0.7058
100	67.35	138.00	-72.62	95.18	0.4880	0.7629
130	28.68	138.00	-31.58	95.18	0.2078	0.3318
140	27.07	138.00	-29.95	95.18	0.1962	0.3146

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	6.17	138.00		95.18	0.0447	
20	33.74	138.00	-25.85	101.41	0.2445	0.2549
30	37.84	138.00	-21.75	101.41	0.2742	0.2144
60	32.24	138.00	-18.03	103.50	0.2336	0.1742
60	11.80	138.00	-10.49	103.50	0.0855	0.1013
70	14.33	138.00	-12.13	101.41	0.1038	0.1196
100	16.47	138.00	-9.88	95.18	0.1193	0.1038
130	17.34	138.00	-8.39	95.18	0.1256	0.0881
140	17.35	138.00	-8.37	95.18	0.1257	0.0880

Absolute Maximum of the all of the Stress Ratio's

0.9018

Governing Element: Shell-3

Governing Load Case 10 : EP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Bottom head	-2.0	...	...	...	...	+ Tens
Bottom head	-2.0	...	...	...	-2.0	- Comp
Shell-1	-1.3	...	29.8	...	28.6	+ Tens
Shell-1	-1.3	...	-29.8	...	-31.0	- Comp
Shell-2	-1.3	4.2	29.8	1.3	34.0	+ Tens
Shell-2	-1.3	4.2	-29.8	-1.3	-28.3	- Comp



Shell-3	-1.1	2.7	25.1	3.0	29.7	+ Tens
Shell-3	-1.1	2.7	-25.1	-3.0	-26.5	- Comp
Shell-3	-1.1	-1.5	11.1	79.6	88.2	+ Tens
Shell-3	-1.1	-1.5	-11.1	-79.6	-93.3	- Comp
Shell-4	-1.3	-1.4	13.2	69.6	80.2	+ Tens
Shell-4	-1.3	-1.4	-13.2	-69.6	-85.5	- Comp
Shell-5	-2.0	-0.7	13.2	67.3	77.8	+ Tens
Shell-5	-2.0	-0.7	-13.2	-67.3	-83.1	- Comp
Shell-6	-2.0	0.5	12.9	17.8	29.2	+ Tens
Shell-6	-2.0	0.5	-12.9	-17.8	-32.1	- Comp
Top head	-2.0	0.5	12.9	15.9	27.3	+ Tens
Top head	-2.0	0.5	-12.9	-15.9	-30.2	- Comp



Shop/Field Installation Options :

Packing is installed in the Field after being lifted.

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Packing	9038.784 mm
Center of Gravity of the Lining	6398.006 mm
Center of Gravity of the Nozzles	6886.730 mm
Center of Gravity of the Lugs	6455.175 mm
Center of Gravity of the Added Weights (Operating)	4728.394 mm
Center of Gravity of the Added Weights (Empty)	7909.048 mm
Center of Gravity of Bare Shell New and Cold	6778.857 mm
Center of Gravity of Bare Shell Corroded	6805.607 mm
Vessel CG in the Operating Condition	5375.489 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	7069.213 mm
Vessel CG in the Test Condition	6870.923 mm

Warning: Lifting Lug is Higher than top of vessel!

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	12790.47 mm
Total vessel weight (No Liquid)	Twt 60565.29 N
Impact weight multiplication factor	Imp 1.50
Design lifting weight, DWT = Imp * Twt	90847.92 N
Elevation of the Tailing Lug (bottom)	65.00 mm
Elevation of the Lifting Lug (top)	13420.00 mm
Design Reaction force at the tailing lug	43201.48 N
Design Reaction force at the lifting lug	47646.44 N
CG distance during field lifting operation	7069.21 mm
CG distance from tailing lug	7004.21 mm
CG distance from the nearer lifting lug/trunnion	6350.79 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	62.97	10093.65	115.88 (UG-23)
Shear	3.44	12750.48	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	0.0	0.0	9997.9	0.8
4250.00	109903936.0	44.4	9332.2	0.5
5874.00	127206256.0	43.9	32.2	0.0
10071.25	103389600.0	41.8	23340.3	1.2
12461.27	39564656.0	24.2	40191.4	3.1
12740.48	29334686.0	17.9	43116.4	3.3

Unity Check (Actual Stress / Allowable Stress):



Maximum Unity Check is 0.5434 at elevation 10093.6533 mm - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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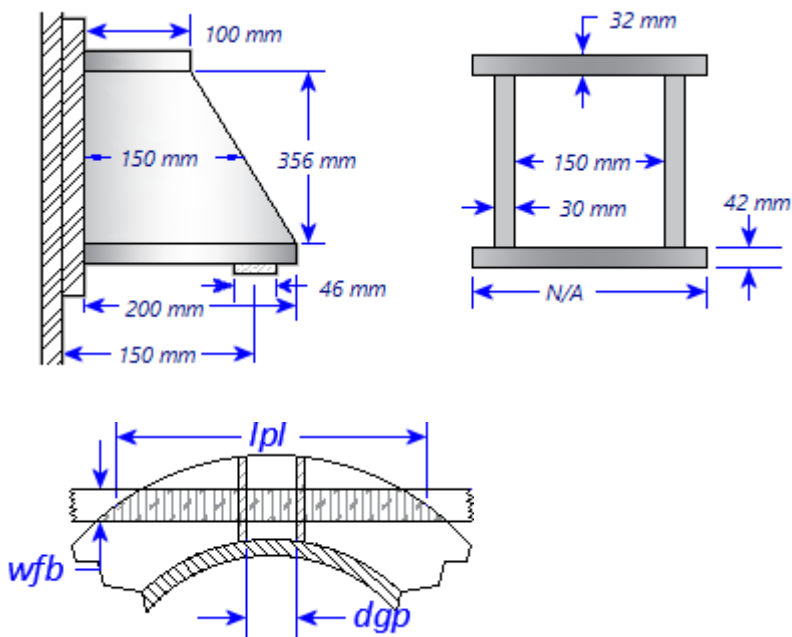


Support Lug Calculations: Operating Case

Input Data: Support Lugs

Type of Geometry :	Gussets with Continuous Top Ring
Number of Support Lugs	Nlug 4
Distance from Vessel OD to Lug Contact Point	Dlug 150.0000 mm
Lug Support Force Bearing Width	Wfb 46.0000 mm
Lug Material	SA-516 70 +IT
Lug Yield Stress	221.00 MPa
Radial Width of bottom Support Lug Plate	Wpl 200.0000 mm
Effective Force Bearing Length	Lpl 204.0000 mm
Thickness of bottom Support Lug Plate	Tpl 42.0000 mm
Distance between Gussets	Dgp 150.0000 mm
Mean Width of Gusset Plate	Wgp 150.0000 mm
Height of Gusset Plate	Hgp 356.0000 mm
Thickness of Gusset Plate	Tgp 30.0000 mm
Radial Width of Reinforcing Ring	Wtp 100.0000 mm
Thickness of Top Plate	Ttp 32.0000 mm

Bolt Material	SA-193 B7
Bolt Allowable Stress at Design Temperature	172.00 MPa
Thread Series	TEMA Metric
Bolt Diameter	56.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	275605632.0 N-mm
Weight Load at the top of one Lug	28810.4 N

Force on one Lug, Operating Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (115242/4 + 0.27561E+09/(417.0 \cdot 4/2))$$

$$= 359138.81 \text{ N}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-31696/4 + 0.19303E+09/(417.0 \cdot 4/2))$$

$$= 223438.03 \text{ N}$$



Ring Girder Calculations Per Bednar pp.157-159:

Force P acting in the plane of the ring [P]:

$$= \text{Flug} \cdot \text{Dlug} / (\text{Hgp} + \text{Tpl})$$

$$= 359139 \cdot 150.0 / (356.0 + 42.0)$$

$$= 135353.83 \text{ N}$$

Angle [α]:

$$= \pi / \text{Nlug}$$

$$= \pi / 4$$

$$= 45.0 \text{ deg}$$

Maximum bending moment (at the load points) [M1]:

$$= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha))$$

$$= 0.5 \cdot 135353.83 \cdot 317.0(1/0.79 - \cos(45.0)/\sin(45.0))$$

$$= 5864342.50 \text{ N-mm}$$

Tangential (axial) thrust induced in the ring [T1]:

$$(\text{Top ring in compression, bottom ring in tension})$$

$$= 0.5 \cdot P \cdot \cos(\alpha) / \sin(\alpha)$$

$$= 0.5 \cdot 135353.83 \cdot \cos(45.0) / \sin(45.0)$$

$$= 67676.91 \text{ N}$$

Resulting combined stress at the load point [S1]:

$$= T1 / (\text{Ttp} \cdot \text{Wtp}) + M1 / (\text{Ttp} \cdot \text{Wtp}^2 / 6)$$

$$= 67677 / (32.0 \cdot 100.0) + 5864343 / (32.0 \cdot 100.0^2 / 6)$$

$$= 131.07 \text{ MPa}$$

Bending moment at the midpoints between the loads [M2]:

$$= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\sin(\alpha) - 1/\alpha)$$

$$= 0.5 \cdot 135353.83 \cdot 317.0(1/\sin(45.0) - 1/0.79)$$

$$= 3025624.50 \text{ N-mm}$$

Axial thrust [T2]:

$$= 0.5 \cdot P / \sin(\alpha)$$

$$= 0.5 \cdot 135353.83 / \sin(45.0)$$

$$= 95709.61 \text{ N}$$

Combined maximum stress [S2]:

$$= T2 / (\text{Wtp} \cdot \text{Ttp}) + M2 / (\text{Ttp} \cdot \text{Wtp}^2 / 6)$$

$$= 95710 / (100.0 \cdot 32.0) + 3025625 / (32.0 \cdot 100.0^2 / 6)$$

$$= 86.62 \text{ MPa}$$

Ring Allowable Stress [Sa]:

$$= 2/3 \cdot \text{Ylug}$$

$$= 2/3 \cdot 221$$

$$= 147.33 \text{ MPa}$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$= (\text{Flugemp} / 2 \cdot \text{Dgp} / 2) / ((\text{Wpl} / 6) \cdot \text{Tpl}^2) \text{ (Uplift Condition)}$$

$$= (223438 / 2 \cdot 150.0 / 2) / ((200.0 / 6) \cdot 42.0^2)$$

$$= 142.51 \text{ MPa}$$

Bottom Plate Required Thickness (Point Load)

41.3069 mm

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUpl]:

$$= 0.72 \cdot \sqrt{ (\text{Flugemp} \cdot \text{Dgp} / ((\text{Wpl} - \text{dt}) \cdot \text{Spa})) }$$

$$= 0.72 \cdot \sqrt{ (223438.0 \cdot 150.0 / ((200.0 - 59.18) \cdot 147.333)) }$$

$$= 28.94 \text{ mm}$$

Bearing Area [Ba]:

$$= \text{Lpl} \cdot \text{Wfb}$$

$$= 204.0 \cdot 46.0$$

$$= 9384.00 \text{ mm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta 1 \cdot \text{Flug} / \text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$



$$= 2.105 \cdot 359138.8/9384.0 \cdot 46.0^2/42.0^2$$

$$= 96.65 \text{ MPa}$$

Bottom Plate Required Thickness (Uniform Load) 34.0165 mm

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(F_{lug} / (L_{pl} \cdot W_{fb}) / S_{pa})}$$

$$= 0.71 \cdot 150.0 \cdot \sqrt{(359139 / (204.0 \cdot 46.0) / 147.333)}$$

$$= 54.282 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 221$$

$$= 147.33 \text{ MPa}$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (F_{lug} / 2) / (W_{gp} \cdot T_{gp})$$

$$= (359139 / 2) / (150.0 \cdot 30.0)$$

$$= 39.91 \text{ MPa}$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 14.4284 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (K_{lr})^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K_{lr}) / (8 \cdot C_c) - (K_{lr})^3 / (8 \cdot C_c^3))$$

$$= (1 - (47.47)^2 / (2 \cdot 130.56^2)) 221 / (5/3 + 3 \cdot (47.47) / (8 \cdot 130.56) - (47.47)^3 / (8 \cdot 130.56^3))$$

$$= 114.85 \text{ MPa}$$

Dimension [Gtp]:

$$= 2 \cdot W_{gp} - W_{pl}$$

$$= 2 \cdot 150.0 - 200.0$$

$$= 100.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((W_{pl} - G_{tp}) / H_{gp})$$

$$= \pi / 2 - \text{atan}((200.0 - 100.0) / 356.0)$$

$$= 74.3 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= F_{lug} (3 \cdot D_{lug} - W_{pl}) / (T_{gp} \cdot W_{pl}^2 \cdot (\sin(\alpha_G))^2)$$

$$= 359139 (3 \cdot 150.0 - 200.0) / (30.0 \cdot 200.0^2 \cdot (\sin(74.31))^2)$$

$$= 80.73 \text{ MPa}$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha_G) / (0.289 \cdot T_{gp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (356.0 / \sin(74.31) / (0.289 \cdot 30.0))^2)$$

$$= 112.72 \text{ MPa}$$

Determine the required bolt area due to Uplift [Abr]:

$$= F_{lug,ten} / S_{abolt}$$

$$= 301518.031 / 171.997$$

$$= 1753.19 \text{ mm}^2$$

The Area available in a single bolt : 1862.725 mm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

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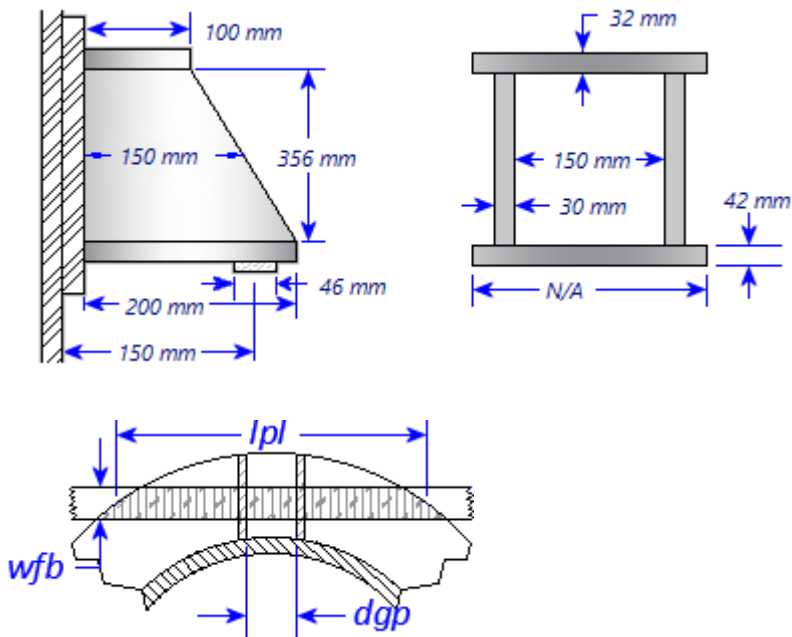


Support Lug Calculations: Test Case

Input Data: Support Lugs

Type of Geometry :	Gussets with Continuous Top Ring
Number of Support Lugs	Nlug 4
Distance from Vessel OD to Lug Contact Point	Dlug 150.0000 mm
Lug Support Force Bearing Width	Wfb 46.0000 mm
Lug Material	SA-516 70 +IT
Lug Yield Stress	260.00 MPa
Radial Width of bottom Support Lug Plate	Wpl 200.0000 mm
Effective Force Bearing Length	Lpl 204.0000 mm
Thickness of bottom Support Lug Plate	Tpl 42.0000 mm
Distance between Gussets	Dgp 150.0000 mm
Mean Width of Gusset Plate	Wgp 150.0000 mm
Height of Gusset Plate	Hgp 356.0000 mm
Thickness of Gusset Plate	Tgp 30.0000 mm
Radial Width of Reinforcing Ring	Wtp 100.0000 mm
Thickness of Top Plate	Ttp 32.0000 mm

Bolt Material	SA-193 B7
Bolt Allowable Stress at Design Temperature	172.00 MPa
Thread Series	TEMA Metric
Bolt Diameter	56.00 mm



Results for Support Lugs:

Description: Support Lugs

Overturning Moment at Support Lug	147259280.0 N-mm
Weight Load at the top of one Lug	14563.0 N

Force on one Lug, Test Condition [Flug]:

$$= (W/Nlug + Mlug/(Rlug \cdot Nlug/2))$$

$$= (58252/4 + 0.14726E+09/(417.0 \cdot 4/2))$$

$$= 191061.38 \text{ N}$$

Force on one lug in the Empty Condition [Flugemp]:

$$= (-Wemp/Nlug + Mlugemp/(Rlug \cdot Nlug/2))$$

$$= (-31696/4 + 0.19303E+09/(417.0 \cdot 4/2))$$

$$= 223438.03 \text{ N}$$



Ring Girder Calculations Per Bednar pp.157-159:

Force P acting in the plane of the ring [P]:

$$= \text{Flug} \cdot \text{Dlug} / (\text{Hgp} + \text{Tpl})$$

$$= 191061 \cdot 150.0 / (356.0 + 42.0)$$

$$= 72008.05 \text{ N}$$

Angle [α]:

$$= \pi / \text{Nlug}$$

$$= \pi / 4$$

$$= 45.0 \text{ deg}$$

Maximum bending moment (at the load points) [M1]:

$$= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\alpha - \cos(\alpha)/\sin(\alpha))$$

$$= 0.5 \cdot 72008.05 \cdot 317.0(1/0.79 - \cos(45.0)/\sin(45.0))$$

$$= 3119822.25 \text{ N-mm}$$

Tangential (axial) thrust induced in the ring [T1]:

$$(\text{Top ring in compression, bottom ring in tension})$$

$$= 0.5 \cdot P \cdot \cos(\alpha) / \sin(\alpha)$$

$$= 0.5 \cdot 72008.05 \cdot \cos(45.0) / \sin(45.0)$$

$$= 36004.03 \text{ N}$$

Resulting combined stress at the load point [S1]:

$$= T1 / (\text{Ttp} \cdot \text{Wtp}) + M1 / (\text{Ttp} \cdot \text{Wtp}^2 / 6)$$

$$= 36004 / (32.0 \cdot 100.0) + 3119822 / (32.0 \cdot 100.0^2 / 6)$$

$$= 69.73 \text{ MPa}$$

Bending moment at the midpoints between the loads [M2]:

$$= 0.5 \cdot P(\text{OD}/2 + \text{Wtp}/2) \cdot (1/\sin(\alpha) - 1/\alpha)$$

$$= 0.5 \cdot 72008.05 \cdot 317.0(1/\sin(45.0) - 1/0.79)$$

$$= 1609628.12 \text{ N-mm}$$

Axial thrust [T2]:

$$= 0.5 \cdot P / \sin(\alpha)$$

$$= 0.5 \cdot 72008.05 / \sin(45.0)$$

$$= 50917.39 \text{ N}$$

Combined maximum stress [S2]:

$$= T2 / (\text{Wtp} \cdot \text{Ttp}) + M2 / (\text{Ttp} \cdot \text{Wtp}^2 / 6)$$

$$= 50917 / (100.0 \cdot 32.0) + 1609628 / (32.0 \cdot 100.0^2 / 6)$$

$$= 46.08 \text{ MPa}$$

Ring Allowable Stress [Sa]:

$$= 2/3 \cdot \text{Ylug}$$

$$= 2/3 \cdot 260$$

$$= 173.33 \text{ MPa}$$

Bottom Support Plate Bending Stress (Point Load) [Spl]:

$$= (\text{Flugemp}/2 \cdot \text{Dgp}/2) / ((\text{Wpl}/6) \cdot \text{Tpl}^2) \text{ (Uplift Condition)}$$

$$= (223438/2 \cdot 150.0/2) / ((200.0/6) \cdot 42.0^2)$$

$$= 142.51 \text{ MPa}$$

Bottom Plate Required Thickness (Point Load)

38.0834 mm

Bottom Plate required thickness for Uplift per ADM S 3/4 [trUpl]:

$$= 0.72 \cdot \sqrt{ (\text{Flugemp} \cdot \text{Dgp} / ((\text{Wpl} - \text{dt}) \cdot \text{Spa})) }$$

$$= 0.72 \cdot \sqrt{ (223438.0 \cdot 150.0 / ((200.0 - 59.18) \cdot 173.33)) }$$

$$= 26.68 \text{ mm}$$

Bearing Area [Ba]:

$$= \text{Lpl} \cdot \text{Wfb}$$

$$= 204.0 \cdot 46.0$$

$$= 9384.00 \text{ mm}^2$$

Bending Stress in bottom Plate (Unif. Load) Per Bednar p.156 [Spl2]:

$$= \beta_1 \cdot \text{Flug}/\text{Ba} \cdot \text{Wfb}^2 / \text{Tpl}^2 \text{ per Roark \& Young 5th Ed.}$$



$$= 2.105 \cdot 191061.4/9384.0 \cdot 46.0^2/42.0^2$$

$$= 51.42 \text{ MPa}$$

Bottom Plate Required Thickness (Uniform Load) 22.8748 mm

Bottom Plate Required Thickness based on ADM S 3/4 [trAD]:

$$= 0.71 \cdot D_{gp} \cdot \sqrt{(Flug / (L_{pl} \cdot W_{fb}) / Spa)}$$

$$= 0.71 \cdot 150.0 \cdot \sqrt{(191061 / (204.0 \cdot 46.0) / 173.33)}$$

$$= 36.503 \text{ mm}$$

If using the AD Code recommendations, the force bearing width (Wfb) must be greater than or equal to 1/3 of the bottom plate radial width (Wpl) plus the pad thickness (Padthk), if there is a pad.

Bottom Support Plate Allowable Stress [Spa]:

$$= 2/3 \cdot Y_{lug}$$

$$= 2/3 \cdot 260$$

$$= 173.33 \text{ MPa}$$

Gusset Plate Axial Stress (Force / Gusset Plate Area) [Sgp]:

$$= (Flug/2) / (W_{gp} \cdot T_{gp})$$

$$= (191061/2) / (150.0 \cdot 30.0)$$

$$= 21.23 \text{ MPa}$$

Check required Gusset Plate Thickness:

Required Thickness of Gussets per AISC: 10.7356 mm

Gusset Plate Allowable Stress [Sga]:

$$= (1 - (K_{lr})^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (K_{lr}) / (8 \cdot C_c) - (K_{lr})^3 / (8 \cdot C_c^3))$$

$$= (1 - (47.47)^2 / (2 \cdot 124.06^2)) 260 /$$

$$(5/3 + 3 \cdot (47.47) / (8 \cdot 124.06) - (47.47)^3 / (8 \cdot 124.06^3))$$

$$= 133.64 \text{ MPa}$$

Dimension [Gtp]:

$$= 2 \cdot W_{gp} - W_{pl}$$

$$= 2 \cdot 150.0 - 200.0$$

$$= 100.0 \text{ mm}$$

Angle [αG]:

$$= \pi / 2 - \text{atan}((W_{pl} - G_{tp}) / H_{gp})$$

$$= \pi / 2 - \text{atan}((200.0 - 100.0) / 356.0)$$

$$= 74.3 \text{ deg}$$

Maximum Compressive Gusset Plate Stress per Bednar [SgpB]:

$$= Flug(3 \cdot D_{lug} - W_{pl}) / (T_{gp} \cdot W_{pl}^2 \cdot (\sin(\alpha G))^2)$$

$$= 191061(3 \cdot 150.0 - 200.0) / (30.0 \cdot 200.0^2 \cdot (\sin(74.31))^2)$$

$$= 42.95 \text{ MPa}$$

Gusset Plate Allowable Compressive Stress [SgaB]:

$$= 18000 / (1 + (1/18000) \cdot (H_{gp} / \sin(\alpha G) / (0.289 \cdot T_{gp}))^2)$$

$$= 18000 / (1 + (1/18000) \cdot (356.0 / \sin(74.31) / (0.289 \cdot 30.0))^2)$$

$$= 112.72 \text{ MPa}$$

Determine the required bolt area due to Uplift [Abr]:

$$= Flug_{emp} / S_{bolt}$$

$$= 223438.031 / 171.997$$

$$= 1299.19 \text{ mm}^2$$

The Area available in a single bolt : 1862.725 mm² [OK]

Review notes about nozzle loadings and supports in the Warnings report.

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**Input, Nozzle Desc: N1 2"****From: 10**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	506.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

Flange Material		SA-105	
Flange Type		Long Weld Neck	

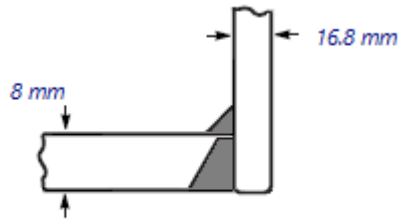
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	196.8500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		650.0	mm
Centerline Distance to Tangent		580.0	mm

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.74 \cdot 508.0 \cdot 0.8966 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.74)$
 $= 6.1911 \text{ mm}$

Reqd thk per UG-37(a) of Elliptical Head, tr [Mapnc]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.08 \cdot 506.0 \cdot 0.9 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.08)$
 $= 6.7490 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.74 / (137.0 \cdot 1.0)] - 1))$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (Sn \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3845 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	D1	101.6000 mm
Parallel to Vessel Wall, opening length	d	50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N1 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.8688 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.8688 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm



Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[5.8, \max(7.8688, 2.5)]$$

$$= 5.8000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(1.7309, 5.8)$$

$$= 5.8000 \text{ mm}$$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	11.2,	Allowable	:	137.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	332.5 MPa	Passed
Occasional	:	2.4,	Allowable	:	182.2 MPa	Passed
Shear	:	9.9,	Allowable	:	95.9 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 8.0, tr = 6.191, c = 1.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.884, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-14 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N1 2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.9289 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 206.7789 mm

Input Echo, WRC107/537 Item 1, Description: N1 2" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.0000 mm
Vessel Diameter	Dv	910.800 mm
Vessel Thickness	Tv	8.000 mm
Design Temperature	T1	225.0 C
Vessel Material		SA-516 70
Vessel UNS Number		K02700
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
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WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	1.0000 mm
Nozzle Diameter	Dn	50.800 mm
Nozzle Thickness	Tn	16.764 mm
Nozzle Material		SA-105
Nozzle UNS Number		K03504
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	137.00 MPa
Design Internal Pressure	Dp	3.741 MPa
Include Pressure Thrust		No

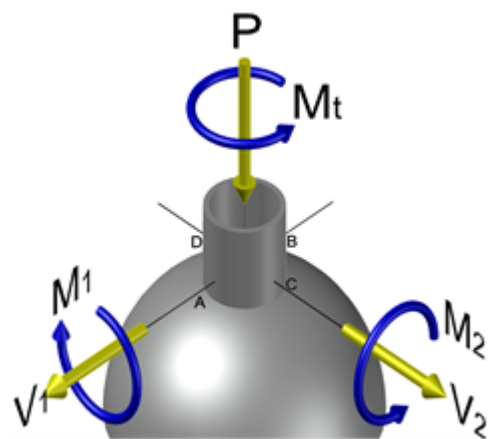
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3750.0	N
Longitudinal Shear (SUS)	(V1) V1	2500.0	N
Circumferential Shear (SUS)	(Vc) V2	2500.0	N
Circumferential Moment (SUS)	(Mc) M1	350000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	350000.0	N-mm
Torsional Moment (SUS)	Mt	500000.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(459.9(8.0 - 1.0))} \\
 &= 271.566 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSained loads:

Radial Load	P	-3750.0	N
Circumferential Shear	(VC) V2	2500.0	N
Longitudinal Shear	(VL) V1	2500.0	N
Circumferential Moment	(MC) M1	350000.0	N-mm
Longitudinal Moment	(ML) M2	350000.0	N-mm
Torsional Moment	MT	500000.0	N-mm



Dimensionless Param: U = 0.74 TAU = 5.00 (2.17) RHO = 0.44

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 1	0.05398	(A,B,C,D)
M(x) / P	SP 1	0.06130	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 1	0.06930	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 1	0.13083	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 1	0.06930	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 1	0.13083	(A,B,C,D)
N(y) * T / P	SP 1	0.08748	(A,B,C,D)
M(y) / P	SP 1	0.01817	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 1	0.08616	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 1	0.04406	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 1	0.08616	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 1	0.04406	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1
Rad. Bend. P		28.2	-28.2	28.2	-28.2	28.2	-28.2	28.2	-28.2
Rad. Memb. MC		0.0	0.0	0.0	0.0	-8.7	-8.7	8.7	8.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-98.8	98.8	98.8	-98.8
Rad. Memb. ML		-8.7	-8.7	8.7	8.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-98.8	98.8	98.8	-98.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-75.2	66.0	139.8	-114.1	-75.2	66.0	139.8	-114.1
Tang. Memb. P		6.7	6.7	6.7	6.7	6.7	6.7	6.7	6.7
Tang. Bend. P		8.3	-8.3	8.3	-8.3	8.3	-8.3	8.3	-8.3
Tang. Memb. MC		0.0	0.0	0.0	0.0	-10.8	-10.8	10.8	10.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-33.3	33.3	33.3	-33.3
Tang. Memb. ML		-10.8	-10.8	10.8	10.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-33.3	33.3	33.3	-33.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-29.1	20.8	59.2	-24.1	-29.1	20.8	59.2	-24.1
Shear VC		2.7	2.7	-2.7	-2.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.7	-2.7	2.7	2.7
Shear MT		6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4
Tot. Shear		9.1	9.1	3.7	3.7	3.7	3.7	9.1	9.1
Str. Int.		77.0	67.8	140.0	114.2	75.5	66.3	140.8	115.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		122.0	122.0	122.0	122.0	122.0	122.0	122.0	122.0
Rad. P1 (SUS)		-4.6	-4.6	12.9	12.9	-4.6	-4.6	12.9	12.9
Rad. Q (SUS)		-70.6	70.6	126.9	-126.9	-70.6	70.6	126.9	-126.9



Long. Pm (SUS)	122.0	122.0	122.0	122.0	122.0	122.0	122.0	122.0
Long. Pl (SUS)	-4.1	-4.1	17.5	17.5	-4.1	-4.1	17.5	17.5
Long. Q (SUS)	-24.9	24.9	41.6	-41.6	-24.9	24.9	41.6	-41.6
Shear Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)	2.7	2.7	-2.7	-2.7	-2.7	-2.7	2.7	2.7
Shear Q (SUS)	6.4	6.4	6.4	6.4	6.4	6.4	6.4	6.4
Pm (SUS)	122.0	122.0	122.0	122.0	122.0	122.0	122.0	122.0
Pm+Pl (SUS)	120.3	120.3	140.8	140.8	120.3	120.3	140.8	140.8
Pm+Pl+Q (Total)	94.6	189.8	261.9	98.1	93.2	188.3	262.8	98.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	121.98	138.00	Passed
Pm+Pl (SUS)	140.75	207.00	Passed
Pm+Pl+Q (TOTAL)	262.79	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N7 2"****Parent: N1 2"**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-105	
Parent Allowable Stress at Temperature	Sv	137.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	50.80	mm
Design Length of Section	L	196.8500	mm
Shell Finished (Minimum) Thickness	t	16.7640	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

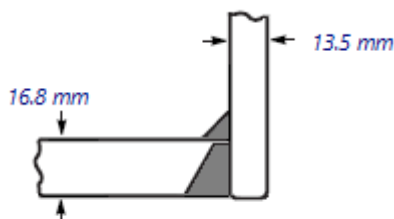
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N7 2"**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per App. 1 of Cylindrical Shell, tr [Int. Press]
 $= R \left(\exp\left(\frac{P}{S_v \cdot E}\right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 26.4 \left(\exp\left(\frac{3.74}{137.0 \cdot 1.0}\right) - 1 \right)$
 $= 0.7309$ mm

Reqd thk per App. 1 of Cylindrical Shell, tr [Mapnc]
 $= R \left(\exp\left(\frac{P}{S_v \cdot E}\right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 25.4 \left(\exp\left(\frac{4.08}{138.0 \cdot 1.0}\right) - 1 \right)$
 $= 0.7618$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 26.4 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 0.7329$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \left(\exp\left(\frac{P}{S_n \cdot E}\right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 25.4 \left(\exp\left(\frac{4.08}{138.0 \cdot 1.0}\right) - 1 \right)$
 $= 0.7618$ mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4056 mm

UG-40, Limits of Reinforcement: [External Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	109.2520	mm
Parallel to Vessel Wall	Rn+tn+t	54.6260	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	31.1550	mm

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7329	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 1.7309	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 2.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(2.5, 2.5)]$
 $= 2.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7329, 2.5)$
 $= 2.5000$ mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A**

Govrn. thk, tg = 13.462, tr = 0.728, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.058$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	1	°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104	°C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Warning: $Rn/R(1.00) > 0.7$ - Performing analysis per Appendix 46-3(a)(1).



Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 12.462 / \ln[1 + 12.462/26.4]$$

$$= 32.230 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 15.764 / \ln[1 + 15.764/26.4]$$

$$= 33.669 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$

$$= 1.5 \cdot 137.0$$

$$= 205.5 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 0.103 \cdot 32.23 \cdot 33.902$$

$$= 113.0 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 0.103 \cdot 33.669 (20.4 + 12.462)$$

$$= 114.4 \text{ N}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 0.103 \cdot 33.669 \cdot 26.4$$

$$= 91.9 \text{ N}$$

Area Resisting External Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (114.4 + 91.9 + 113.0) / 0.103$$

$$= 3088.0 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 205.5 / (2 \cdot 3087.996 / 794.079 - 33.669 / 15.764)$$

$$= 36.4 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / [t / R_{xs}]$$

$$= 137.0 [15.764 / 33.669]$$

$$= 64.1 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(36.425, 64.144)$$

$$= 36.425 \text{ MPa}$$

Average Primary Membrane Stress [σAvg]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (112.997 + 114.42 + 91.92) / 794.079$$

$$= 0.402 \text{ MPa}$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 0.103 \cdot 33.669 / 15.764$$

$$= 0.2 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 0.402 - 0.221, 0.221)$$



= 0.6 MPa

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	205.50	MPa
Local Primary Membrane Stress	PL	0.58	MPa
Maximum Allowable External Pressure	Pmax	36.43	MPa

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[137.0/137.0, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 13.462 - 1.0$$

$$= 12.462 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 52.8/15.764$$

$$= 3.349 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 50.8/2 + 1.0$$

$$= 26.400 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{R_{eff} \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{26.4 \cdot 15.764}, 2 \cdot 26.4)$$

$$= 20.400 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 15.764, 0.0] + \sqrt{26.4 \cdot 12.462}$$

$$= 18.138 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(15.764 + 0.0)$$

$$= 126.112 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[18.138, 250.0, 126.112] + 15.764$$

$$= 33.902 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot frp) / LR)$$

$$= 15.764 + ((0.0 \cdot 1.0) / 20.4)$$

$$= 15.764 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))$$

$$= \min(12, (2 \cdot 26.4 + 12.462) / (\sqrt{(52.8 + 15.764) \cdot 15.764}))$$

$$= 1.985$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :



Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max\left(\left(\frac{\lambda}{5}\right)^{0.85}, 1\right)$$

$$= 15.764 \cdot 20.4 \cdot \max\left(\left(\frac{1.985}{5}\right)^{0.85}, 1\right)$$

$$= 321.589 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 12.462 \cdot 33.902$$

$$= 422.490 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 321.589 + 1.0(422.49 + 0.0) + 50.0 + 0.0 + 0.0$$

$$= 794.079 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$

$$= 12.462 / \ln[1 + 12.462/26.4]$$

$$= 32.230 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{\text{eff}} / \ln[1 + t_{\text{eff}}/R_{\text{eff}}]$$

$$= 15.764 / \ln[1 + 15.764/26.4]$$

$$= 33.669 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 137.0 \cdot 1.0$$

$$= 205.5 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{\text{xn}} \cdot LH$$

$$= 3.741 \cdot 32.23 \cdot 33.902$$

$$= 4087.6 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{\text{xs}}(LR + t_n)$$

$$= 3.741 \cdot 33.669(20.4 + 12.462)$$

$$= 4139.1 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{\text{xs}} \cdot R_{\text{nc}}$$

$$= 3.741 \cdot 33.669 \cdot 26.4$$

$$= 3325.1 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$

$$= (4139.1 + 3325.1 + 4087.6)/3.741$$

$$= 3088.0 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{\text{allow}} / (2 \cdot A_p / AT - R_{\text{xs}} / t_{\text{eff}})$$

$$= 205.5 / (2 \cdot 3087.996 / 794.079 - 33.669 / 15.764)$$

$$= 36.4 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{\text{xs}}]$$

$$= 137.0[15.764/33.669]$$

$$= 64.1 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{\text{max1}}, P_{\text{max2}})$$



$$= \min(36.425, 64.144)$$

$$= 36.425 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (4087.612 + 4139.082 + 3325.147) / 794.079$$

$$= 14.549 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 3.741 \cdot 33.669 / 15.764$$

$$= 8.0 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 14.549 - 7.991, 7.991)$$

$$= 21.1 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	205.50	MPa
Local Primary Membrane Stress	PL	21.11	MPa
Maximum Allowable Working Pressure	Pmax	36.43	MPa

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [k_y]:

$$= (R_{nc} + t_n) / R_{nc}$$

$$= (26.4 + 12.462) / 26.4$$

$$= 1.472 \text{ For set-in Nozzles}$$

Weld Length of Nozzle to Shell Weld [L_T]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (26.4 + 12.462)$$

$$= 61.044 \text{ mm}$$

Weld Throat Dimensions, $(0.7071 \cdot \text{Leg Dimensions})$ [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 12.462, 6.000) \text{ Passed}$$

Weld Load Value [f_{welds}]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(3325 \cdot 1.47, 1.5 \cdot 137.0(422.49 + 0.0), \pi/4 \cdot 3.7 \cdot 26.4^2 \cdot 1.47^2)$$

$$= 4437.257 \text{ N}$$

Weld Stress Value [τ]:

$$= f_{welds} / (L_T(0.49 \cdot L41T + 0.6 \cdot t_{w1} + 0.49 \cdot L43T))$$

$$= 4437.257 / (61.044(0.49 \cdot 7.071 + 0.6 \cdot 6.0 + 0.49 \cdot 0.0))$$

$$= 10.290 < \text{or } = \text{to } 137.000 \text{ Weld Stress Passed}$$

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: M.A.P. Case:

Nozzle Material Factor [f_{rn}]:

$$= \min[S_n/S, 1]$$

$$= \min[138.0/138.0, 1]$$

$$= 1.000$$

Thickness of Nozzle [t_n]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 13.462 - 0.0$$

$$= 13.462 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$



$$= 50.8/16.764$$

$$= 3.030 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= Di/2 + \text{corrosion allowance}$$

$$= 50.8/2 + 0.0$$

$$= 25.400 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot Rn)$$

$$= \min(\sqrt{25.4 \cdot 16.764}, 2 \cdot 25.4)$$

$$= 20.635 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, te] + \sqrt{Rn \cdot tn}$$

$$= \min[1.5 \cdot 16.764, 0.0] + \sqrt{25.4 \cdot 13.462}$$

$$= 18.491 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= Lpr1$$

$$= 250.0$$

$$= 250.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + te)$$

$$= 8(16.764 + 0.0)$$

$$= 134.112 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \text{ for set-in nozzles}$$

$$= \min[18.491, 250.0, 134.112] + 16.764$$

$$= 35.255 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR)$$

$$= 16.764 + ((0.0 \cdot 1.0) / 20.635)$$

$$= 16.764 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff}))$$

$$= \min(12, (2 \cdot 25.4 + 13.462) / (\sqrt{(50.8 + 16.764) \cdot 16.764}))$$

$$= 1.909$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 16.764 \cdot 20.635 \cdot \max((1.909/5)^{0.85}, 1)$$

$$= 345.926 \text{ mm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= tn \cdot LH$$

$$= 13.462 \cdot 35.255$$

$$= 474.609 \text{ mm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot Leg41^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 50.000 \text{ mm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43$$

$$= 345.926 + 1.0(474.609 + 0.0) + 50.0 + 0.0 + 0.0$$

$$= 870.535 \text{ mm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$

$$= 13.462 / \ln[1 + 13.462/25.4]$$



$$= 31.655 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$

$$= 16.764 / \ln[1 + 16.764/25.4]$$

$$= 33.077 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$

$$= 1.5 \cdot 138.0 \cdot 1.0$$

$$= 207.0 \text{ MPa}$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H$$

$$= 4.078 \cdot 31.655 \cdot 35.255$$

$$= 4550.9 \text{ N}$$

Force acting on the Shell [fS]:

$$= P \cdot R_{xs} (L_R + t_n)$$

$$= 4.078 \cdot 33.077 (20.635 + 13.462)$$

$$= 4599.0 \text{ N}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 4.078 \cdot 33.077 \cdot 25.4$$

$$= 3426.0 \text{ N}$$

Area Resisting Internal Pressure [Ap]:

$$= (f_S + f_Y + f_N) / P$$

$$= (4599.0 + 3426.0 + 4550.9) / 4.078$$

$$= 3084.0 \text{ mm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff})$$

$$= 207.0 / (2 \cdot 3084.008 / 870.535 - 33.077 / 16.764)$$

$$= 40.5 \text{ MPa}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / t / R_{xs}$$

$$= 138.0 [16.764 / 33.077]$$

$$= 69.9 \text{ MPa}$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(40.491, 69.941)$$

$$= 40.491 \text{ MPa}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / A_T$$

$$= (4550.917 + 4599.048 + 3425.98) / 870.535$$

$$= 14.447 \text{ MPa}$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 4.078 \cdot 33.077 / 16.764$$

$$= 8.0 \text{ MPa}$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 14.447 - 8.047, 8.047)$$

$$= 20.8 \text{ MPa}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	207.00	MPa
Local Primary Membrane Stress	PL	20.85	MPa
Maximum Allowable Working Pressure	Pmax	40.49	MPa

Weld Size Calculations, Description: N7 2"



Intermediate Calc. for nozzle/shell Welds Tmin 12.4620 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 36 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

Approximate Maximum Allowable Pressure New & Cold: 40 MPa

The Drop for this Nozzle is : 25.4000 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 292.1640 mm

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**Input, Nozzle Desc: N104"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 1550.00 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm

Flange Material SA-105

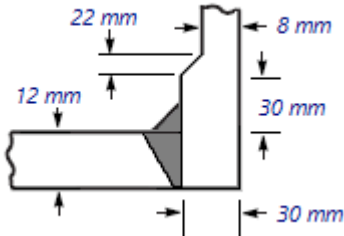
Hub Height of Integral Nozzle	h	30.0000	mm
Height of Beveled Transition	L'	22.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N104"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	4.000 in.
Actual Thickness Used in Calculation	0.315 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 51.8 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 1.4381 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 50.8 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 1.5283 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5573 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22.0 / 0.5773$
 $= 38.1051 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	207.2000 mm
Parallel to Vessel Wall, opening length	d	103.6000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 22.0 / 0.5773$
 $= 38.1051 \text{ mm}$



Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit) D1 203.2000 mm

Parallel to Vessel Wall, opening length d 101.6000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 30.0000 mm

Weld Strength Reduction Factor [fr1]:

= min(1, Sn/Sv)

= min(1, 137.0/138.0)

= 0.993

Weld Strength Reduction Factor [fr2]:

= min(1, Sn/Sv)

= min(1, 137.0/138.0)

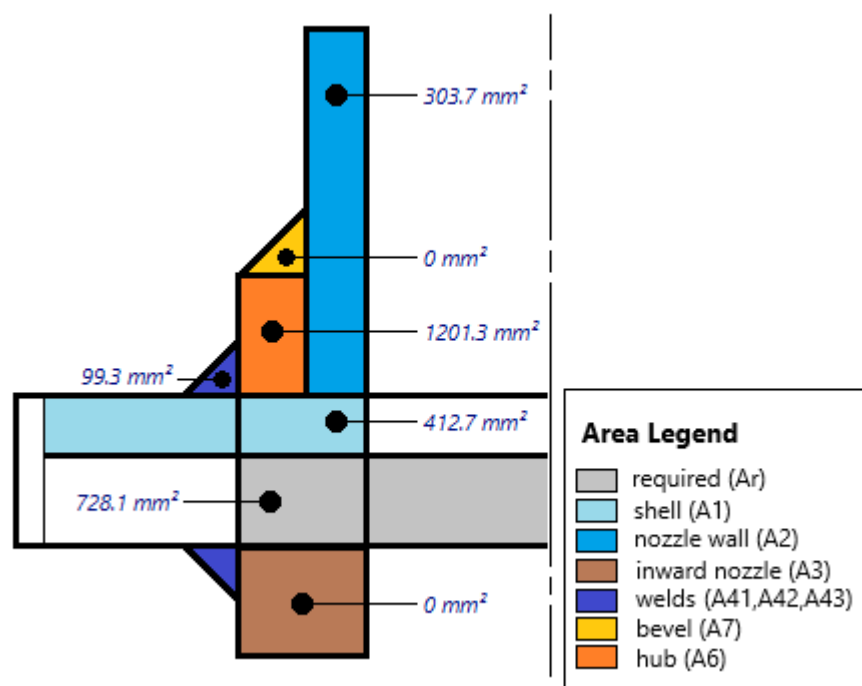
= 0.993

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.993, 1.0)

= 0.993



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	728.133	229.477	773.351
Area in Shell	A1	412.724	679.735	445.849
Area in Nozzle Wall	A2	303.693	351.786	388.300
Area in Inward Nozzle	A3	0.000	0.000	0.000
Area in Welds	A41+A42+A43	99.278	99.278	100.000
Area in Element	A5	0.000	0.000	0.000
Area in Hub	A6	1201.258	1201.258	1320.000
TOTAL AREA AVAILABLE	Atot	2016.953	2332.057	2254.150

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.



Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (103.6 \cdot 7.0 \cdot 1.0 + 2 \cdot 29.0 \cdot 7.0 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 728.133 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 103.6 (1.0 \cdot 11.0 - 1.0 \cdot 7.0) - 2 \cdot 29.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.0) \cdot (1 - 0.993)$$

$$= 412.724 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$

$$= (2 \cdot 27.5) (7.0 - 1.44) \cdot 0.9928$$

$$= 303.693 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9928 + (0.0)^2 \cdot 0.9928$$

$$= 99.278 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(27.5, 300.0, 30.0)) \cdot (30.0 - 8.0) \cdot 0.993$$

$$= 1201.258 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 2.4381 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 6.7300 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[6.73, \max(8.0, 2.5)]$$

$$= 6.7300 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(2.4381, 6.73)$$

$$= 6.7300 \text{ mm}$$

Available Nozzle Neck Thickness = 8.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 41.1,	Allowable	: 137.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 302.7 MPa	Passed
Occasional	: 13.0,	Allowable	: 182.2 MPa	Passed
Shear	: 23.1,	Allowable	: 95.9 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: A**

Govrn. thk, tg = 8.0, tr = 1.428, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.204, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, $t_g = 12.0$, $t_r = 7.0$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.74/10.20 = 0.367$

Weld Size Calculations, Description: N104"

Intermediate Calc. for nozzle/shell Welds $T_{min} \quad 11.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (728.1332 - 412.7238 + 2 \cdot 29.0 \cdot 0.9928 \cdot (1.0 \cdot 11.0 - 7.0)) \cdot 138)$
 $= 75303.15 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1504.9513 + 99.2775 - 0.0 \cdot 0.99) \cdot 138$
 $= 221359.95 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1504.9513 + 0.0 + 99.2775 + (633.3907)) \cdot 138$
 $= 308758.53 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1504.9513 + 0.0 + 99.2775 + 0.0 + (633.3907)) \cdot 138$
 $= 308758.53 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 161.6 \cdot 10.0 \cdot 0.49 \cdot 137$
 $= 170389. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 66.3) \cdot (30.0 - 1.0) \cdot 0.7 \cdot 137$
 $= 579219. \text{ N}$



Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 161.6 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 155517. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (170389 + 579219) = 749608 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (170389 + 0 + 155517 + 0) = 325905 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (170389 + 155517 + 0) = 325905 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 749608 N , must exceed W = 75303 N or W1 = 221359 N

Path 2-2 = 325905 N , must exceed W = 75303 N or W2 = 308758 N

Path 3-3 = 325905 N , must exceed W = 75303 N or W3 = 308758 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Approximate Maximum Allowable Pressure New & Cold: 6 MPa

Note: The M.A.P.(NC) of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.2494 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 325.2494 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 103.60 mm

Input Echo, WRC297 Item 1, Description: N104" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	506.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	101.600	mm
Nozzle Thickness	30.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	137.00 MPa



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	3.74	MPa
Radial Load	P	-7500.00	N
Circumferential Shear	Vc	5000.00	N
Longitudinal Shear	Vl	5000.00	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.387) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 161.6 + 2 \cdot 1.65 \cdot \sqrt{(259.5(12.0 - 1.0))}$$

$$= 337.911 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	519.000 mm
Nozzle Outside Diameter	(d) =	161.600 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	29.000 mm
T / t	=	0.379
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.139$$

Nr/P	=	0.067
Mr/P	=	0.136
M0/P	=	0.040
N0/P	=	0.087
MrD/Mc	=	0.283
NrDL/Mc	=	0.096
M0d/Mc	=	0.087
N0DL/Mc	=	0.075
MrD/Ml	=	0.107
NrDL/Ml	=	0.071
M0D/Ml	=	0.033
N0DL/Ml	=	0.111

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-14	14	14	-14
Normal Pressure Stress	84	88	84	88
Outplane Stress Summary	82	85	127	72

**Vessel Stresses (MPa):**

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	50	-50	50	-50
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	42	42	42	42
Inplane Stress Summary	45	36	148	-45

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	6	6	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	83	85	148	117

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-37	37	37	-37
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42
Outplane Stress Summary	19	64	105	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	50	-50	50	-50
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-121	121	121	-121
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	88	84	88
Inplane Stress Summary	11	156	267	-72

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4

Shear Stress Summary	2	2	6	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	20	156	267	74
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	10	6	10

Hoop Stress Summary	3	8	19	24
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-4	4	4	-4
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	0	-1	17	-2
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	5	10	20	26
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0



Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	10	6	10

Hoop Stress Summary	6	10	17	21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	5	-5	5	-5
Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-14	14	14	-14
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-9	8	27	-12
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	0	0	2	2
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	15	11	28	34
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	88	84	88	84	88	84	88
Circ. Pl (SUS)	-2	-2	12	12	-2	-2	10	10
Circ. Q (SUS)	0	0	28	-28	-71	71	171	-171
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-1	-1	9	9	0	0	10	10
Long. Q (SUS)	4	-4	96	-96	-23	23	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	84.0	88.0	84.0	88.0	84.0	88.0	84.0	88.0
Pm+Pl (SUS)	82.0	86.0	96.0	100.0	82.0	86.0	94.0	98.0
Pm+Pl+Q (Total)	82.7	86.5	147.4	117.2	20.0	157.1	265.2	74.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	100.02	207.00	Passed
Pm+Pl+Q (TOTAL)	265.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	10	6	10	6	10	6	10
Circ. Pl (SUS)	-2	-2	12	12	0	0	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-4	-4	4	4	-4	-4	4	4
Long. Q (SUS)	1	-1	9	-9	-9	9	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	6.0	10.0	6.0	10.0	6.0	10.0	6.0	10.0
Pm+Pl (SUS)	5.0	9.0	18.0	22.0	7.0	11.0	16.0	20.0
Pm+Pl+Q (Total)	4.5	10.2	18.4	24.1	16.1	10.4	26.1	32.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	137.00	Passed
Pm+Pl (SUS)	22.00	205.50	Passed
Pm+Pl+Q (TOTAL)	32.06	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N2 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

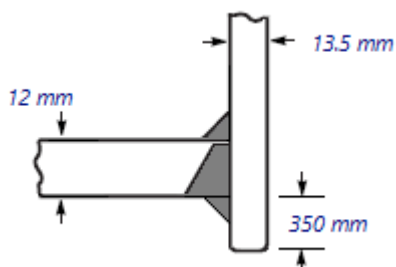
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	350.0000	mm
Weld leg size, Inside Element to Shell	Wi	10.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N2 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 26.4 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 0.7329 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1) - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1) - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000	mm
Parallel to Vessel Wall, opening length	d	52.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm
Normal to Vessel Wall, Inward		27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.7240	mm
Parallel to Vessel Wall	Rn+tn+t	50.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm
Normal to Vessel Wall, Inward		30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N2 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7329	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7329, 5.8)$
 $= 5.8000 \text{ mm}$



Available Nozzle Neck Thickness = 13.4620 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	15.4,	Allowable	:	137.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	328.4 MPa	Passed
Occasional	:	3.2,	Allowable	:	182.2 MPa	Passed
Shear	:	12.6,	Allowable	:	95.9 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N2 2"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000 mm
Intermediate Calc. for Inward Weld	TminIns	11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	6.0700 = 0.7 * Wi-Can mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.0025 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5500 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.80 mm

Input Echo, WRC297 Item 1, Description: N2 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000 mm	
Shell External Corrosion Allowance	0.0000 mm	
Shell Diameter	506.000 mm	
Shell Thickness	12.0000 mm	
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000 mm	
Nozzle Diameter	50.800 mm	
Nozzle Thickness	13.4620 mm	
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa



Nozzle Hot S.I. Allowable SNmh 137.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.74	MPa
Radial Load	P	-3750.00	N
Circumferential Shear	Vc	2500.00	N
Longitudinal Shear	Vl	2500.00	N
Circumferential Moment	Mc	350000.00	N-mm
Longitudinal Moment	Ml	350000.00	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.774) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(259.5(12.0 - 1.0))}$$

$$= 254.035 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	519.000 mm
Nozzle Outside Diameter	(d) =	77.724 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	12.462 mm
T / t	=	0.883
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.029$$

Nr/P	=	0.120
Mr/P	=	0.194
M0/P	=	0.064
N0/P	=	0.122
MrD/Mc	=	0.285
NrDL/Mc	=	0.133
M0d/Mc	=	0.105
N0DL/Mc	=	0.038
MrD/Ml	=	0.195
NrDL/Ml	=	0.106
M0D/Ml	=	0.071
N0DL/Ml	=	0.118

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-4	-4	4	4
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	84	88	84	88



Outplane Stress Summary 80 91 120 68

Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	42	42	42	42

Inplane Stress Summary 34 49 129 -29

Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary 6 6 2 2

Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 81 92 130 98

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary 33 56 82 12

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-63	63	63	-63
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	88	84	88

Inplane Stress Summary 55 114 192 -2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress normal to

	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	55	114	193	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12
Hoop Stress Summary	7	11	15	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-24	24	24	-24
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-9	0	58	-29

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	2	2

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	20	15	59	49

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12

Hoop Stress Summary	10	14	12	17
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-38	38	38	-38
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-23	14	72	-42
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4

Shear Stress Summary	2	2	5	5
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	33	17	72	60

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	88	84	88	84	88	84	88
Circ. Pl (SUS)	-1	-1	7	7	-1	-1	7	7
Circ. Q (SUS)	-4	4	26	-26	-28	28	98	-98
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	0	0	6	6	2	2	4	4
Long. Q (SUS)	-8	8	78	-78	-12	12	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	84.0	88.0	84.0	88.0	84.0	88.0	84.0	88.0
Pm+Pl (SUS)	83.0	87.0	91.0	95.0	83.0	87.0	91.0	95.0
Pm+Pl+Q (Total)	79.5	91.6	126.9	99.2	55.4	115.2	189.2	18.0

Vessel Stress Summation Comparison (MPa):

Type of	Max. S.I.	S.I. Allowable	Result
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Stress Int.			
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	95.02	207.00	Passed
Pm+Pl+Q (TOTAL)	189.23	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	12	7	12	7	12	7	12
Circ. Pl (SUS)	-1	-1	7	7	2	2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-5	5	43	-43	-19	19	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	7.0	12.0	7.0	12.0	7.0	12.0	7.0	12.0
Pm+Pl (SUS)	11.2	16.1	14.6	19.2	14.1	19.1	13.4	16.3
Pm+Pl+Q (Total)	18.9	14.9	56.2	49.4	33.5	17.0	70.4	60.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	137.00	Passed
Pm+Pl (SUS)	19.16	205.50	Passed
Pm+Pl+Q (TOTAL)	70.42	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: N8 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		150.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

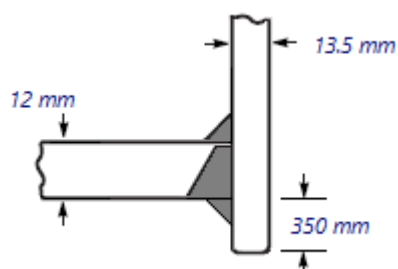
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	350.0000	mm
Weld leg size, Inside Element to Shell	Wi	12.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N8 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 26.9 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 0.7468 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm
Normal to Vessel Wall, Inward		26.1550	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.7240	mm
Parallel to Vessel Wall	Rn+tn+t	50.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm
Normal to Vessel Wall, Inward		30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N8 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2468	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(8.0, 3.0)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(2.2468, 6.3)$
 $= 6.3000 \text{ mm}$



Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N8 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Intermediate Calc. for Inward Weld TminIns 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Inward Weld	6.0000 = Min per Code	6.9840 = 0.7 * Wi-Can mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.0025 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 571.5500 mm

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**Input, Nozzle Desc: N3 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

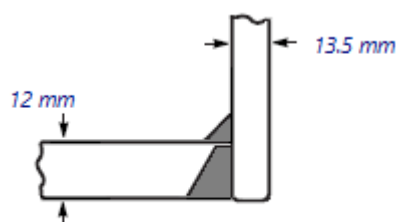
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.11	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N3 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 26.4 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 0.7329 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.7240 mm
Parallel to Vessel Wall	Rn+tn+t	50.8620 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N3 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7329 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7329, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK



Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	15.4,	Allowable	:	137.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	328.4 MPa	Passed
Occasional	:	3.2,	Allowable	:	182.2 MPa	Passed
Shear	:	12.6,	Allowable	:	95.9 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: A

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.636, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-23 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N3 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.0025 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.5525 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.80 mm

Input Echo, WRC297 Item 1, Description: N3 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	506.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	13.4620	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	137.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	3.74	MPa
Radial Load	P	-3750.00	N
Circumferential Shear	Vc	2500.00	N
Longitudinal Shear	Vl	2500.00	N
Circumferential Moment	Mc	350000.00	N-mm
Longitudinal Moment	Ml	350000.00	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.774) is < 10
check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(259.5(12.0 - 1.0))}$$

$$= 254.035 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	519.000 mm
Nozzle Outside Diameter	(d) =	77.724 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	12.462 mm
T / t	=	0.883
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.029$$

Nr/P	=	0.120
Mr/P	=	0.194
M0/P	=	0.064
N0/P	=	0.122
MrD/Mc	=	0.285
NrDL/Mc	=	0.133
M0d/Mc	=	0.105
N0DL/Mc	=	0.038
MrD/Ml	=	0.195
NrDL/Ml	=	0.106
M0D/Ml	=	0.071
N0DL/Ml	=	0.118

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-4	-4	4	4
Outplane Bending (Ml)	-15	15	15	-15
Normal Pressure Stress	84	88	84	88
Outplane Stress Summary	80	91	120	68

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses parallel to longitudinal plane)

	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	42	42	42	42

Inplane Stress Summary	34	49	129	-29
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Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE
(Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	81	92	130	98
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-23	23	23	-23
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary	33	56	82	12
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-63	63	63	-63
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	88	84	88

Inplane Stress Summary	55	114	192	-2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
(Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1



Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	55	114	193	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12
Hoop Stress Summary	7	11	15	20

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-24	24	24	-24
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-9	0	58	-29

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	5	5	2	2

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	20	15	59	49

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-1	-1	1	1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	7	12	7	12
Hoop Stress Summary	10	14	12	17

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-38	38	38	-38
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-23	14	72	-42
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	2	2	5	5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	33	17	72	60

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	88	84	88	84	88	84	88
Circ. Pl (SUS)	-1	-1	7	7	-1	-1	7	7
Circ. Q (SUS)	-4	4	26	-26	-28	28	98	-98
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	0	0	6	6	2	2	4	4
Long. Q (SUS)	-8	8	78	-78	-12	12	34	-34
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	84.0	88.0	84.0	88.0	84.0	88.0	84.0	88.0
Pm+Pl (SUS)	83.0	87.0	91.0	95.0	83.0	87.0	91.0	95.0
Pm+Pl+Q (Total)	79.5	91.6	126.9	99.2	55.4	115.2	189.2	18.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	95.02	207.00	Passed



Pm+Pl+Q (TOTAL)	189.23	413.99	Passed
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Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	12	7	12	7	12	7	12
Circ. Pl (SUS)	-1	-1	7	7	2	2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-5	5	43	-43	-19	19	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	7.0	12.0	7.0	12.0	7.0	12.0	7.0	12.0
Pm+Pl (SUS)	11.2	16.1	14.6	19.2	14.1	19.1	13.4	16.3
Pm+Pl+Q (Total)	18.9	14.9	56.2	49.4	33.5	17.0	70.4	60.8

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	137.00	Passed
Pm+Pl (SUS)	19.16	205.50	Passed
Pm+Pl+Q (TOTAL)	70.42	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: K1A 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

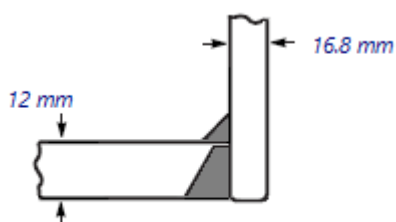
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K1A 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	106.3280	mm
Parallel to Vessel Wall	Rn+tn+t	53.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1A 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K1A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K2A 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

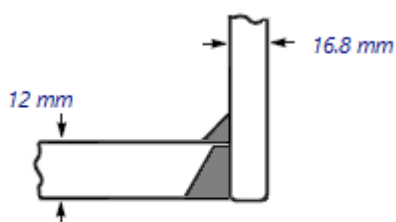
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K2A 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	106.3280	mm
Parallel to Vessel Wall	Rn+tn+t	53.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2A 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K3A 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 1050.00 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

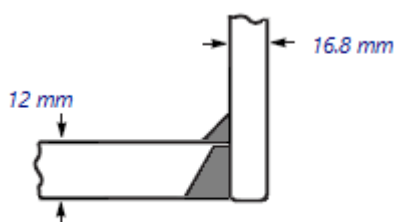
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K3A 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	106.3280	mm
Parallel to Vessel Wall	Rn+tn+t	53.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K3A 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K4A 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

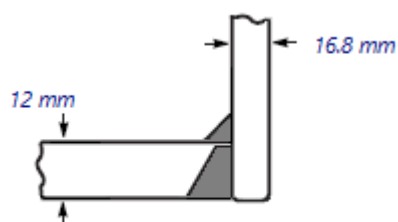
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K4A 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7447 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K4A 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2447	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(8.0, 3.0)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(2.2447, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$ **Weld Size Calculations, Description: K4A 2"**

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K1B 2****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

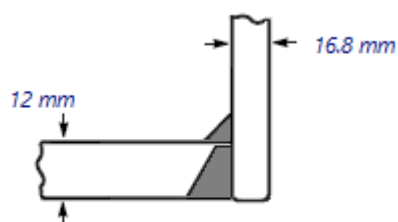
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K1B 2**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7447 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	107.6000 mm
Parallel to Vessel Wall, opening length	d	53.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280 mm
Parallel to Vessel Wall	Rn+tn+t	54.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K1B 2.

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2447 mm
Wall Thickness per UG16(b),	tr16b	= 3.0000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000 mm
Wall Thickness per table UG-45	tb3	= 6.3000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(8.0, 3.0)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(2.2447, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K1B 2

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K2B 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

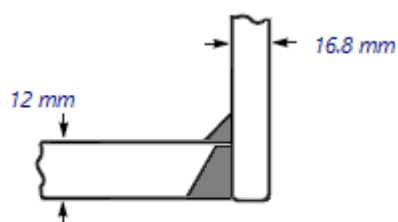
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K2B 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	106.3280 mm
Parallel to Vessel Wall	Rn+tn+t	53.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280 mm
Parallel to Vessel Wall	Rn+tn+t	54.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K2B 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K2B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K3B2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		240.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

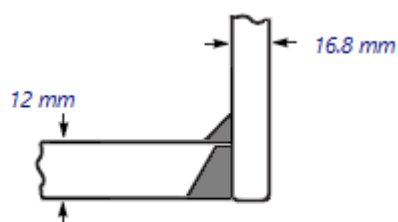
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K3B2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.9 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7447 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	107.6000	mm
Parallel to Vessel Wall, opening length	d	53.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K3B2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 2.2447	mm
Wall Thickness per UG16(b),	tr16b	= 3.0000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.0000	mm
Wall Thickness per table UG-45	tb3	= 6.3000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.3, \max(8.0, 3.0)]$
 $= 6.3000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(2.2447, 6.3)$
 $= 6.3000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K3B2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: K4B 2"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

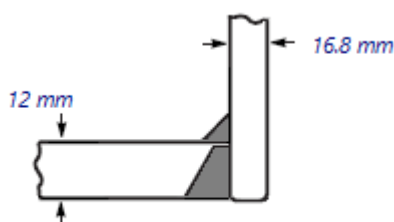
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	6.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K4B 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	106.3280	mm
Parallel to Vessel Wall	Rn+tn+t	53.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K4B 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: A**

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C

Min Metal Temp. at Required thickness (UCS 66.1) -23 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: K4B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

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**Input, Nozzle Desc: N118"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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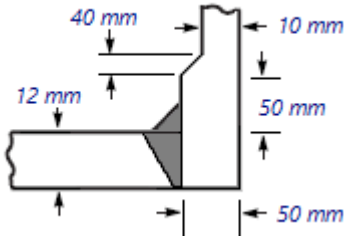
Hub Height of Integral Nozzle	h	50.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N118"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 102.6 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 2.8485 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 101.6 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 3.0567 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6919 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	410.4000 mm
Parallel to Vessel Wall, opening length	d	205.2000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$



Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit) D1 406.4000 mm

Parallel to Vessel Wall, opening length d 203.2000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 30.0000 mm

Weld Strength Reduction Factor [fr1]:

= min(1, Sn/Sv)

= min(1, 137.0/138.0)

= 0.993

Weld Strength Reduction Factor [fr2]:

= min(1, Sn/Sv)

= min(1, 137.0/138.0)

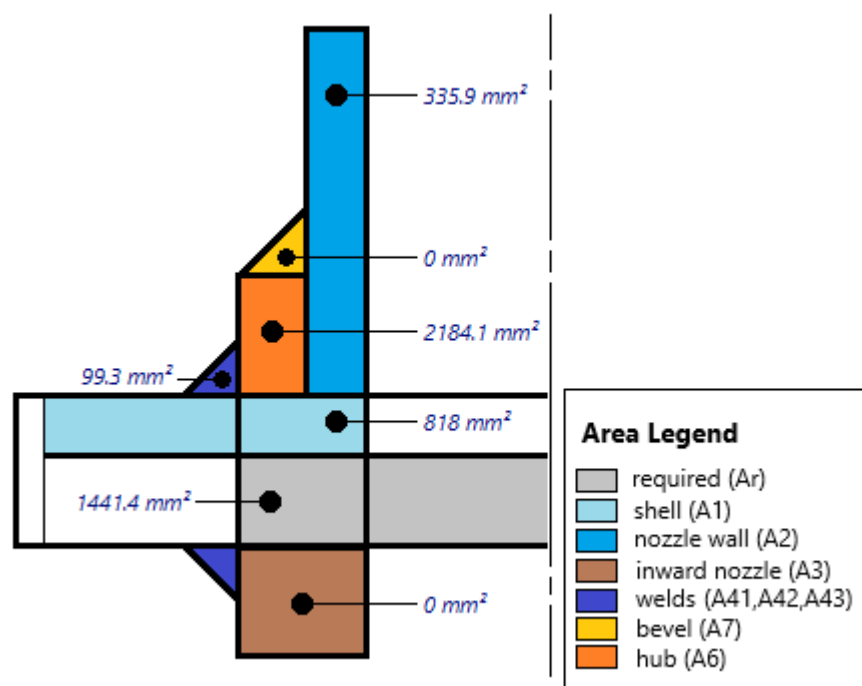
= 0.993

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.993, 1.0)

= 0.993



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1441.356	454.255	1546.701
Area in Shell	A1	817.968	1347.151	891.699
Area in Nozzle Wall	A2	335.889	453.646	416.601
Area in Inward Nozzle	A3	0.000	0.000	0.000
Area in Welds	A41+A42+A43	99.278	99.278	100.000
Area in Element	A5	0.000	0.000	0.000
Area in Hub	A6	2184.105	2184.105	2400.000
TOTAL AREA AVAILABLE	Atot	3437.240	4084.180	3808.299

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.



Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (205.2 \cdot 7.0 \cdot 1.0 + 2 \cdot 49.0 \cdot 7.0 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 1441.356 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 205.2 (1.0 \cdot 11.0 - 1.0 \cdot 7.0) - 2 \cdot 49.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.0) \cdot (1 - 0.993)$$

$$= 817.968 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$

$$= (2 \cdot 27.5) (9.0 - 2.85) \cdot 0.9928$$

$$= 335.889 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9928 + (0.0)^2 \cdot 0.9928$$

$$= 99.278 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(27.5, 206.4, 50.0)) \cdot (50.0 - 10.0) \cdot 0.993$$

$$= 2184.105 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.8485 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.1026 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.103, \max(8.0, 2.5)]$$

$$= 8.0000 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.8485, 8.0)$$

$$= 8.0000 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	43.3,	Allowable	:	137.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	300.4 MPa	Passed
Occasional	:	20.4,	Allowable	:	182.2 MPa	Passed
Shear	:	18.3,	Allowable	:	95.9 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: A**

Govern. thk, tg = 10.0, tr = 2.827, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.314, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C



Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: A

Govrn. thk, tg = 12.0, tr = 7.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.636$, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Weld Size Calculations, Description: N118"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (1441.3563 - 817.9678 + 2 \cdot 49.0 \cdot 0.9928 \cdot (1.0 \cdot 11.0 - 7.0)) \cdot 138)$
 $= 139717.86 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2519.9946 + 99.2775 - 0.0 \cdot 0.99) \cdot 138$
 $= 361420.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (2519.9946 + 0.0 + 99.2775 + (1070.2118)) \cdot 138$
 $= 509094.41 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (2519.9946 + 0.0 + 99.2775 + 0.0 + (1070.2118)) \cdot 138$
 $= 509094.41 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 303.2 \cdot 10.0 \cdot 0.49 \cdot 137$
 $= 319690. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 127.1) \cdot (50.0 - 1.0) \cdot 0.7 \cdot 137$
 $= 1876174. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 303.2 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 486310. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

**Strength of Failure Paths:**

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (319690 + 1876174) = 2195864 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (319690 + 0 + 486310 + 0) = 806000 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (319690 + 486310 + 0) = 806000 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2195864 N , must exceed W = 139717 N or W1 = 361420 N

Path 2-2 = 806000 N , must exceed W = 139717 N or W2 = 509094 N

Path 3-3 = 806000 N , must exceed W = 139717 N or W3 = 509094 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Approximate Maximum Allowable Pressure New & Cold: 6 MPa

Note: The M.A.P.(NC) of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 50.4502 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 268.8252 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 205.20 mm

Input Echo, WRC297 Item 1, Description: N118" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	506.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	203.200	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-105	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	137.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:**External Forces and Moments in WRC 107/537 Convention:**

These loads are assumed to be SUstained loads.

Design Internal Pressure Dp 3.74 MPa



Radial Load	P	-14999.99	N
Circumferential Shear	Vc	9999.99	N
Longitudinal Shear	Vl	9999.99	N
Circumferential Moment	Mc	0.560E+07	N-mm
Longitudinal Moment	Ml	0.560E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Dv (.572) is > 0.5
check the limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (6.064) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 303.2 + 2 \cdot 1.65 \cdot \sqrt{259.5(12.0 - 1.0)}$$

$$= 479.511 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	519.000 mm
Nozzle Outside Diameter	(d) =	303.200 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	49.000 mm
T / t	=	0.224
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 4.013$$

Nr/P	=	0.049
Mr/P	=	0.081
M0/P	=	0.026
N0/P	=	0.055
MrD/Mc	=	0.255
NrDL/Mc	=	0.090
M0d/Mc	=	0.076
N0DL/Mc	=	0.085
MrD/Ml	=	0.068
NrDL/Ml	=	0.047
M0D/Ml	=	0.021
N0DL/Ml	=	0.058

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-8	-8	8	8
Outplane Bending (Ml)	-18	18	18	-18
Normal Pressure Stress	84	88	84	88
Outplane Stress Summary	82	85	138	65

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6



Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	42	42	42	42

Inplane Stress Summary	39	43	178	-67
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary	6	6	3	3
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	83	87	178	133
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-69	69	69	-69
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary	-14	86	151	-27
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-13	-13	13	13
Inplane Bending (Mc)	-233	233	233	-233
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	88	84	88

Inplane Stress Summary	-95	253	398	-185
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary	3	3	6	6
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**Vessel Stresses (MPa):**CIRCUMFERENTIAL PLANE
(Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	95	253	398	186

Nozzle Stresses (MPa):LONGITUDINAL PLANE
(Stresses in the
hoop direction)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12

Hoop Stress Summary 5 10 23 27

Nozzle Stresses (MPa):LONGITUDINAL PLANE
(Stresses Normal to
pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-2	2	2	-2
Axial Pressure Stress	3	3	3	3

Axial Stress Summary 1 1 10 2

Nozzle Stresses (MPa):LONGITUDINAL PLANE
(Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary 1 1 0 0

Nozzle Stresses (MPa):LONGITUDINAL PLANE
(Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	6	10	23	27

Nozzle Stresses (MPa):CIRCUMFERENTIAL PLANE
(Stresses in the
hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12

Hoop Stress Summary 1 6 27 31

**Nozzle Stresses (MPa):**

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	2	-2	2	-2
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-5	9	18	-5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	0	0	1	1

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	7	9	27	37

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	88	84	88	84	88	84	88
Circ. Pl (SUS)	-2	-2	14	14	-7	-7	19	19
Circ. Q (SUS)	1	-1	37	-37	-173	173	293	-293
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-1	-1	13	13	-6	-6	18	18
Long. Q (SUS)	-2	2	122	-122	-50	50	88	-88
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	84.0	88.0	84.0	88.0	84.0	88.0	84.0	88.0
Pm+Pl (SUS)	82.0	86.0	98.0	102.0	77.0	81.0	103.0	107.0
Pm+Pl+Q (Total)	83.8	85.8	177.4	132.2	96.2	254.1	396.1	186.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	107.02	207.00	Passed
Pm+Pl+Q (TOTAL)	396.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	12	7	12	7	12	7	12
Circ. Pl (SUS)	-2	-2	14	14	-6	-6	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	4	-4	-7	7	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	7.0	12.0	7.0	12.0	7.0	12.0	7.0	12.0
Pm+Pl (SUS)	5.0	10.0	21.0	26.0	1.0	6.0	25.0	30.0
Pm+Pl+Q (Total)	5.2	10.1	21.1	26.0	7.3	8.4	25.1	36.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	137.00	Passed
Pm+Pl (SUS)	30.00	205.50	Passed
Pm+Pl+Q (TOTAL)	36.06	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: A3 8"****From: 20**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	5092.1665	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 2550.00 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material SA-105

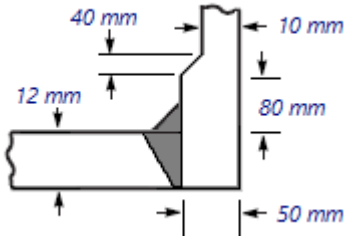
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: A3 8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 102.6 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 2.8485 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 101.6 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 3.0567 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7525 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	410.4000 mm
Parallel to Vessel Wall, opening length	d	205.2000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$



Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit) D1 406.4000 mm

Parallel to Vessel Wall, opening length d 203.2000 mm

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 30.0000 mm

Weld Strength Reduction Factor [fr1]:

= min(1, Sn/Sv)

= min(1, 137.0/138.0)

= 0.993

Weld Strength Reduction Factor [fr2]:

= min(1, Sn/Sv)

= min(1, 137.0/138.0)

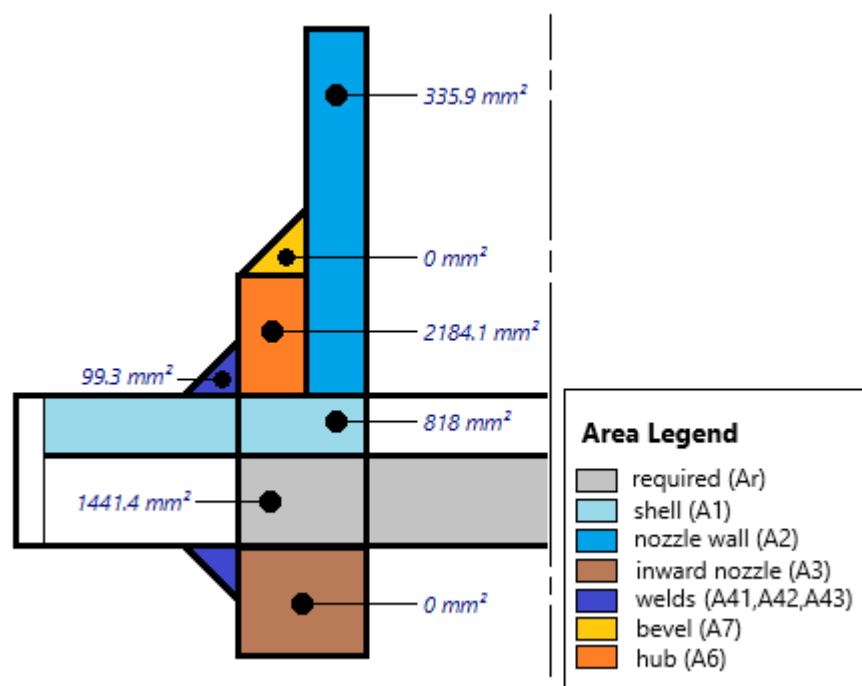
= 0.993

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(0.993, 1.0)

= 0.993



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1441.356	454.255	1546.701
Area in Shell	A1	817.968	1347.151	891.699
Area in Nozzle Wall	A2	335.889	450.335	416.601
Area in Inward Nozzle	A3	0.000	0.000	0.000
Area in Welds	A41+A42+A43	99.278	99.278	100.000
Area in Element	A5	0.000	0.000	0.000
Area in Hub	A6	2184.105	2184.105	2400.000
TOTAL AREA AVAILABLE	Atot	3437.240	4080.869	3808.299

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.



Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (205.2 \cdot 7.0 \cdot 1.0 + 2 \cdot 49.0 \cdot 7.0 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 1441.356 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 205.2(1.0 \cdot 11.0 - 1.0 \cdot 7.0) - 2 \cdot 49.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 7.0) \cdot (1 - 0.993)$$

$$= 817.968 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.5)(9.0 - 2.85) \cdot 0.9928$$

$$= 335.889 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9928 + (0.0)^2 \cdot 0.9928$$

$$= 99.278 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(27.5, 250.0, 80.0)) \cdot (50.0 - 10.0) \cdot 0.993$$

$$= 2184.105 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

$$\text{Wall Thickness for Internal/External pressures} \quad ta = 3.8485 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 3.8485 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: A**

Govrn. thk, tg = 10.0, tr = 2.827, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.314$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Weld Size Calculations, Description: A3 8"

Intermediate Calc. for nozzle/shell welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr))Sv)$$



$$= \max(0, (1441.3563 - 817.9678 + 2 \cdot 49.0 \cdot 0.9928 \cdot (1.0 \cdot 11.0 - 7.0)) \cdot 138)$$

$$= 139717.86 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot \text{fr2}) \cdot S_v$$

$$= (2519.9946 + 99.2775 - 0.0 \cdot 0.99) \cdot 138$$

$$= 361420.94 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S_v$$

$$= (2519.9946 + 0.0 + 99.2775 + (1070.2118)) \cdot 138$$

$$= 509094.41 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot \text{fr1})) \cdot S$$

$$= (2519.9946 + 0.0 + 99.2775 + 0.0 + (1070.2118)) \cdot 138$$

$$= 509094.41 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 303.2 \cdot 10.0 \cdot 0.49 \cdot 137$$

$$= 319690. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 127.1) \cdot (50.0 - 1.0) \cdot 0.7 \cdot 137$$

$$= 1876174. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 303.2 \cdot (8.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 389048. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (319690 + 1876174) = 2195864 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (319690 + 0 + 389048 + 0) = 708738 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (319690 + 389048 + 0) = 708738 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2195864 N	, must exceed W = 139717 N	or W1 = 361420 N
Path 2-2 = 708738 N	, must exceed W = 139717 N	or W2 = 509094 N
Path 3-3 = 708738 N	, must exceed W = 139717 N	or W3 = 509094 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Approximate Maximum Allowable Pressure New & Cold: 6 MPa



Note: The M.A.P.(NC) of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 50.4502 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.4502 mm

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 73.05 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 205.20 mm

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**Input, Nozzle Desc: K6 2"****From: 60**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	4707.6504	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 6777.18 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

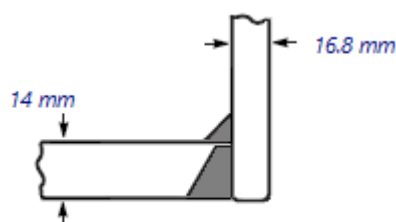
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K6 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9018 and the shell thk. is 14.0000 mm

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.74 / (137.0 \cdot 1.0)] - 1))$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.3280	mm
Parallel to Vessel Wall	Rn+tn+t	55.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	112.3280	mm
Parallel to Vessel Wall	Rn+tn+t	56.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	35.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K6 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:



$= \max(t_a, t_b)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: K6 2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.9132 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 55.16 mm

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**Input, Nozzle Desc: P1A 2"****From: 60**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	4707.6504	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 6377.18 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

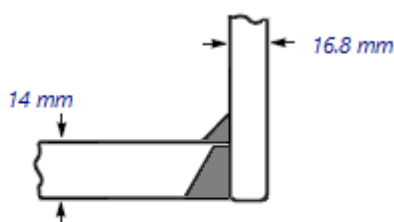
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: P1A 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	110.3280 mm
Parallel to Vessel Wall	Rn+tn+t	55.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	32.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	112.3280 mm
Parallel to Vessel Wall	Rn+tn+t	56.1640 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	35.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P1A 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: P1A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 223.9132 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 55.16 mm

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**Input, Nozzle Desc: P1B 2"****From: 70**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	4707.6504	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9377.16	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

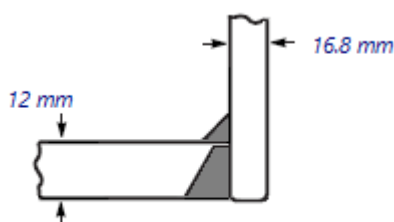
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: P1B 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9018 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot (\exp([3.74 / (137.0 \cdot 1.0)] - 1))$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	106.3280	mm
Parallel to Vessel Wall	Rn+tn+t	53.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	108.3280	mm
Parallel to Vessel Wall	Rn+tn+t	54.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P1B 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:



$= \max(t_a, t_b)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: P1B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.9132 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 53.16 mm

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**Input, Nozzle Desc: N4 6"****From: 70**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	4707.6504	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		9677.17	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
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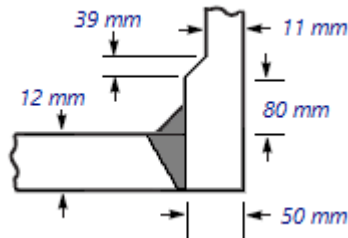
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	39.0272	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	350.0000	mm
Weld leg size, Inside Element to Shell	Wi	10.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N4 6"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9018 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 74.16 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 2.0590 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 73.16 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 2.2012 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.6364 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 39.027 / 0.5773$
 $= 67.5971 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	296.6588	mm
Parallel to Vessel Wall, opening length	d	148.3294	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm
Normal to Vessel Wall, Inward		22.4320	mm

Intermediate Hub Nozzle Calculations:



Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 39.027 / 0.5773$
 $= 67.5971 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	D1	292.6588	mm
Parallel to Vessel Wall, opening length	d	146.3294	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm
Normal to Vessel Wall, Inward		27.4320	mm

Weld Strength Reduction Factor [fr1]:

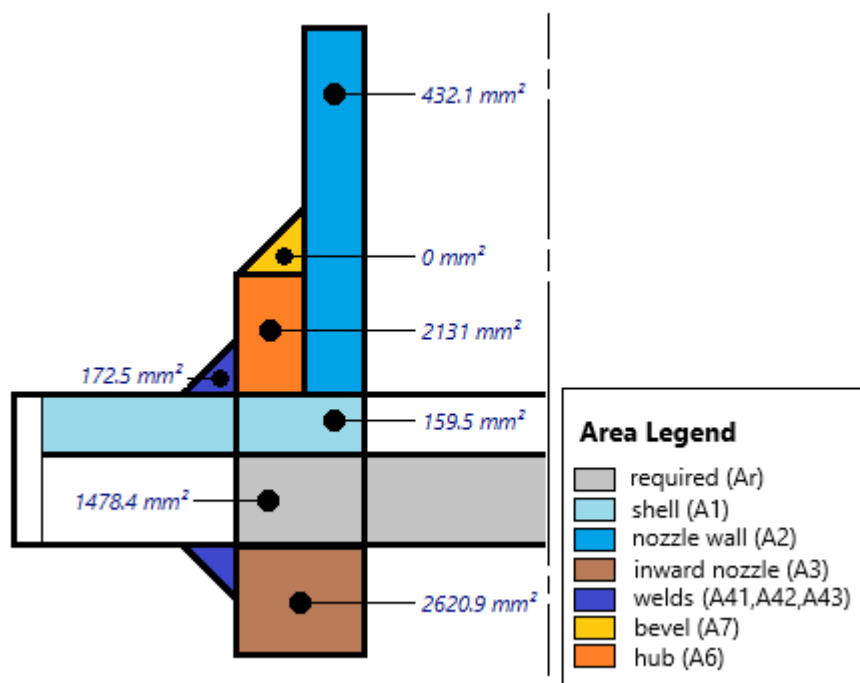
$= \min(1, S_n/S_v)$
 $= \min(1, 137.0/138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n/S_v)$
 $= \min(1, 137.0/138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.993, 1.0)$
 $= 0.993$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1478.359	322.104	1113.818
Area in Shell	A1	159.523	985.748	642.135
Area in Nozzle Wall	A2	432.113	509.794	526.298
Area in Inward Nozzle	A3	2620.927	2620.927	3000.000
Area in Welds	A41+A42+A43	172.460	172.460	200.000
Area in Element	A5	0.000	0.000	0.000
Area in Hub	A6	2130.988	2130.988	2341.632



TOTAL AREA AVAILABLE	Atot	5516.011	6419.917	6710.064
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The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (148.3294 \cdot 9.9194 \cdot 1.0 + 2 \cdot 49.0 \cdot 9.9194 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 1478.359 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 148.329(1.0 \cdot 11.0 - 1.0 \cdot 9.919) - 2 \cdot 49.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 9.919) \cdot (1 - 0.993)$$

$$= 159.523 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.5)(9.97 - 2.06) \cdot 0.9928$$

$$= 432.113 \text{ mm}^2$$

Area Available in Inward Nozzle [A3]:

$$= 2 \cdot ti \cdot \min(h, TI, 2.5 \cdot ti) \cdot fr2$$

$$= 2 \cdot 48.0 \cdot (27.5) \cdot 0.9928$$

$$= 2620.927 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9928 + (8.5858)^2 \cdot 0.9928$$

$$= 172.460 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(27.5, 250.0, 80.0)) \cdot (50.0 - 11.0) \cdot 0.993$$

$$= 2130.988 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.0590 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 7.2200 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[7.22, \max(8.0, 2.5)]$$

$$= 7.2200 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.059, 7.22)$$

$$= 7.2200 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	34.8,	Allowable :	137.0 MPa	Passed
Expansion :	0.0,	Allowable :	309.0 MPa	Passed
Occasional :	13.0,	Allowable :	182.2 MPa	Passed



Shear : 17.6, Allowable : 95.9 MPa Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.601, tr = 2.044, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.238$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $3.74/10.20 = 0.367$

Weld Size Calculations, Description: N4 6"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (1478.3586 - 159.5227 + 2 \cdot 49.0 \cdot 0.9928 \cdot (1.0 \cdot 11.0 - 9.9194)) \cdot 138)$
 $= 196487.09 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (2563.1006 + 172.4604 - 73.7155 \cdot 0.99) \cdot 138$
 $= 367368.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (2563.1006 + 2620.9268 + 172.4604 + (1070.2118)) \cdot 138$
 $= 886789.81 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (2563.1006 + 2620.9268 + 172.4604 + 0.0 + (1070.2118)) \cdot 138$
 $= 886789.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis



Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 246.3294 \cdot 10.0 \cdot 0.49 \cdot 137$$

$$= 259726. \text{ N}$$

Shear, Inward Nozzle Weld [Sinw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 246.3294 \cdot 8.5858 \cdot 0.49 \cdot 137$$

$$= 222995. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 98.6647) \cdot (50.0 - 1.0) \cdot 0.7 \cdot 137$$

$$= 1456430. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 246.3294 \cdot (12.0 - 1.0) \cdot 0.74 \cdot 138$$

$$= 434604. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (259726 + 1456430) = 1716156 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (259726 + 0 + 434604 + 222995) = 917325 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (259726 + 434604 + 222995) = 917325 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1716155 N , must exceed W = 196487 N or W1 = 367368 N
 Path 2-2 = 917324 N , must exceed W = 196487 N or W2 = 886789 N
 Path 3-3 = 917324 N , must exceed W = 196487 N or W3 = 886789 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.0035 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 612.0000 mm

Input Echo, WRC297 Item 1, Description: N4 6" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	506.000	mm
Shell Thickness	12.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	50.0000	mm
Nozzle Stress Concentration Factor	1.000	



Nozzle Material	SA-105
Nozzle Cold S.I. Allowable	SNmc 138.00 MPa
Nozzle Hot S.I. Allowable	SNmh 137.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	3.74	MPa
Radial Load	P	-11250.01	N
Circumferential Shear	Vc	7500.00	N
Longitudinal Shear	Vl	7500.00	N
Circumferential Moment	Mc	0.315E+07	N-mm
Longitudinal Moment	Ml	0.315E+07	N-mm
Torsional Moment	Mt	0.450E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (4.927) is < 10
check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 246.329 + 2 \cdot 1.65 \cdot \sqrt{(259.5(12.0 - 1.0))}$$

$$= 422.640 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	519.000 mm
Nozzle Outside Diameter	(d) =	246.329 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	49.000 mm
T / t	=	0.224
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.260$$

Nr/P	=	0.056
Mr/P	=	0.099
M0/P	=	0.030
N0/P	=	0.062
MrD/Mc	=	0.263
NrDL/Mc	=	0.091
M0d/Mc	=	0.079
N0DL/Mc	=	0.086
MrD/Ml	=	0.079
NrDL/Ml	=	0.058
M0D/Ml	=	0.023
N0DL/Ml	=	0.072

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-7	-7	7	7
Outplane Bending (Ml)	-14	14	14	-14



Normal Pressure Stress	84	88	84	88
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Outplane Stress Summary	84	84	129	70
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-50	50	50	-50
Inplane Pressure Stress	42	42	42	42

Inplane Stress Summary	46	36	158	-51
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary	6	6	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	85	84	159	121
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-49	49	49	-49
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary	5	72	123	-9
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	55	-55	55	-55
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-166	166	166	-166
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	88	84	88

Inplane Stress Summary	-31	195	320	-118
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Vessel Stresses (MPa):



CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	2	2	6	6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	37	195	321	118

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	9	5	9
Hoop Stress Summary	3	7	18	22

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-2	-2	2	2
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	0	0	8	1

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	0	0

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	3	8	18	22

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside



Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	9	5	9

Hoop Stress Summary	1	6	19	24
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-2	-2	2	2
Axial Bending (Mc)	-6	6	6	-6
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-4	5	14	-3
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	1	1
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	6	6	20	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	88	84	88	84	88	84	88
Circ. Pl (SUS)	-2	-2	12	12	-4	-4	14	14
Circ. Q (SUS)	2	-2	30	-30	-111	111	221	-221
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	-1	-1	11	11	-4	-4	14	14
Long. Q (SUS)	5	-5	105	-105	-33	33	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	84.0	88.0	84.0	88.0	84.0	88.0	84.0	88.0
Pm+Pl (SUS)	82.0	86.0	96.0	100.0	80.0	84.0	98.0	102.0
Pm+Pl+Q (Total)	84.6	84.5	158.3	122.1	36.5	195.1	319.1	119.2

Vessel Stress Summation Comparison (MPa):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	88.00	138.00	Passed
Pm+Pl (SUS)	102.02	207.00	Passed
Pm+Pl+Q (TOTAL)	319.13	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	9	5	9	5	9	5	9
Circ. Pl (SUS)	-2	-2	12	12	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-2	-2	2	2	-2	-2	2	2
Long. Q (SUS)	0	0	2	-2	-5	5	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	5.0	9.0	5.0	9.0	5.0	9.0	5.0	9.0
Pm+Pl (SUS)	3.0	7.0	17.0	21.0	1.0	5.0	19.0	23.0
Pm+Pl+Q (Total)	3.0	7.0	17.0	21.0	6.0	5.0	19.0	26.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	137.00	Passed
Pm+Pl (SUS)	23.00	205.50	Passed
Pm+Pl+Q (TOTAL)	26.00	411.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: A2 8"****From: 70**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	4707.6504	mm
Shell Finished (Minimum) Thickness	t	12.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 9077.17 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material SA-105

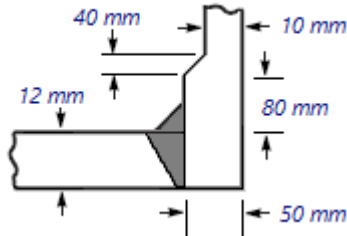
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: A2 8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 8.000 in.
 Actual Thickness Used in Calculation 0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

The Longitudinal Stress Governs over the Hoop Stress on the shell course where this nozzle is located. The Maximum stress ratio times the Shell thickness will be used in the calculation of the Area required.

The Stress Ratio is 0.9018 and the shell thk. is 12.0000 mm

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 102.6 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 2.8485 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 101.6 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 3.0567 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7525 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	410.4000	mm
Parallel to Vessel Wall, opening length	d	205.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):



Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0/0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	D1	406.4000	mm
Parallel to Vessel Wall, opening length	d	203.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	30.0000	mm

Weld Strength Reduction Factor [fr1]:

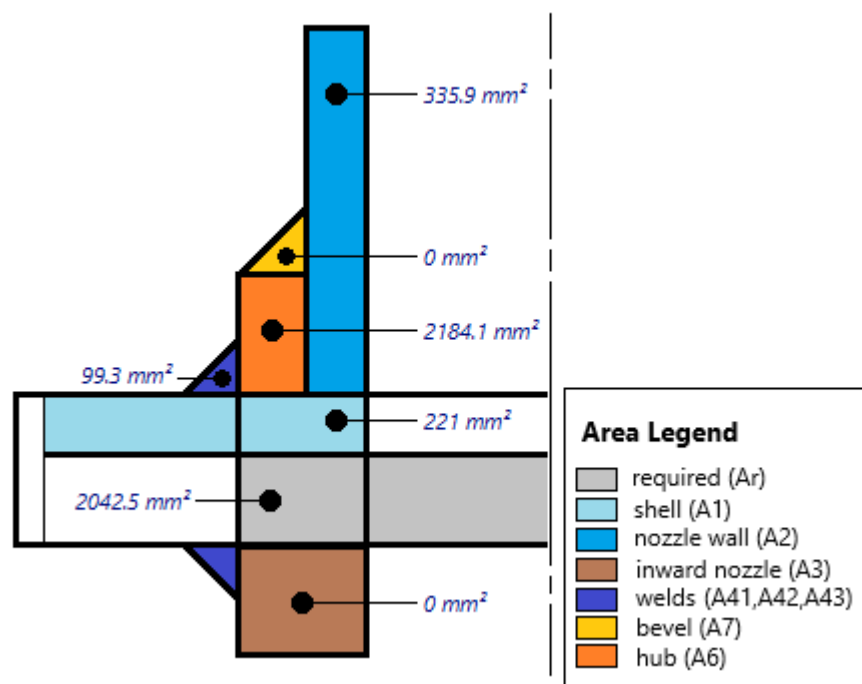
$= \min(1, S_n/S_v)$
 $= \min(1, 137.0/138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n/S_v)$
 $= \min(1, 137.0/138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.993, 1.0)$
 $= 0.993$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2042.479	445.014	1546.701
Area in Shell	A1	220.978	1365.504	891.699
Area in Nozzle Wall	A2	335.889	450.335	416.601
Area in Inward Nozzle	A3	0.000	0.000	0.000
Area in Welds	A41+A42+A43	99.278	99.278	100.000
Area in Element	A5	0.000	0.000	0.000
Area in Hub	A6	2184.105	2184.105	2400.000
TOTAL AREA AVAILABLE	Atot	2840.250	4099.222	3808.299



The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (205.2 \cdot 9.9194 \cdot 1.0 + 2 \cdot 49.0 \cdot 9.9194 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 2042.479 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 205.2(1.0 \cdot 11.0 - 1.0 \cdot 9.919) - 2 \cdot 49.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 9.919) \cdot (1 - 0.993)$$

$$= 220.978 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.5)(9.0 - 2.85) \cdot 0.9928$$

$$= 335.889 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9928 + (0.0)^2 \cdot 0.9928$$

$$= 99.278 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tlnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(27.5, 250.0, 80.0)) \cdot (50.0 - 10.0) \cdot 0.993$$

$$= 2184.105 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 3.8485 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 3.8485 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 10.0$, $t_r = 2.827$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.314$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -21 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = 3.74/5.10 = 0.734

Weld Size Calculations, Description: A2 8"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (2042.4795 - 220.9783 + 2 \cdot 49.0 \cdot 0.9928 \cdot (1.0 \cdot 11.0 - 9.9194)) \cdot 138)$
 $= 265847.50 \text{ N}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2519.9946 + 99.2775 - 0.0 \cdot 0.99) \cdot 138$
 $= 361420.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (2519.9946 + 0.0 + 99.2775 + (1070.2118)) \cdot 138$
 $= 509094.41 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (2519.9946 + 0.0 + 99.2775 + 0.0 + (1070.2118)) \cdot 138$
 $= 509094.41 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 303.2 \cdot 10.0 \cdot 0.49 \cdot 137$
 $= 319690. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 127.1) \cdot (50.0 - 1.0) \cdot 0.7 \cdot 137$
 $= 1876174. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 303.2 \cdot (12.0 - 1.0) \cdot 0.74 \cdot 138$
 $= 534941. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (319690 + 1876174) = 2195864 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (319690 + 0 + 534941 + 0) = 854631 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (319690 + 534941 + 0) = 854631 \text{ N}$



Summary of Failure Path Calculations:

Path 1-1 = 2195864 N , must exceed W = 265847 N or W1 = 361420 N

Path 2-2 = 854631 N , must exceed W = 265847 N or W2 = 509094 N

Path 3-3 = 854631 N , must exceed W = 265847 N or W3 = 509094 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 50.4502 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 312.4502 mm

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**Input, Nozzle Desc: N5 2"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		12012.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

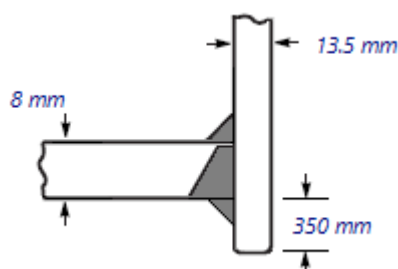
Flange Material		SA-105	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	350.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N5 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 26.4 / (118 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 0.8532 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (118.0 \cdot 1.0)] - 1))$
 $= 0.8932 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000	mm
Parallel to Vessel Wall, opening length	d	52.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm
Normal to Vessel Wall, Inward		17.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm
Normal to Vessel Wall, Inward		20.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N5 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.8532	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.8532, 5.8)$
 $= 5.8000 \text{ mm}$



Available Nozzle Neck Thickness = 13.4620 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	15.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	279.6 MPa	Passed
Occasional	:	3.2,	Allowable	:	156.9 MPa	Passed
Shear	:	12.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, tg = 8.0, tr = 7.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 1.0, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 3.74/10.20 = 0.367

Weld Size Calculations, Description: N5 2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.0025 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 567.5500 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.80 mm

Input Echo, WRC297 Item 1, Description: N5 2" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	506.000	mm
Shell Thickness	8.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	50.800	mm



Nozzle Thickness		13.4620	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	3.74	MPa
Radial Load	P	-3750.00	N
Circumferential Shear	Vc	2500.00	N
Longitudinal Shear	Vl	2500.00	N
Circumferential Moment	Mc	350000.00	N-mm
Longitudinal Moment	Ml	350000.00	N-mm
Torsional Moment	Mt	500000.00	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

**Warning - The ratio, D_n/T_n (5.774) is < 10
check the Limitations of WRC bulletin 297.**

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{(257.5(8.0 - 1.0))}$$

$$= 217.828 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	515.000	mm
Nozzle Outside Diameter	(d) =	77.724	mm
Vessel Thickness used	(T) =	7.000	mm
Nozzle Thickness used	(t) =	12.462	mm
T / t	=	0.562	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.295$$

Nr/P	=	0.092
Mr/P	=	0.186
M0/P	=	0.055
N0/P	=	0.119
MrD/Mc	=	0.299
NrDL/Mc	=	0.102
M0d/Mc	=	0.095
N0DL/Mc	=	0.051
MrD/Ml	=	0.163
NrDL/Ml	=	0.084
M0D/Ml	=	0.054
N0DL/Ml	=	0.131

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0



Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-29	29	29	-29
Normal Pressure Stress	133	137	133	137

Outplane Stress Summary	126	139	210	103
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	85	-85	85	-85
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-90	90	90	-90
Inplane Pressure Stress	66	66	66	66

Inplane Stress Summary	61	70	257	-93
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	7	7	7	7

Shear Stress Summary	10	10	4	4
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	128	140	257	197
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-52	52	52	-52
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66

Outplane Stress Summary	44	98	158	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	85	-85	85	-85
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-164	164	164	-164
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	133	137	133	137

Inplane Stress Summary	52	214	400	-96
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**Vessel Stresses (MPa):**

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-2	-2	2	2
Torsional Shear (Mt)		7	7	7	7
Shear Stress Summary		4	4	10	10

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		54	215	401	101

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		9	9	9	9
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-12	-12	12	12
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		7	12	7	12
Hoop Stress Summary		4	9	28	33

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		1	1	1	1
Axial Bending (P)		20	-20	20	-20
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-9	-9	9	9
Axial Bending (ML)		-21	21	21	-21
Axial Pressure Stress		3	3	3	3
Axial Stress Summary		-5	-3	56	-26

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		1	1	-1	-1
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		4	4	4	4
Shear Stress Summary		5	5	2	2

Nozzle Stresses (MPa):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		15	17	56	59

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the		Left	Left	Right	Right



hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-4	-4	4	4
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	12	7	12
Hoop Stress Summary	12	16	21	25

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	20	-20	20	-20
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-43	43	43	-43
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-27	18	78	-48

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	2	2	5	5

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	39	20	78	75

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	137	133	137	133	137	133	137
Circ. Pl (SUS)	-3	-3	21	21	-2	-2	16	16
Circ. Q (SUS)	-4	4	54	-54	-79	79	249	-249
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	0	0	14	14	5	5	13	13
Long. Q (SUS)	-5	5	175	-175	-27	27	77	-77
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	7	7	7	7	7	7	7	7
Pm (SUS)	133.0	137.0	133.0	137.0	133.0	137.0	133.0	137.0
Pm+Pl (SUS)	130.1	134.1	154.1	158.1	131.1	135.1	149.1	153.1
Pm+Pl+Q (Total)	127.2	139.2	255.5	199.3	54.4	214.2	398.3	99.6

**Vessel Stress Summation Comparison (MPa):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	158.05	207.00	Passed
Pm+Pl+Q (TOTAL)	398.33	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	12	7	12	7	12	7	12
Circ. Pl (SUS)	-3	-3	21	21	5	5	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-8	-8	10	10	-8	-8	10	10
Long. Q (SUS)	-1	1	41	-41	-23	23	63	-63
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	7.0	12.0	7.0	12.0	7.0	12.0	7.0	12.0
Pm+Pl (SUS)	9.2	14.1	28.1	33.0	17.1	22.1	20.1	25.1
Pm+Pl+Q (Total)	14.1	16.4	54.3	61.3	40.4	20.5	76.4	75.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	33.05	177.00	Passed
Pm+Pl+Q (TOTAL)	76.44	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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**Input, Nozzle Desc: K5A 2"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10312.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

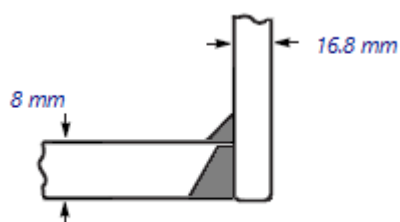
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K5A 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000 mm
Parallel to Vessel Wall, opening length	d	50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K5A 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: K5A 2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.9132 mm

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**Input, Nozzle Desc: K5B 2"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10312.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

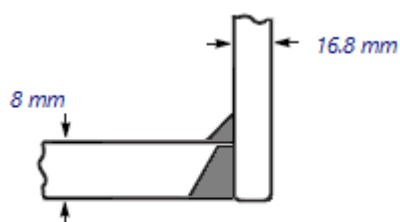
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K5B 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000 mm
Parallel to Vessel Wall, opening length	d	50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K5B 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: K5B 2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.9132 mm

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**Input, Nozzle Desc: K5C 2"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10312.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

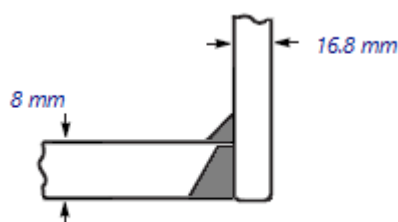
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K5C 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000 mm
Parallel to Vessel Wall, opening length	d	50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K5C 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: K5C 2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.9132 mm

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**Input, Nozzle Desc: P1C 2"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		10312.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

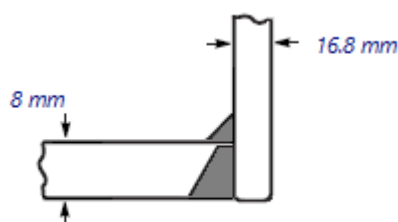
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: P1C 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000 mm
Parallel to Vessel Wall, opening length	d	50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P1C 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: P1C 2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.9132 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.80 mm

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**Input, Nozzle Desc: P1D 2"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		11812.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

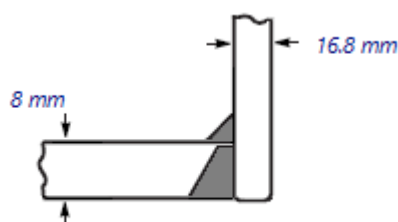
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: P1D 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000 mm
Parallel to Vessel Wall, opening length	d	52.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000 mm
Parallel to Vessel Wall, opening length	d	50.8000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000 mm

Taking a UG-36(c)(3)(a) exemption for nozzle: P1D 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 8.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 8.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 5.8000 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(8.0, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :**

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: P1D 2"

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 3.5382 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 217.9132 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 52.80 mm

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**Input, Nozzle Desc: A1 8"****From: 100**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	506.00	mm
Design Length of Section	L	2210.8201	mm
Shell Finished (Minimum) Thickness	t	8.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		12012.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
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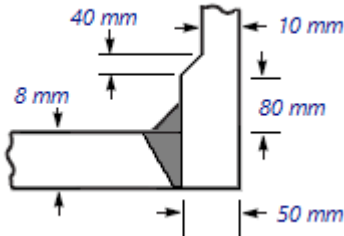
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	40.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		50.0000	mm

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: A1 8"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	8.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 254.0 / (138 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 7.0000 \text{ mm}$

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Mapnc]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 253.0 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 7.6117 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 102.6 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 2.8485 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 101.6 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 3.0567 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7525 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	410.4000 mm
Parallel to Vessel Wall, opening length	d	205.2000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 40.0 / 0.5773$
 $= 69.2821 \text{ mm}$



Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	D1	406.4000	mm
Parallel to Vessel Wall, opening length	d	203.2000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 137.0/138.0)$$

$$= 0.993$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n/S_v)$$

$$= \min(1, 137.0/138.0)$$

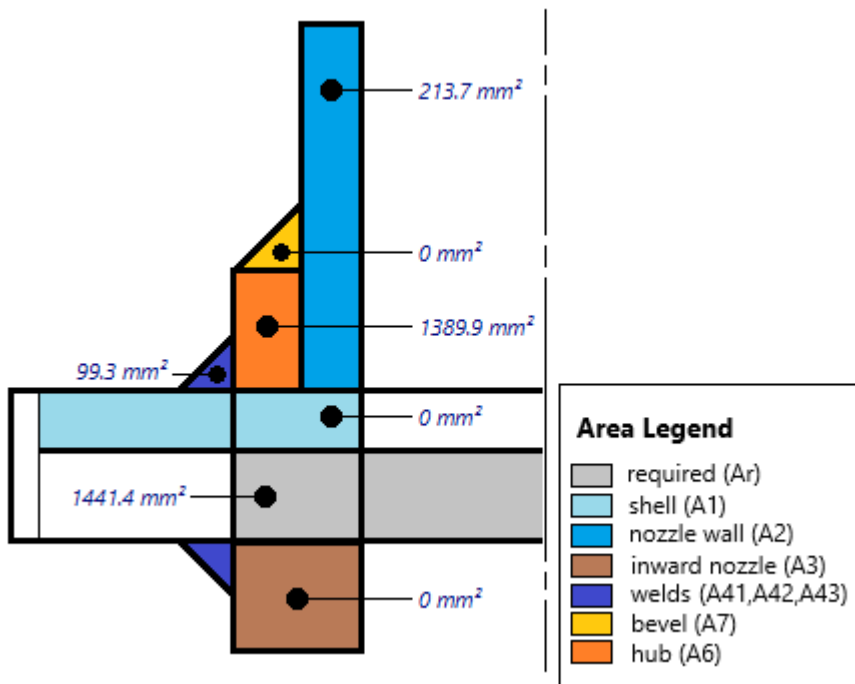
$$= 0.993$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(0.993, 1.0)$$

$$= 0.993$$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1441.356	314.439	1546.701
Area in Shell	A1	0.000	806.890	78.899
Area in Nozzle Wall	A2	213.748	286.577	277.734
Area in Inward Nozzle	A3	0.000	0.000	0.000
Area in Welds	A41+A42+A43	99.278	99.278	100.000
Area in Element	A5	0.000	0.000	0.000
Area in Hub	A6	1389.885	1389.885	1600.000
TOTAL AREA AVAILABLE	Atot	1702.910	2582.630	2056.632

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.



Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (205.2 \cdot 7.0 \cdot 1.0 + 2 \cdot 49.0 \cdot 7.0 \cdot 1.0 \cdot (1 - 0.99))$$

$$= 1441.356 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 205.2(1.0 \cdot 7.0 - 1.0 \cdot 7.0) - 2 \cdot 49.0$$

$$(1.0 \cdot 7.0 - 1.0 \cdot 7.0) \cdot (1 - 0.993)$$

$$= 0.000 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 17.5)(9.0 - 2.85) \cdot 0.9928$$

$$= 213.748 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.9928 + (0.0)^2 \cdot 0.9928$$

$$= 99.278 \text{ mm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(Tnp, ho, Hubht)) \cdot (Hubtk - tn) \cdot fr2$$

$$= (2 \cdot \min(17.5, 250.0, 80.0)) \cdot (50.0 - 10.0) \cdot 0.993$$

$$= 1389.885 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

$$\text{Wall Thickness for Internal/External pressures} \quad ta = 3.8485 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= ta$$

$$= 3.8485 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: A**

Govrn. thk, tg = 10.0, tr = 2.827, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.314$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 3.74/5.10 = 0.734$$

Weld Size Calculations, Description: A1 8"Intermediate Calc. for nozzle/shell welds $T_{min} = 7.0000 \text{ mm}$

**Results Per UW-16.1:**

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (1441.3563 - 0.0 + 2 \cdot 49.0 \cdot 0.9928 \cdot (1.0 \cdot 7.0 - 7.0)) \cdot 138)$$

$$= 198885.92 \text{ N}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - C_{an} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (1603.6328 + 99.2775 - 0.0 \cdot 0.99) \cdot 138$$

$$= 234976.55 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (1603.6328 + 0.0 + 99.2775 + (681.0438)) \cdot 138$$

$$= 328950.56 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (1603.6328 + 0.0 + 99.2775 + 0.0 + (681.0438)) \cdot 138$$

$$= 328950.56 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 303.2 \cdot 10.0 \cdot 0.49 \cdot 137$$

$$= 319690. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 127.1) \cdot (50.0 - 1.0) \cdot 0.7 \cdot 137$$

$$= 1876174. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 303.2 \cdot (8.0 - 1.0) \cdot 0.74 \cdot 138$$

$$= 340417. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (319690 + 1876174) = 2195864 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (319690 + 0 + 340417 + 0) = 660107 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (319690 + 340417 + 0) = 660107 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2195864 N	, must exceed W = 198885 N	or W1 = 234976 N
Path 2-2 = 660107 N	, must exceed W = 198885 N	or W2 = 328950 N
Path 3-3 = 660107 N	, must exceed W = 198885 N	or W3 = 328950 N



Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Approximate Maximum Allowable Pressure New & Cold: 4 MPa

Note: The M.A.P.(NC) of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 50.4502 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 308.4502 mm

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**Input, Nozzle Desc: N6 12"****From: 140**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	506.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

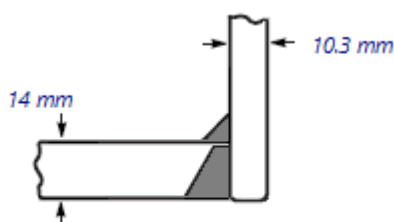
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N6 12"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	11.938 in.
Actual Thickness Used in Calculation	0.406 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.74 \cdot 508.0 \cdot 0.995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.74)$
 $= 6.8688 \text{ mm}$

Reqd thk per UG-37(a) of Elliptical Head, tr [Mapnc]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.08 \cdot 506.0 \cdot 1.0 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.08)$
 $= 7.4989 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 152.61 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 4.2370 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Mapnc]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 4.08 \cdot 151.61 / (138 \cdot 1.0 - 0.6 \cdot 4.08)$
 $= 4.5613 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9294 mm

The diameter limit is greater than the head diameter.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	610.4504 mm
Parallel to Vessel Wall, opening length	d	305.2252 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	23.2810 mm

The diameter limit is greater than the head diameter.

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	D1	606.4504 mm
Parallel to Vessel Wall, opening length	d	303.2252 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	25.7810 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.0 / 138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 137.0 / 138.0)$
 $= 0.993$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.993, 1.0)$
 $= 0.993$

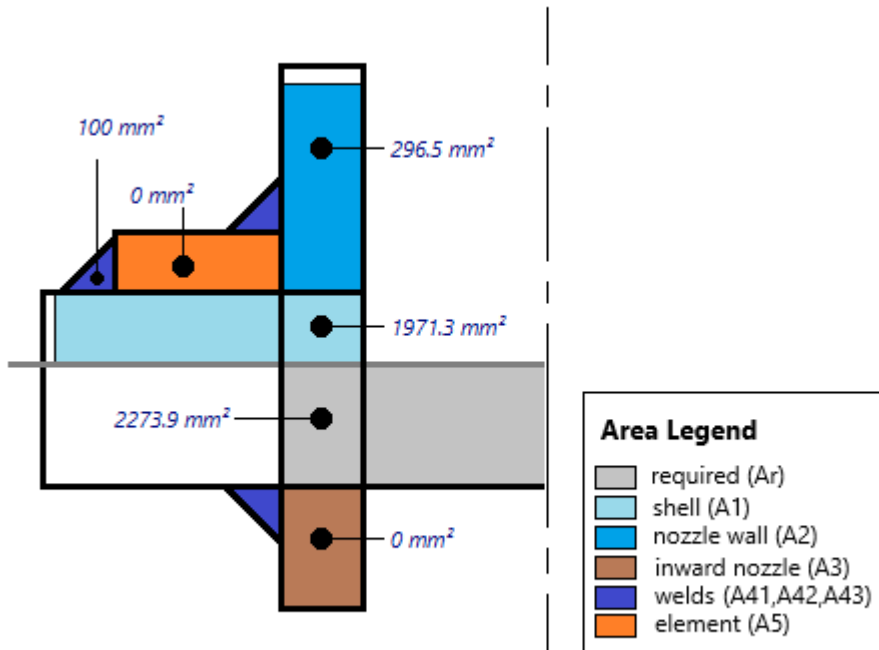


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2097.454	206.421	2273.851
Area in Shell	A1	1870.573	3553.699	1971.302
Area in Nozzle Wall	A2	234.614	387.506	296.539
Area in Inward Nozzle	A3	0.000	0.000	0.000
Area in Welds	A41+A42+A43	99.277	99.277	100.000
Area in Element	A5	0.000	0.000	0.000
TOTAL AREA AVAILABLE	Atot	2204.464	4040.482	2367.841

The MAP(nc) Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (303.2252 \cdot 7.4989 \cdot 1.0 + 2 \cdot 10.3124 \cdot 7.4989 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 2273.851 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 303.225 (1.0 \cdot 14.0 - 1.0 \cdot 7.499) - 2 \cdot 10.312$$

$$(1.0 \cdot 14.0 - 1.0 \cdot 7.4989) \cdot (1 - 1.0)$$

$$= 1971.302 \text{ mm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp) (tn - trn) fr2$$

$$= (2 \cdot 25.78) (10.31 - 4.56) 1.0$$

$$= 296.539 \text{ mm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can / 0.707)^2 \cdot fr2$$



$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 100.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 5.2370 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.8688 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.8688 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 9.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.331, \max(7.8688, 2.5)]$$

$$= 7.8688 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(5.237, 7.8688)$$

$$= 7.8688 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.312 = 9.023 \text{ mm}$ -> OK**UG-45 Minimum Nozzle Neck Thickness Requirement: [MAPnc]**

Wall Thickness for Internal/External pressures	ta = 4.5613 mm
Wall Thickness per UG16(b),	tr16b = 1.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 7.4989 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 7.4989 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 1.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.331, \max(7.4989, 1.5)]$$

$$= 7.4989 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.5613, 7.4989)$$

$$= 7.4989 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 10.312 = 9.023 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 52.6,	Allowable	: 137.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 291.1 MPa	Passed
Occasional	: 29.7,	Allowable	: 182.2 MPa	Passed
Shear	: 18.0,	Allowable	: 95.9 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, Curve: A**

Govrn. thk, tg = 9.023, tr = 4.206, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.524$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.023, tr = 4.206, c = 1.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.524$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: N6 12"

Intermediate Calc. for nozzle/shell Welds Tmin 9.3124 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (2273.8508 - 1971.302 + 2 \cdot 10.3124 \cdot 0.9928 \cdot (1.0 \cdot 14.0 - 7.4989)) \cdot 138)$$

$$= 60115.41 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (296.5386 + 0.0 + 100.0 - 0.0 \cdot 0.99) \cdot 138$$

$$= 54716.68 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (296.5386 + 0.0 + 100.0 + (286.66)) \cdot 138$$

$$= 94271.69 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (296.5386 + 0.0 + 100.0 + 0.0 + (286.66)) \cdot 138$$

$$= 94271.69 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 323.85 \cdot 10.0 \cdot 0.49 \cdot 137$$

$$= 341463. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 156.7688) \cdot (10.3124 - 0.0) \cdot 0.7 \cdot 137$$

$$= 487025. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 323.85 \cdot (8.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 415547. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:

$$= (\text{SONW} + \text{SNW}) = (341463 + 487025) = 828488 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (341463 + 0 + 415547 + 0) = 757010 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw}) \\ = (341463 + 415547 + 0) = 757010 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 828487 N , must exceed W = 60115 N or W1 = 54716 N

Path 2-2 = 757009 N , must exceed W = 60115 N or W2 = 94271 N

Path 3-3 = 757009 N , must exceed W = 60115 N or W3 = 94271 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 4 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Approximate Maximum Allowable Pressure New & Cold: 4 MPa

The Drop for this Nozzle is : 28.9976 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 292.9976 mm

Input Echo, WRC107/537 Item 1, Description: N6 12" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm
Vessel Diameter	Dv	910.800	mm
Vessel Thickness	Tv	14.000	mm
Design Temperature	T1	225.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	1.0000	mm
Nozzle Diameter	Dn	303.225	mm
Nozzle Thickness	Tn	10.312	mm
Nozzle Material		SA-105	
Nozzle UNS Number		K03504	
Nozzle Cold S.I. Allowable	SNmc	138.00	MPa
Nozzle Hot S.I. Allowable	SNmh	137.00	MPa
Design Internal Pressure	Dp	3.741	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-22500.0	N
Longitudinal Shear (SUS)	(V1) V1	15000.0	N
Circumferential Shear (SUS)	(Vc) V2	15000.0	N
Circumferential Moment (SUS)	(Mc) M1	12600002.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	12600002.0	N-mm
Torsional Moment (SUS)	Mt	18000002.0	N-mm



Use Interactive Control

No

WRC107 Version

Version March 1979

Include Pressure Stress Indices per Div. 2

No

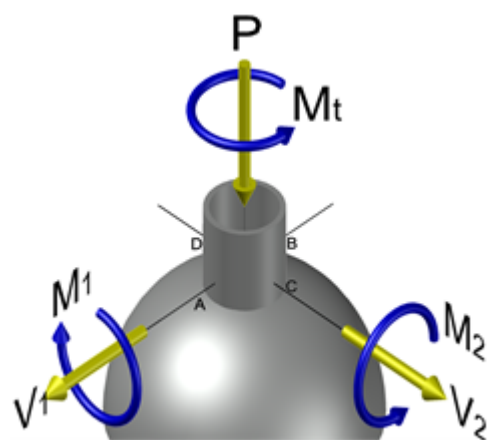
Compute Pressure Stress per WRC-368

No

Local Loads applied at end of Nozzle/Attachment

No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(462.9 (14.0 - 1.0))}$$

$$= 579.844 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-22500.0	N
Circumferential Shear	(VC) V2	15000.0	N
Longitudinal Shear	(VL) V1	15000.0	N
Circumferential Moment	(MC) M1	12600002.0	N-mm
Longitudinal Moment	(ML) M2	12600002.0	N-mm
Torsional Moment	MT	18000002.0	N-mm

Dimensionless Param: U = 2.09 TAU = 16.89 RHO = 4.00 (1.40)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 7	0.02288	(A,B,C,D)
M(x) / P	SP 7	0.00238	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.02063	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.00207	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.02063	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.00207	(A,B,C,D)
N(y) * T / P	SP 7	0.03985	(A,B,C,D)
M(y) / P	SP 7	0.00475	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.03635	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.00400	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.03635	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.00400	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00



Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Rad. Bend. P		1.9	-1.9	1.9	-1.9	1.9	-1.9	1.9	-1.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-19.8	-19.8	19.8	19.8
Rad. Bend. MC		0.0	0.0	0.0	0.0	-11.9	11.9	11.9	-11.9
Rad. Memb. ML		-19.8	-19.8	19.8	19.8	0.0	0.0	0.0	0.0
Rad. Bend. ML		-11.9	11.9	11.9	-11.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-26.8	-6.7	36.7	9.0	-26.8	-6.7	36.7	9.0
Tang. Memb. P		5.3	5.3	5.3	5.3	5.3	5.3	5.3	5.3
Tang. Bend. P		3.8	-3.8	3.8	-3.8	3.8	-3.8	3.8	-3.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-34.9	-34.9	34.9	34.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-23.1	23.1	23.1	-23.1
Tang. Memb. ML		-34.9	-34.9	34.9	34.9	0.0	0.0	0.0	0.0
Tang. Bend. ML		-23.1	23.1	23.1	-23.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-48.9	-10.3	67.1	13.4	-48.9	-10.3	67.1	13.4
Shear VC		2.3	2.3	-2.3	-2.3	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-2.3	-2.3	2.3	2.3
Shear MT		8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Tot. Shear		10.7	10.7	6.1	6.1	6.1	6.1	10.7	10.7
Str. Int.		53.2	21.6	68.3	17.7	50.5	14.9	70.5	22.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		65.7	65.7	65.7	65.7	65.7	65.7	65.7	65.7
Rad. Pl (SUS)		-16.8	-16.8	22.9	22.9	-16.8	-16.8	22.9	22.9
Rad. Q (SUS)		-10.0	10.0	13.8	-13.8	-10.0	10.0	13.8	-13.8
Long. Pm (SUS)		65.7	65.7	65.7	65.7	65.7	65.7	65.7	65.7
Long. Pl (SUS)		-29.6	-29.6	40.2	40.2	-29.6	-29.6	40.2	40.2
Long. Q (SUS)		-19.3	19.3	26.9	-26.9	-19.3	19.3	26.9	-26.9
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		2.3	2.3	-2.3	-2.3	-2.3	-2.3	2.3	2.3
Shear Q (SUS)		8.4	8.4	8.4	8.4	8.4	8.4	8.4	8.4
Pm (SUS)		65.7	65.7	65.7	65.7	65.7	65.7	65.7	65.7
Pm+Pl (SUS)		49.3	49.3	106.2	106.2	49.3	49.3	106.2	106.2
Pm+Pl+Q (Total)		43.2	68.0	134.0	83.4	40.5	63.6	136.2	87.8

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.71	138.00	Passed
Pm+Pl (SUS)	106.23	207.00	Passed



Pm+Pl+Q (TOTAL)	136.18	413.99	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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**Input, Nozzle Desc: N9 2"****From: 140**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	506.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	200.0000	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm

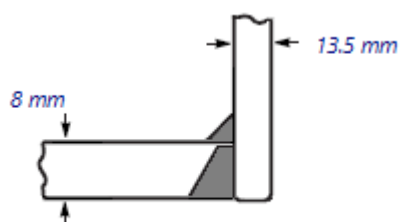
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	209.5500	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: N9 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.74 \cdot 508.0 \cdot 0.995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.74)$
 $= 6.8688 \text{ mm}$

Reqd thk per UG-37(a) of Elliptical Head, tr [Mapnc]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.08 \cdot 506.0 \cdot 1.0 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.08)$
 $= 7.4989 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 3.74 \cdot 26.4 / (137 \cdot 1.0 - 0.6 \cdot 3.74)$
 $= 0.7329 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([4.08 / (138.0 \cdot 1.0)] - 1))$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3775 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000	mm
Parallel to Vessel Wall, opening length	d	52.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: N9 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7329	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.8688	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.8688	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(7.8688, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:



$= \max(t_a, t_b)$
 $= \max(1.7329, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govern. thk, $t_g = 8.0$, $t_r = 6.869$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r * E^* / (t_g - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$

Weld Size Calculations, Description: N9 2"

Intermediate Calc. for nozzle/shell Welds T_{min} 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.6381 mm

The Cut Length for this Nozzle is, Drop + H_o + H + T : 219.1881 mm

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**Input, Nozzle Desc: K7 2"****From: 140**

Pressure for Reinforcement Calculations	P	3.7412	MPa
Temperature for Internal Pressure	Temp	225	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	150	°C
Maximum Allowable Pressure New & Cold		4.08	MPa

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Elliptical Head	D	506.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	8.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	200.0000	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

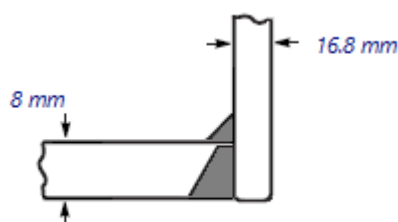
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	206.3750	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



**Reinforcement CALCULATION, Description: K7 2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 3.74 \cdot 508.0 \cdot 0.995 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 3.74)$
 $= 6.8688 \text{ mm}$

Reqd thk per UG-37(a) of Elliptical Head, tr [Mapnc]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 4.08 \cdot 506.0 \cdot 1.0 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 4.08)$
 $= 7.4989 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 26.4 \cdot \exp([3.74 / (137.0 \cdot 1.0)] - 1)$
 $= 0.7309 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Mapnc]
 $= R \cdot \exp([P / (S_n \cdot E)] - 1)$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot \exp([4.08 / (138.0 \cdot 1.0)] - 1)$
 $= 0.7618 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.3925 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	105.6000	mm
Parallel to Vessel Wall, opening length	d	52.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm

UG-40, Limits of Reinforcement: [Mapnc]

Parallel to Vessel Wall (Diameter Limit)	Dl	101.6000	mm
Parallel to Vessel Wall, opening length	d	50.8000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	20.0000	mm

Taking a UG-36(c)(3)(a) exemption for nozzle: K7 2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.7309	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 7.8688	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 7.8688	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 5.8000	mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[5.8, \max(7.8688, 2.5)]$
 $= 5.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(1.7309, 5.8)$
 $= 5.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:****Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B**

Govrn. thk, tg = 8.0, tr = 6.869, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -33 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $3.74/5.10 = 0.734$ Weld Size Calculations, Description: K7 2"

Intermediate Calc. for nozzle/shell Welds Tmin 9.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.9289 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.3039 mm

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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